On The Role of Trademarks: From Micro Evidence to Macro Outcomes

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Abstract

What are the effects of trademarks on the U.S. economy? Evidence from comprehensive micro data on trademark registrations and outcomes for U.S. employer firms suggests that trademarks protect firm value and are linked to higher firm growth and marketing activity. Motivated by this evidence, trademarks are introduced in a general equilibrium framework to quantify their aggregate effects. Firms invest in product quality and engage in both informative and persuasive advertising to build a customer base subject to depreciation. Persuasive advertising induces a perception of higher quality. Firms can register trademarks to reduce customer depreciation and enhance product awareness. The model's predictions about trademark registrations, firm growth, and advertising expenditures align with the empirical evidence. The analysis shows that, compared to the counterfactual economy without trademarks, the U.S. economy with trademarks generates higher average product quality but lower variety, ultimately resulting in greater welfare and higher industry concentration. While informative advertising improves welfare, persuasive advertising reduces it. Nevertheless, the positive welfare impact of trademarks outweighs the negative effects of persuasive advertising.

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1 Introduction

Nearly all firms strive to build a reputation and achieve broader consumer recognition. Trademarks enhance this recognition by formally and uniquely associating words, marks, pictures, or even a scent with a firm's goods or services, and by legally protecting the firm against imitation and infringement. This protection is crucial, as the damage to commerce from trademark infringement is economically significant. For instance, in 2019 the global trade in infringing products was 2.5% of world trade, and the businesses most affected were in OECD countries, including the United States.[1] The strength of institutions and enforceability matter for the extent of this damage. A U.S. International Trade Commission survey found that U.S. firms doing business abroad incur significant revenue loss due to intellectual property infringement, and sustain damages to their brand names and reputation.[2] Trademark infringement was the most common form of infringement cited: nearly 92% of firms that manufacture consumer goods reported material losses due to this type of infringement.

In the United States, trademark protections were introduced by The Lanham (Trademark) Act of 1946. The Lanham Act prohibits activities such as trademark infringement and dilution, and false advertising. Trademark registrations in the United States and other countries have been on the rise in recent decades (Figure 1).[3] Firms' marketing activities, measured by advertising expenditures, have also grown substantially (Figure 2). At the same time, reallocation towards productive and large firms has resulted in increasing concentration of economic activity.[4] These trends, taken together, suggest a connection between firms' marketing and brand protection efforts on one hand, and consumer reallocation and concentration, on the other. What role do trademarks play in this connection?

This paper studies the effects of *federally registered* trademarks in the United States, starting with firm-level evidence and building up towards the macroeconomic implications in a general equilibrium framework.[5] Empirically, this paper is the first to analyze the incidence and timing of trademark registrations over the life-cycles of employer businesses in the United States, based on the comprehensive matched trademark-firm data set introduced in Dinlersoz et al. (2018). It documents new facts about the characteristics of firms with trademarks and assesses the role of trademarks in firm growth by analyzing the effects on firms of the first trademark registration, a change in trademark law that strengthened trademark protection, and a trade shock that increased potentially infringing competition. Theoretically, it incorporates trademarks into a general equilibrium framework to study the key effects of trademarks on the U.S. economy. The framework emphasizes the fundamental roles of trademarks in increasing consumer awareness about firms and protecting firms from infringement and imitation. Consumer awareness and customer acquisition are achieved by both informative and persuasive advertising, with distinct and potentially conflicting contributions of these two types of marketing activity to the welfare effects of trademarks. The model is calibrated based on firm-level data and analyzed to quantify the significance

[1] See the 2021 OECD-EUIPO report Global Trade in Fakes: A Worrying Threat.

[2] See, e.g., the 2011 USITC report China: Effects of Intellectual Property Infringement and Indigenous Innovation Policies on the U.S. Economy.

[3] See also Dinlersoz et al. (2018) and Graham et al. (2018) for recent trends in trademark activity.

[4] See, e.g., Autor et al. (2020).

[5] Whenever there is no ambiguity, the words "trademark" or "registered trademark" will be used to refer to a "federally registered trademark" for the rest of the paper. Firms can have trademarks without registering them at the federal level, though such trademark ownership does not confer the broader benefits of federally registered trademarks that are the focus of this paper.

of the trademark system for welfare, concentration, quality provision, and product variety.

The main data set used in this paper contains information on trademark registrations from the USPTO's Trademark Case File Database (TCFD), combined with firm-level information from the Longitudinal Business Database (LBD), allowing the observation of key events in a firm's life cycle, such as firm birth, subsequent growth, first trademark registration, and firm death. Data on firm innovation (patenting), broader market expansion (exporting activity), and advertising expenditures are also added. Empirical analysis characterizes firms with registered trademarks. Firms that are born large, innovate early, grow rapidly, and expand geographically are more likely to register trademarks, and tend to do so earlier in their life-cycles. Firms with trademarks are in general much larger, and exhibit higher growth rates compared to those without trademarks. Furthermore, firms with trademarks engage in more intense marketing. They have, on average, higher advertising expenditures before trademark registration, and these expenditures grow after registration.

The empirical analysis offers new evidence on the extent registered trademarks matter in protecting and enhancing a firm's value. Measuring the effects of trademarks on firms is complicated by potential selection bias. To explore the link between trademark registrations and firm outcomes, the analysis relies on three exercises. First, it compares firm outcomes before and after the first trademark registration for firms with trademarks and a set of matched control firms with no trademarks. Firms that register a trademark exhibit significantly higher growth following the registration compared to their matched counterparts. Second, it studies the effects of the 1984 Trademark Counterfeiting Act, which substantially strengthened the protections against trademark infringement and counterfeiting. After the Trademark Counterfeiting Act, firms with registered trademarks as of 1984 performed significantly better compared to a set of matched control firms that did not have trademarks, and the effects were stronger for firms that likely have higher exposure to infringement: firms with more established trademarks, large firms with trademarks, and trademark firms in historically high-infringement industries. Third, the analysis exploits a trade shock that increased foreign competition for U.S. firms and the likelihood of depreciation in their customer base. The entrance of China into the World Trade Organization (WTO) and the resulting rise in U.S. imports from China had an adverse effect on some U.S. firms, especially on those with a higher degree of exposure to import competition. The findings indicate that firms with trademarks registered prior to the shock fared better relative to a matched control group without trademarks, and the protective effects were larger for firms with greater exposure to the shock. In other words, registered trademarks were instrumental in protecting domestic firms against a shock from foreign competition.

Motivated by the empirical evidence, a model is proposed where consumer information about firms is central. Firms first invest in product quality, and engage in a persuasive advertising campaign, which, if successful, enhances consumers' perception of the quality of the firm's product. Firms then gradually grow by building a customer base through informative marketing. However, they also experience customer depreciation for a variety of reasons, including imitative or infringing competition. At any point in their life-cycles, firms can choose to register a trademark to mitigate customer depreciation. Building on the insights by Landes and Posner (Landes and Posner (1987, 1988, 2003)), trademarks play two fundamental roles in the model: (i) reduce informational frictions between consumers and firms by helping consumers uniquely identify a firm and its product, thereby making it easier for firms to reach and attract consumers, and (ii) protect a firm's customer base from depreciation through imitation and infringement. The first

role is only possible because of the second, since it is the protection from imitation that prevents consumer confusion and allows identification of a firm's product by consumers. These roles are critical for the provision of quality and marketing. By reducing customer depreciation, trademarks protect firm value and encourage investment in quality and marketing. At the same time, trademarks distinguish firms and increase consumer awareness, helping firms more effectively disseminate information and increase the efficiency of their marketing.

The model has predictions on which types of firms register trademarks, and when they do so in their life cycles, as well as the effect of trademarks on firm growth and marketing. Firm-level data on trademark registrations and firm outcomes are used to discipline the model. To assess the effects of trademarks, the model economy with trademarks is compared to a counterfactual economy with no trademarks. The economy with trademarks exhibits higher average quality but lower product variety. In particular, high-quality firms with trademarks engage in more intense marketing, acquire more consumers, and can better retain their customers due to the protection provided by trademarks. As a result, they grow faster and achieve larger sizes, at the expense of smaller, lower quality firms–leading to lower product variety. Industry concentration is higher in the economy with trademarks, as more consumers, and inputs of production, are allocated towards higher quality, larger firms with trademarks. Overall, there are significant welfare gains from the trademark system in the United States due to the more efficient allocation of production and consumption towards higher-quality firms, even though overall consumer information and product variety are lower. Further analysis also indicates that trademark protection mitigates the adverse effects of an exogenous increase in imitative and infringing activity on product quality and welfare. The analysis also indicates that stronger trademark protection results in higher welfare but increases concentration. Finally, removing persuasive advertising also leads to an increase in both welfare and concentration. While persuasive advertising acts to dampen the welfare gains from trademarks, it cannot, under reasonable parameter values, entirely erase these gains.

There is a growing empirical literature on the connection between trademarks and firm outcomes. Due to a lack of comprehensive data linking firm outcomes with registered trademarks in the United States, existing empirical literature has mainly developed around the data for other countries. In particular, research with European and Australian data has documented a number of empirical regularities.[6] Prior empirical work using U.S. data has been mainly limited to publicly-traded firms.[7] Systematic evidence on the timing and incidence of trademark registrations, and on the connection between firm dynamics and trademark ownership has been lacking for the broader set of U.S. firms. This paper provides new evidence on these fronts.

The implications of trademark protection for the aggregate economy have not been studied in a general equilibrium context. This paper fills this gap by using a macroeconomic framework with firm dynamics informed by micro data moments based on the matched trademark-firm data for all employer businesses in the United States. The modeling approach is influenced by early seminal work on the

[6] Schautschick and Greenhalgh (2016) survey provides an excellent review of this literature. In the case of China, Alfaro et al. (2022) analyze various effects of the China's Trademark Law of 1923 from a historical perspective.

[7] For instance, Heath and Mace (2020) use the introduction of the 2006 Federal Trademark Dilution Act (FTDA) and find that the additional protection offered by FTDA to publicly-traded firms with famous trademarks have economically significant effects for these firms. Again using publicly-traded firms, Kost et al. (2019) provide some evidence on the effects of trademark acquisitions and sales on firm markups and profits. Similarly, Pearce and Wu (2022) use a match of the trademark data to brands in RMS NielsenIQ scanner data for a subset of products and firms in the economy to study brand reallocation among firms.

economics of trademarks (Landes and Posner (1987, 1988, 2003) and Economides (1988, 1998)). These studies emphasize the role of trademarks in reducing consumer search costs and protecting a firm against imitation and infringement, thereby enhancing the provision of product quality. This earlier body of work neither studies the connection between trademarks and firm dynamics, nor offers models to assess the macroeconomic implications of trademarks. The model is also related to recent work on the role of customer capital in firm and industry dynamics.[8] Here, the specific role of trademarks in building and protecting a customer base is highlighted. The role of marketing in reducing informational frictions between consumers and firms is similar to that in Dinlersoz and Yorukoglu (2012), which emphasizes the informative role of advertising.[9] Here, persuasive advertising is also included, and its role in welfare effects of a trademark system is analyzed. Cavenaile et al. (2024) also study persuasive advertising in a macroeconomic model to explore how firms' investments in advertising and innovation impact competition and growth. While advertising can boost allocative efficiency by channeling resources toward more efficient firms, it may also crowd out R&D investment and reduce overall welfare, consistent with the effects of persuasive advertising in this study.

The rest of the paper is organized as follows. Empirical evidence is presented in Section 2. The model is introduced in Section 3. Section 4 studies the properties of the stationary equilibrium. Section 5 discusses calibration and the properties of the benchmark economy. The significance of the U.S. trademark system based on the calibrated model is quantified in Section 6. Section 7 concludes.

2 Firm-level evidence on the effects of trademark registrations

This section presents evidence on the connection between trademarks registered at the federal level and various firm outcomes. Summary statistics are presented of the relationship between trademarking, firm characteristics, and the firm life-cycle. Then three complimentary analyses are considered. The first measures the effects of a firm's first trademark registration using a matched control sample. The second measures the role of the increased protection afforded to existing trademarks by the Trademark Counterfeiting Act of 1984. Finally, the third considers the protective effects of existing trademarks in the face of an exogenous shock to competition via China's entrance to the WTO in 2002.

2.1 Data

The analysis utilizes the trademark-firm database introduced in Dinlersoz et al. (2018). The database pertains to the period 1976-2014, and contains information on trademarks from the USPTO Trademark Case Files Dataset (TCFD), which covers all federal trademark applications and registrations in the U.S. since 1870.[10] The information on trademarks includes the name and location of the registering business, dates of application and registration, trademark classification, and other events associated with a trademark. Dinlersoz et al. (2018) link the TCFD with the U.S. Census Bureau's Business Register

[8] See, e.g., Dinlersoz and Yorukoglu (2012); Gourio and Rudanko (2014); Roldan and Gilbukh (2018). Ignaszak and Sedláček (2022) analyze economic growth implications of innovation and gradual customer acquisition by firms.

[9] See, for instance, Butters (1977); Grossman and Shapiro (1984); Stegeman (1991) for foundational analyses of informative advertising. See also Dinlersoz and Yorukoglu (2008) for an analysis of informative advertising in a competitive industry where firms differ in production efficiency. In a recent paper Greenwood et al. (2024) develop a similar information-based model where traditional and digital advertising finance the provision of free media goods and affect price competition.

[10] See Graham et al. (2013) for details on the construction and contents of the TCFD.

(BR), administrative data for the universe of non-farm employer businesses in the United States. The TCFD and BR were matched at the firm-level using name and address matching techniques. About 83% of the trademarks match with firms for the sample period, most with a high degree of precision – see Dinlersoz et al. (2018).

Information from the Longitudinal Business Database (LBD) was added to analyze firm outcomes in relation to trademark activity. The LBD provides a longitudinally-linked version of the BR at the establishment level starting in 1976, and enables identification of entry and exit of firms and their establishments (Jarmin and Miranda, 2002; Chow et al., 2021). Key firm-level variables from the LBD include employment, payroll, and revenue, as well as geography and industry classifications.[11] The trademark-firm data thus allows tracking of key events in a firm's life-cycle: firm birth and growth, first trademark application and registration, firm geographic expansion, and firm death. This data is combined with information on firm innovation and global engagement: patenting activity from USPTO, and exporting activity from Longitudinal Firm Trade Transactions Database (LFTTD). In addition, data on firm-level exposure to trade is used to analyze the protective role of trademarks against an exogenous change in the competition faced by firms. The trademark-firm data was further supplemented with data on firm-level advertising expenditures (cost of advertising) reported in the U.S. Census Bureau's Economic Census (EC) and Annual Survey of Manufacturers (ASM). While the data on advertising expenditures mainly pertain to manufacturing firms, it offers a rare opportunity to compare the advertising expenditures by firms that have registered trademarks versus those without, and the intensity of marketing activity before and after a firm registers its first trademark.

The analysis is restricted to U.S. employer firms and their trademark activity.[12] The focus is on the critical event of the first trademark registration by a firm at the federal-level.[13] By simply engaging in commerce, firms in the United States gain limited common law protection for their names and brands in a specific local geography. Firms can also register trademarks at the state level, which provides protection in a given state. A federal registration, however, offers much stronger and broader protection and benefits. Among the key benefits are nationwide protection, validity and eventual incontestability of a trademark (5 years after registration), protection against infringing imports, ability to defend the trademark in federal courts, and enhanced remedies in case of infringement. The general equilibrium effects of these benefits is the main concern of this paper.

Registering a trademark at the federal level requires an application to be filed with the USPTO, which determines whether the trademark in consideration is legally protectable and if there is a "likelihood of confusion" with a previously registered trademark. Filing an application entails a relatively small upfront fee, ranging from $175 to $375 over the analysis period. This fee, however, reflects only a fraction of the total costs associated with registering, maintaining, and defending a trademark. For

[11] Firm-level industry is determined by iteratively selecting the classification code with the largest employment first at the 2-digit-level, then the 3-digit, and so on to the 6-digit-level. Firm-level geography is determined by identifying the geography code with the most employment.

[12] Trademark filings by foreign entities in the United States are excluded. There is no detailed information available about the characteristics and dynamics of these entities.

[13] In contrast, Dinlersoz et al. (2018) focus only on trademark applications, which signal an intent to register, but may not turn into actual registrations for reasons such as the failure to successfully launch a brand or a product and the associated lack of commercial activity. The analysis here considers the trademarks that are granted federal registration, for which the proof of the presence of commercial activity associated with the applied-for trademark is a necessary condition for registration. The trademark activity after the first registration is not independent of the initial registration and it is more challenging to assess the separate contribution of these additional registrations to firm dynamics.

example, an extensive search by an attorney is typically needed to ensure that the proposed trademark is sufficiently distinguished from the existing registered marks. Post-registration, the periodic renewal of registration and monitoring for potential infringement can be costly. Moreover, the legal costs of defending a trademark against infringement can be very large.

2.2 Trademark registrations and firm life-cycle

Table 1 contains descriptive statistics about firms with and without registered trademarks in 2014.[14] While only about 4% of firms had a registered trademark, these firms accounted for nearly half of the non-farm private sector employment. The average size of firms with a registered trademark was about 285 employees, 23 times the average size (12) for firms without trademarks. These differences in size are even more stark at the right tails (top 1%) of the firm size distributions for firms with vs. without registered trademarks.

While firms are more likely to obtain a trademark early in their life-cycle, among newborn firms trademark activity is still a rare event. About 0.9% of firms born in 2014 registered a trademark in that year, and they accounted for 1.5% of the total employment of newborn firms in 2014. Firms that own a trademark as of 2014 registered their first trademark about 6 to 7 years after birth on average, but the distribution of age at first trademark is highly skewed towards the first few years of a firm's existence. Overall, trademark registration is uncommon. Still, it is much more prevalent than other key firm outcomes, such as R&D, patenting, or exporting – less than 1% of U.S. firms export or have patents, while around 2.5% perform R&D (Foster et al. (2020)).

Recent literature on firm growth suggests that early firm characteristics and initial conditions are relevant for subsequent firm dynamics (Sedláček and Sterk (2017), Bayard et al. (2018), Guzman and Stern (2020), Sterk et al. (2021)). If trademark registrations reflect underlying superior firm characteristics (e.g., higher product quality or growth potential), registrations should be positively correlated with early indicators of firm success. Furthermore, a federal trademark registration is more likely to benefit firms that grow fast, innovate, and expand beyond their local markets – firms that are more likely to be exposed to infringement and imitation, and need the broader protection provided by federal registration.

Observable dimensions of this selection into trademark registrations are explored in Table 2. The first column presents estimates from a linear probability model that relates the indicator of a trademark registration within the first five years after a firm enters to initial firm characteristics.[15] Firms that are larger at birth and grow faster within their first year are more likely to register a trademark. Firms in the top 5 percent of the distributions of initial size and first-year growth rate have 2.2 and 2.3 percentage points higher likelihood of trademark registration, respectively. Firms that innovate or expand to broader markets early on are also more likely to register. Notably, having a patent upon birth is associated with a 16.8 percentage points higher likelihood of trademark registration, and early exporting activity is associated with a 3 percentage points increase in trademark propensity. All of these estimates are economically significant, given the small overall likelihood of registered trademark ownership (4%) and

[14]The final year of analysis is set to 2014 in order to be able to use the subsequent years in the LBD to study early firm dynamics and its relation to trademark activity.

[15]This analysis excludes firms born before 1976 as their birth year is not known and their initial conditions cannot be measured. This estimation also makes no claim to causality and simply identifies early firm characteristics associated with early trademark behavior.

the likelihood of trademark registrations at birth ($< 1\%$).

The second column in Table 2 uses the duration (in years) between firm birth and first trademark registration as the dependent variable (conditional on registration). The estimated coefficients indicate that firms born larger and grow faster register their first trademark earlier. Similarly, firms with patents and those that expand beyond local boundaries (multi-state and exporting firms) in their first year tend to register earlier.

Early growth patterns are markedly different for firms that have a trademark registration at some point in their life-cycle. Figure 3 shows the average employment growth rates for the first five years of a firm.[16] Conditional on survival (Panel (a)), firms with trademarks grow twice as fast in their first year with the gap between growth rates increasing over time. The growth rate of trademark firms in the fifth year is about 4.2 times that of non-trademark firms. Large differences are also observed in unconditional growth rates (Panel (b)).

Beyond the first five years, the size gap between the two types of firms continues to expand — see Table 3. At birth (age zero), firms with no trademark have an average size of 5.6 employees, compared to 15.2 employees for firms with a trademark, implying a relative average size of 2.7. The relative size grows over time, conditional on survival, reaching about 20 by age 30. By age 30, the average size of trademarking firms is about 320 employees versus 16 for firms without a trademark. Firms with a trademark achieve a much larger average size over their life-cycles.

2.3 Firm outcomes before and after first trademark registration

Consider next firm outcomes before and after a firm's first trademark registration. To control for the selection into registered trademark ownership and examine the effects associated with registered trademarks, firms with a trademark registration are matched to a set of control firms that *never* register a trademark in their lifetime. Firms are matched based upon employment size deciles in the year of first trademark registration (t) and the prior four years ($t-1$ to $t-4$). Trademarking and control firms are also required to have, in the year of first trademark registration, the same 4-digit NAICS industry, location (state), and firm age. A single matched control firm is selected among candidate control firms that meet these criteria by minimizing the difference in employment in each year from t to $t-4$.[17]

The differences between trademark firms and their matched controls are analyzed based on the following specification:

$$y_{it} = \alpha + \sum_{j \neq -1} \beta^{TM} TM_i + \sum_{j \neq -1} \beta^j d_j + \sum_{j \neq -1} \gamma^j (TM_i \times d_j) + \lambda_i + \varepsilon_{it}, \tag{1}$$

where y_{it} is the firm employment or revenue in log, d_j ($j \in \{-5, ..., 5\}$) indicate years before and after the first registration event ($j = 0$), TM_i is an indicator of treatment, and λ_i are event year (year of first

[16] The analysis of growth rates is restricted to firms born after 1976. Organic firm-level growth rates are computed as the difference in employment divided by average employment in t and $t-1$. This measure, first developed by Törnqvist et al. (1985), has become standard in the firm dynamics literature (Davis et al., 1998).

[17] Specifically, the Euclidean distance in employment between trademark and non-trademark firms for the entire pre-registration period is minimized. For each candidate treated-control pair, (i, j), the pre-treatment employment distance is computed $EmpDiff_{i,j} = \sum_{k=-4}^{0} (Emp_{i,k} - Emp_{j,k})^2$,where $k = 0$ denotes the year of registration and values of $k < 0$ denote years before registration (up to 4 years). Ties are broken randomly. Once a control firm is selected for a given trademark firm, it cannot be selected as a control for any other trademark firm.

trademark) fixed effects.

The estimates for γ^j based on specification (1) measure the difference in y_{it} between trademarking and control firms in each year before and after the year of first trademark registration. The estimates of γ^j are shown in Figure 4.[18] As shown in Figure 4a, the trademark firms and their controls have similar trends before the registration, but trademark firms reach much higher employment levels following the first registration. Based on the estimated semi-elasticities γ^j, five years after the first registration the trademark firms' employment is about 23% higher – about 4.2% in annualized terms. Despite differences in levels, Figure 4b also indicates no systematic difference in pre-trends for log revenue even though revenue was not used in matching. The treated firms' average revenue is higher by 32% five years after the registration event relative to the control group, corresponding to about 5.6% in annualized terms.

Overall, the evidence in this section suggests trademark registrations contribute significantly to firm growth. The next section exploits a change in trademark law that strengthened trademark protections to provide additional evidence on the treatment effects associated with registered trademarks.

2.4 Evidence from Trademark Counterfeiting Act of 1984

The Trademark Counterfeiting Act of 1984 was introduced to address the problem of trafficking in counterfeit trademark goods – a severe form of trademark infringement. The Act amended the 1946 Lanham Act to create stronger remedies in cases involving the intentional use of a counterfeit trademark. It established penalties of up to five years' imprisonment and/or substantial monetary fines for trafficking in counterfeit goods or services.[19] Critically, it allowed the injured owner of a registered trademark to sue for increased (treble) fines based on the damages, and enabled the Federal district courts to issue orders for the seizure and destruction of counterfeit goods. All of these provisions of the Act substantially strengthened trademark protection.

It is important to emphasize that the proof of the presence of "counterfeiting" and many of the remedies provided by the Act require that the infringed trademark be federally registered. The provisions of the Act therefore disproportionately benefit firms with registered trademarks by credibly deterring the infringement activity these firms are exposed to. In addition, firms with more established trademarks and larger firms with trademarks stand to benefit more, since their exposure to infringement is likely greater. Similarly, the effects are expected to be larger for trademark firms in industries with historically high infringement activity, as these firms likely gain disproportionately from the greater deterrence provided by the Act.[20]

To assess how firms with trademarks performed before and after the Act, we compare firms that owned registered trademarks as of 1984 with a set of matched controls with no such trademarks, using a similar matching strategy as in Section 2.3. Consider a specification that follows treated and control

[18] For the set full set of estimates from specification (1), see Appendix Table A1.

[19] The fine was $250,000, but much higher ($1,000,000) for a corporation or other legal entity.

[20] See Heath and Mace (2020) for a similar analysis based on the Federal Trademark Dilution Act (FTDA) of 1996, which provided additional protection against the dilution of "famous" trademarks. What constitutes a famous trademark is not defined precisely by FTDA – Heath and Mace (2020) use trademark age to designate famous trademarks. They study the effects of FTDA on large publicly traded firms with famous trademarks. Compared to FTDA, Trademark Counterfeiting Act is a broader law applicable to all registered trademarks, with provisions to directly address counterfeiting and infringement.

firms in the years before and after the Act

$$y_{it} = \alpha + \sum_{j \neq 1983} \beta^j d_j + \sum_{j \neq 1983} \gamma^j (TM_i \times d_j) + \tau_t + f_i + \varepsilon_{it}, \quad (2)$$

where y_{it} is the firm employment, payroll, or establishment count (all expressed in logarithms). Firm revenue is not included in these analyses as it is only available in the LBD starting in 1997. The dummies d_j ($j \in \{1981, ..., 1989\}$) indicate years before and after the 1984 Act, and TM_i is an indicator of treatment (trademark presence) as of 1984, and τ_t and f_i are year and firm fixed effects. Another version of (2) with treatment as of 1981 (instead of 1984) is also considered to assess whether firms with more established trademarks experienced larger effects.[21]

Panels (a) and (b) of Figure 5 shows the evolution of employment and payroll around the Act for firms with trademarks as of 1984 and 1981, respectively. Panels (c) and (d) show differences in establishment counts before and after the Act for firms with trademarks as of 1984 and 1981, respectively. Figure 5 indicates that pre-trends are similar for the treated and control groups, but by the end of the post-period the treated firms have 10%, 10%, and 5% higher employment, payroll, and establishment count, respectively. These patterns suggest that the Act benefited firms with registered trademarks as of 1984. When the firms that registered trademarks earlier (as of 1981) are considered, the results are more pronounced for employment and payroll, with the corresponding magnitudes of 11% and 13%, respectively.[22] This suggests that firms that had more established trademarks with potentially higher exposure to customer depreciation benefited more from the Act. Note also that the effects for employment are smaller than the ones obtained in Section 2.3. The Act introduced additional protection to trademark holders. The analysis in Section 2.3, in contrast, measured the extensive margin effect of having trademark protection by identifying the year of first trademark registration.

Next, further analysis is implemented for treated and control firms by incorporating the potential heterogeneity in the effects of the Act with respect to firm size and exposure to infringement in the industry a firm belongs to. The specification we estimate is

$$y_{it} = \alpha + \beta^{PT} \times Post_t \times TM_i + \beta^{PC} \times Post_t \times Category_i + \beta^{PTC} \times Post_t \times TM_i \times Category_i + \tau_{it} + f_i + \varepsilon_{it}, \quad (3)$$

where y_{it} is firm i's outcome in year t, TM_i is an indicator of trademark ownership by firm i as of 1984, $Category_i$ is an indicator of firm category, and $Post_t$ is an indicator of years $t \geq 1984$. Industry×year and firm fixed effects, τ_{it} and f_i, are also included. Firm category indicates either the large firms as of 1984, $Category_i = Large_i$, or the firms in a high infringement industry, $Category_i = High_i$. We consider a threshold of 250 employees to define large firms.[23] High infringement industries were identified based on the historical volumes of infringement activity provided by the Customs and Border Patrol.[24]

[21] Firm outcome measures in the Longitudinal Business Database start in 1976, with no annual history for firms born before 1976 - which we omit. 1981 was thus chosen to strike a balance between obtaining a longer trademark history and a large enough sample size that contains firms born after 1976 for which we can confidently determine trademark registration.

[22] Note that the smaller sample size for firms with trademarks as of 1981 result in larger confidence bands.

[23] Higher thresholds were also used and the results were generally robust – however, very large size classes have only a small number of firms with trademarks so the precision of the estimates is lower for such classes.

[24] The following industries were classified as high infringement based on the monetary value of seizures in the IPR Annual Seizure Statistics released by the CBP annually over the period 2003-2019: Watches/Jewelry – Naics 33991, 42394, 448; Wearing Apparel/Accessories – Naics 315, 4243, 44815, 44819; Handbags/Wallets – Naics 316, 33991; Footwear – Naics 316,

Table 4 shows the results from the estimation of (3). There is substantial heterogeneity in outcomes. Trademark firms that are large and trademark firms in industries exposed to higher infringement have better outcomes in the post-period. The evidence in this section suggests that stronger trademark protection leads to better outcomes for firms with registered trademarks, in support of trademarks' role in protecting firm value by deterring and reducing infringement. For further evidence, the next section uses a different approach: an exogenous shock to import competition to assess how firms with trademarks registered before the shock fared compared to those without such trademarks.

2.5 Evidence from a trade shock

Do trademarks protect firm value in the presence of rising competition that can increase customer depreciation? The analysis here considers a trade shock, China's entrance to WTO in 2002, which has been used to assess the effects of an exogenous shock to competition for U.S. firms (See Acemoglu et al. (2016); Autor et al. (2013, 2016)). Some of the competition associated with this shock is potentially imitating and infringing in nature. A key benefit of a federally registered trademark is the ability to record the registration with U.S. Customs and Border Protection and the International Trademark Commission to deter or stop importation of infringing foreign goods. In 2020 alone, U.S. authorities seized 26,503 shipments of goods infringing intellectual property—the value of the seized goods exceeded $1.3 B.[25] The credible threat of protection against infringing imports provided by trademarks can help maintain firm value in the presence of increased import competition. The analysis focuses on firms with trademarks registered before the shock and investigates whether this pre-determined protection was instrumental in preventing the erosion of firm value.

The post-shock performance of firms with trademarks registered before the shock is compared with that of matched control firms with no such trademarks.[26] The main specification is

$$y_{it} = \alpha + \beta^{C} \times IC_{it} + \beta^{PT} \times Post_t \times TM_i + \beta^{PC} \times Post_t \times IC_{it} + \beta^{PTC} \times Post_t \times TM_i \times IC_{it} + \tau_t + f_i + \varepsilon_{it}, \quad (4)$$

where y_{it} is firm i's outcome in year t, TM_i is an indicator of trademark ownership by firm i before the shock ($t < 2001$), IC_{it} is a firm-level import competition measure, and $Post_t$ is an indicator of years after the shock ($t \geq 2001$). Year and firm fixed effects, τ_t and f_i, are also included. The import competition measure IC_{it} is defined as

$$IC_{it} = \sum_j s_{ijT} \frac{m_{jt}}{o_{jt} + x_{jt} - m_{jt}}, \quad (5)$$

where s_{ijt} is firm i's employment share in industry j in pre-shock year $T = 2000$, m_{jt} is Chinese imports in industry j in year t, o_{jt} is domestic output in industry j in year t, and x_{jt} is exports in industry j in year t.[27]

To account for the potential endogeneity associated with the import penetration measure, an instrumental variables approach is used. The instrument, IV_{it}, for IC_{it} is identical to the one used in Autor

45111, 42434; Consumer Electronics – 3341, 3343, 3346, 334511, 334512, 33521, 33522, 443, 4326, 42362; Cigarettes – 31223, 33999, 32599, 42494, 453991; Pharmaceuticals/personal care – 3254, 4242, 446, 32562.

[25] See IPR Annual Seizure Statistics.

[26] A matching strategy similar to that in sections 2.3 and 2.4 was used.

[27] The sample is restricted to firms that have non-zero value for IC_{it} to exclude firms that are not subject to the shock – these firms are largely outside of manufacturing.

et al. (2013, 2016) and is obtained by replacing m_{jt} in (5) with Chinese imports in other developed countries, $\widetilde{m}_{jt}$.[28] Table 5 presents the second-stage results from the estimation of the model (4) using a two-stage least squares IV approach.[29] Three different firm outcomes are used as the dependent variable: employment, revenue, and labor productivity (revenue per worker), all log-transformed. Two versions of (4) are estimated for each outcome. The first version (columns 1, 3, and 5) measures existing registered trademark ownership as of 2001. The second version (columns 2, 4, 6) instead uses 1999 as the year for this measurement, as a robustness check to account for the possibility that some trademarks may have been registered close to 2001 in anticipation of the shock.

The time series for average outcomes for the treated and control firms do not indicate any systematic differences in trends before 2001 – see Appendix Figure A1. In Table 5, β^{PC} is negative and significant in specifications for employment and revenue – a higher import penetration in the post period implies worse employment and revenue outcomes for firms without a trademark. For these firms, a one standard deviation increase in the import competition measure in the post-period is associated with a 11.2% decline in employment, a 9.5% decline in revenue, and a 0.4% decline in labor productivity. However, the positive and significant estimate for β^{PTC} in all specifications suggests that pre-existing trademark ownership mitigated the adverse effects of import competition: firms with a trademark experience less negative outcomes as the degree of exposure to imports increases in the post period. Based on specifications 1, 3, and 5, a one standard deviation increase in the import competition measure is associated with a 1.1 pp lower reduction in employment and 6.1 pp lower reduction in revenue, resulting in a 3.5 pp higher labor productivity on average in the post-period for trademark firms, compared to their matched controls. Overall, the estimated effects suggest an important role for pre-existing registered trademarks in protecting firm revenue and employment in the wake of an exogenous shock to the competitive environment.

The estimates in this section do not measure the full benefits from registered trademarks to firms, as registered trademarks offer broader benefits than protection against import competition. Thus, it is not surprising that the estimates in this section are relatively small compared to those obtained in the exercises in sections 2.3 and 2.4, which arguably capture broader effects of trademarks. Moreover, due to the scope of the trade shock, the estimates in this section pertain primarily to manufacturing firms, and may not be informative about other sectors such as services and retail.

The estimates provided sections 2.3, 2.4, and 2.5 do not reflect the general equilibrium effects of trademark registrations. For instance, reallocation across firms and changes in concentration induced by trademark registrations cannot be measured by the estimates at the firm level only. Similarly, these estimates do not measure the magnitude of the pecuniary externalities imposed by trademark firms on others; for instance, through advertising. The model developed below quantifies the broader effects of trademarks by analyzing the key general equilibrium channels at work.

2.6 Trademark registrations and advertising

U.S. aggregate advertising spending increased substantially over time (Figure 2). However, advertising's share of GDP moved in a narrow range, hovering slightly above 2% for much of the post-WWII

[28] The instrument IV_{it} is used to construct instruments for all three endogenous variables involving IC_{it}. That is, the instruments are IV_{it}, $Post_t \times IV_{it}$, and $Post_t \times TM_i \times IV_{it}$.

[29] See Appendix Table A2 for statistics on the first stage, which suggest that the instruments do indeed well in predicting the three explanatory variables that involve the endogenous variable (5) in the model (4).

period. This relatively stable aggregate pattern likely masks significant heterogeneity at the firm level. A key benefit of trademarks is to increase the appropriability of a firm's marketing activities. Because registered trademarks can help firms build a customer base and prevent customer depreciation, returns to advertising are likely larger for firms with trademarks.

Table 6 contains statistics on advertising intensity (advertising expenditures as a percentage of revenue) for firms in the Census of Manufactures (CMF) 2007 and 2012.[30] Advertising expenditures are likely interpreted by survey respondents as formal outlays to media and advertising agencies. It is not surprising that many firms do not formally incur such expenses, and hence, report zero advertising. However, nearly all firms undertake some form of passive or active informal advertising. For instance, a firm's name posted on its premises, or its website can serve as advertising. Word-of-mouth advertising by consumers works in a similar way. Much of this informal or indirect advertising is unmeasured. Thus, the reported advertising expenditures likely underestimate the true extent of information dissemination by firms to consumers.

In Table 6, average advertising expenditures and advertising intensity are shown by trademark registration status.[31] When zero advertising instances are included, average advertising expenditure for trademark firms is about 12 times that for non-trademark firms. This large difference in expenditure is partially explained by the fact that, as shown in Table 1, firms with a trademark are much larger than firms without trademarks. However, advertising intensity is also larger for trademarking firms. When only nonzero advertising cases are considered, this difference diminishes to some extent since zero advertising cases are more concentrated in non-trademark firms – consistent with the fact that trademark firms engage in more advertising activity in general.

Because trademark firms tend to be larger and older, it is important to examine the differences in advertising activity between the two types of firms after controlling for differences in life-cycle stage. Table 7 examines the variation in advertising intensity using regression analysis. The sample is an unbalanced annual panel of firms available from the Annual Survey of Manufactures (ASM) and CMF over the relatively short period of 2007-2014.[32] Measures of life-cycle stage (firm size and age) are included as controls, in addition to firm, industry, and year fixed effects, depending on the specification. Some specifications also include a "registration" dummy that indicates firms with a trademark registration during the sample period, and a "post-registration" binary that is equal to one in periods after the registration of the firm's first trademark.

The baseline specifications in columns 1 and 4 indicate that advertising intensity is negatively related to firm size (revenue). Having a trademark registration, on the other hand, is associated with a higher advertising intensity. Furthermore, firms in their post-registration period have an even higher advertising intensity. The fixed effects specification in column 4 suggests that a 1% increase in firm size (revenue)

[30] The 2007 and 2012 surveys contain much higher fractions of firms with non-missing information on advertising expenditures compared to other years. Therefore, these two surveys are used to provide some broad evidence on advertising intensity by trademark versus non-trademark firms in the manufacturing sector. Available data on firm-level advertising pertains to manufacturing firms, and leaves out firms in other sectors, particularly retail and services, which may engage in higher advertising activity.

[31] Because advertising intensity is a ratio, some misreported values of advertising expenditures especially for small firms can lead to outliers. To mitigate the influence of potential major outliers, advertising intensity is winsorized at the top 0.1%.

[32] In all specifications, only the observations with nonzero advertising expenditures are considered. The focus on nonzero advertising is driven by the fact that in the model studied below all firms advertise every period. Thus, the model's implications apply to firms that advertise.

is associated with a 0.063 percentage point decline in advertising intensity. Based on column 4, after a firm registers its first trademark, advertising intensity is higher on average by about 0.066 percentage points, equivalent to about 9.4% of the pre-registration mean advertising intensity of firms that register a trademark (see Table 8).

One concern with specifications in columns 1 and 4 of Table 7 is division bias, since the revenue variable used in the denominator of the independent variable (advertising intensity) also appears as an explanatory variable. To mitigate this potential bias, two approaches are considered. First, firm employment is used as an instrument for revenue. Employment and revenue are highly correlated at the firm level, and to the extent their measurement errors are uncorrelated, the bias should be lower. Columns 2 and 5 present the instrumental variable estimation results. The fixed effects specification in column 5 implies that a 1% increase in firm size is associated with a 0.023 percentage point decline in advertising intensity. The second approach directly replaces revenue with employment as an explanatory variable (columns 3 and 6). This approach also indicates that advertising intensity declines as firm size increases. Overall, the findings in Table 7 point to a decline in advertising intensity as a firm grows.

Finally, Table 8 provides average advertising intensity measures for the firms included in the regression samples in Table 7. Firms with no trademark registration have an average advertising intensity of about 0.56%. On the other hand, pre-registration average advertising intensity for firms with trademarks is about 0.70%. These figures are consistent with selection documented in Table 7. Would-be registrants have better growth prospects, which is associated with higher advertising intensity even before registration compared with firms that never register a trademark. Advertising intensity of trademark firms increase to 0.79% in the post-registration period, consistent with trademark registrations boosting the incentives to advertise.

The evidence overall suggests that firms with trademarks engage in more advertising activity both before and after their first trademark registration, compared to firms without trademarks. Note that the relationship between advertising intensity and trademark activity is not necessarily causal – both can be driven by unobserved firm characteristics. Next, we turn to a model that features firm product quality as the fundamental driver of firm outcomes, including trademark registration and marketing activity.

3 The model

The model builds on the industry dynamics framework in Dinlersoz and Yorukoglu (2012), where firms grow gradually by acquiring customers through informative advertising. The framework is extended here to general equilibrium. In addition, product quality, persuasive advertising, and trademark decisions are introduced, while some other dimensions, such as the determination of prices, are simplified.

3.1 The Setup

Consider an economy with a continuum of symmetric industries (product varieties) indexed by $i \in [0, 1]$. In each industry, a continuum of firms offer quality-variants of the same product. Time, t, is discrete. There is free entry, with a large mass (continuum) of potential entrants in every period. Upon paying a sunk entry cost, κ, an entrant learns its efficiency in product development, e, a random variable with density $h(e)$. This efficiency is time-invariant and determines the firm's technology for developing a

product with a given quality.

After observing e, a firm can exit at no cost without developing a product. If the firm stays, it develops its product with quality $q^o > 0$, which remains constant for the rest of the firm's life.[33] During product development, firms also engage in a one-time persuasive advertising campaign to improve consumers' perception of the quality of their products. If the campaign is successful, the firm's product acquires a *perceived* quality, $q = q^o + \Delta$, $\Delta > 0$; otherwise, $q = q^o$. A firm exits in any subsequent period with probability, χ.

There is a unit mass of households (consumers), who live for several periods. Consumers are not aware of all firms selling a given product variety. To grow, a firm must inform consumers about itself and build a customer base over time. This is done through informative and persuasive advertising (or marketing): every period, firms send ads to consumers, who make their purchases based on the ads they receive.

A firm can register a trademark at any period to increase consumer awareness. Once registered, a trademark stays with the firm until the firm exits. A trademark firm can more easily attract consumers because consumers recognize the trademark and uniquely associate it to the firm without confusion, since the trademark is registered and legally protected against infringement. This key benefit of trademarks is captured by a more efficient informative advertising technology for trademark firms compared to firms with no trademark.

Competition in marketing implies that a firm can lose customers to firms that provide a higher surplus. A firm's customer base is also subject to further depreciation because customers can be diverted to other firms through imitation and infringement. The rate of this *customer depreciation (or diversion)* is given by $\delta_\Gamma \in (0,1), \Gamma \in \{N, T\}$, where Γ denotes whether the firm has a trademark (T) or not (N). A trademark firm has a lower customer depreciation rate: $\delta_T < \delta_N$. The assumption that firms with trademarks are less exposed to customer depreciation captures the role of registered trademarks in deterring imitation and infringement.

The timing of events within a period is given in Figure 6. Each period entrants choose their quality and undertake a persuasive advertising campaign. At that point, entrants decide whether to stay. Then, the outcome of the persuasive advertising campaign is revealed. Entrants, and incumbents without an existing trademark, then decide whether to register a trademark. After trademark registration decisions, firms choose informative advertising expenditures, prices, and employment.

3.2 Households

A household supplies one unit of labor inelastically, and receives utility

$$U = \int_0^1 u(q_i, x_i) di$$

where $u(q_i, x_i)$ is the utility from consuming x_i amount of product variety $i \in [0,1]$ at quality level q_i (actual or perceived)

$$u(q_i, x) = q_i^{\xi} x_i^{\theta}, \quad \xi, \theta \in (0,1).$$

[33] For simplicity, quality upgrading is not considered. It is possible to introduce quality upgrading by allowing, for instance, firms to make investments over time to improve their product development efficiency.

While it is the actual quality, q^o, that ultimately matters for the utility from consumption, households make purchase decisions based on the perceived quality, q, as discussed below.

3.2.1 Consumer information

Firm information relevant for consumer decision making is conveyed through informative advertising. Consumers receive ads randomly and independently. Within a period, each consumer is equally likely to receive an ad, no consumer receives more than one ad from the same firm, and each ad is received by some consumer.[34] Let I_t^i denote a consumer's period- t information about firms selling variety i. The set I_t^i is either empty ($\varnothing$), implying that the consumer has no information, or contains elements corresponding to the firms the consumer has received ads from.

Each element ι_{jt}^i of I_t^i is a vector of variables that describe firm j's attributes that help a consumer identify a firm and evaluate the surplus from patronizing it. Because consumers live for several periods, I_t^i may contain information about the set of firms the consumer was informed of in the past. For simplicity, firms can inform their existing customers at no cost about their current payoff-relevant information. Therefore, consumers know the current period payoff-relevant information of the firm that they purchased from in the previous period, unless that firm exits.

Next, consider the contents of $\iota_{jt}^i \in I_t^i$. An ad reveals the perceived quality, q_j^i.[35] Price, on the other hand, is not included in the ad; prices are flexible enough that firms cannot commit to them in the ads. However, consumers know the distribution of prices. In a dynamic environment where consumers can purchase from a firm repeatedly, the firm attributes that matter for firm survival are also payoff relevant because consumers value the survival of the highest-surplus providing firm they purchased from in the previous period – current-period information about that firm will be in I_t^i only if that firm survives. One such attribute is firm size, m_{jt}^i—measured by the number of customers the firm has at the beginning of period t. For instance, if exit depends on firm size, firm size matters for the expected surplus from a firm. We assume that size is included in an ad or it is public information. Similarly, the information in an ad includes whether the firm has a trademark, which is relevant for growth and survival if trademarks increase firm value. To sum up, consumer information about a firm consists of $\iota_{jt}^i = \{\Gamma_{jt}^i, q_j^i, m_{jt}^i\}$, $\Gamma_{jt}^i \in \{N, T\}$. The dimension of ι_{jt}^i will be reduced given the additional structure imposed on the model below.

3.2.2 Household's problem

If the consumer has no information about firms selling variety i (i.e. $I_t^i \equiv \varnothing$), the consumer does not purchase that variety in period t. Otherwise, the consumer purchases from the firm in I_t^i that provides the highest net expected perceived surplus. Let $W_t^{\Gamma_{jt}^i}(q_j^i, m_{jt}^i)$ be the surplus from purchasing variety i from firm j, where q_j^i denotes the perceived quality.

Consider now a household with information $\mathbf{I}_t = \{I_t^i\}_{i\in[0,1]}$ and assets a_t. Suppose that the next period's assets, a_{t+1}, is given. The household then solves a static problem that determines for each

[34] These assumptions are compatible with the fact that there is a very large number (continuum) of consumers.

[35] The analysis abstracts from imperfect information about (perceived) quality. While perfect information is a strong assumption, it simplifies the analysis substantially by ruling out potential multiple equilibria when quality is unknown and firms can adopt strategies to manipulate consumers' beliefs.

variety the optimal quality and quantity to consume

$$\max_{\left\{x^i_{j(i),t}\right\}_{j(i)\in\ I^i_t}} \int_0^1 u(q^i_{j(i)}, x^i_{j(i),t})di \tag{6}$$

s.t.

$$\begin{aligned} j(i) &= \arg\max_j \left\{ W_t^{\Gamma^i_{jt}}(q^i_j, m^i_{jt}) \right\}_{j\in I^i_t}, \text{ for all } i \in [0,1], \\ \int_0^1 p^i_{j(i),t} x^i_{j(i),t} di &= w_t + a_t(1+r_t) - a_{t+1}, \quad x^i_{j(i),t} \geq 0 \text{ for all } (i, j(i)),\ a_{t+1} \text{ given.} \end{aligned}$$

The first constraint states that for each variety i, the household chooses the firm $j(i) \in I^i_t$ that provides the highest net lifetime surplus among the firms the household has information about. The second is the budget constraint. The quantities $x^i_{j(i),t}$ are determined after quality choices.

Denote by $\mathfrak{U}(a_{t+1}, \mathbf{I}_t)$ the maximized value in problem (6) as a function of a_{t+1} and $\mathbf{I}_t$. The household then solves the following dynamic problem

$$U(a_t, \mathbf{I}_t) = \max_{a_{t+1}} \{\mathfrak{U}(a_{t+1}, \mathbf{I}_t) + \beta U(a_{t+1}, \mathbf{I}_{t+1})\} \tag{7}$$

s.t.

$$\int_0^1 p^i_{j(i),t} x^i_{j(i),t} di + a_{t+1} = w_t + a_t(1+r_t) \quad \text{for all } t,$$

In (7), the household picks the optimal amount of total expenditure (or equivalently, a_{t+1}). The information about firms the household carries over to the next period is included in $\mathbf{I}_{t+1}$ and consists of the relevant information for all the firms from which a purchase was made in the current period.

Next, turn to the perceived consumer surplus, $W^\Gamma_t(q,m)$, where the indices i, j, and t are dropped from q and m for ease of notation. Let $W^\varnothing_t$ be the surplus of a consumer with no information from firms selling the variety. Denote by Λ^k_t the probability that a consumer receives $k \geq 0$ ads for variety i. The consumer surplus is

$$\begin{aligned} W^\Gamma_t(q,m) &= \widetilde{u}(q) \\ &\quad +\beta \left\{ \Lambda^0_{t+1} \left[(1-\chi)(1-\delta_\Gamma) W^C_{t+1}(q,m') + (1-\chi)\delta_\Gamma W^D_{t+1}(q^D, m^D) + \chi W^\varnothing_{t+1} \right] \right. \\ &\quad + \sum_{k=1}^{\infty} \Lambda^k_{t+1} [(1-\chi)(1-\delta_\Gamma) \int \max\{W, W^C_{t+1}(q,m')\} dZ^k_{t+1}(W) \\ &\quad \left. +(1-\chi)\delta_\Gamma W^D_{t+1}(q^D, m^D) + \chi \int W dZ^k_{t+1}(W)] \right\}. \end{aligned} \tag{8}$$

where $\widetilde{u}(q)$ is the current period perceived surplus.[36] The second line on the right hand side of (8) is the expected discounted future surplus if no ad is received next period for that specific variety. It has three components. The first is the future surplus if the consumer's current firm stays in the market and the

[36] Note that $\widetilde{u}(q) = \max_x [u(q,x) - \upsilon p x]$, where q is the perceived quality of a firm that charges price p, and υ represents the Lagrange multiplier associated with the consumer's static problem (6).

consumer is not diverted to another firm. In this case, the consumer obtains a continuation surplus of

$$W^C_{t+1}(q,m') = T_{t+1}(q,m')W^T_{t+1}(q,m') + (1-T_{t+1}(q,m'))W^N_{t+1}(q,m'), \tag{9}$$

where $T_{t+1}(\cdot) = 1$ if the firm has a trademark in period $t+1$. The second term is the future surplus if the firm stays but the consumer is diverted. The consumer's surplus when diverted by another firm is denoted by $W^D_{t+1}(q^D, m^D)$ – the consumer is assumed to know the type of the diverting firm (Γ^D, q^D, m^D). The third term in the second line is the future surplus if the consumer's current firm exits (with probability χ), leaving the consumer with a surplus of

$$W^{\varnothing}_{t+1} = \beta\{\Lambda^0_{t+2}W^{\varnothing}_{t+2} + \sum_{k=1}^{\infty}\Lambda^k_{t+2}\int W dZ^k_{t+2}(W)\}. \tag{10}$$

In this case, the surplus in period $t+1$ is zero, and the following period's surplus depends on the number of ads received. If no ad is received, the surplus is $W^{\varnothing}_{t+2}$, otherwise the consumer chooses the firm that yields the highest surplus. Z^k_t is the period-t cumulative distribution function for consumer surplus if $k > 0$ ads are received.[37]

The third and fourth lines in (8) give the expected future surplus from receiving one or more ads. If the firm stays, the consumer is aware of $k+1$ firms corresponding to k new ads, plus the most recently visited firm. If the firm exits, the consumer has k new ads. In either case, the consumer chooses the firm with the highest surplus.

Next, consider a diverted consumer's surplus, $W^D_{t+1}(q^D, m^D)$. One process of customer depreciation consistent with typical cases in trademark infringement is firms with lower surplus diverting consumers from firms with higher surplus. This mechanism, however, is already achieved by persuasive advertising which allows lower surplus firms to create a perception of higher surplus for consumers. Therefore, to maintain consumer rationality and simplify the customer diversion process, we assume that only the firms that offer identical perceived surplus can divert customers from each other. Thus, $W^D_{t+1}(q^D, m^D) = W^{\Gamma}_{t+1}(q, m')$.

Consumer surplus has the following properties. First, consumers always prefer to purchase from some firm compared to not consuming a product at all, i.e. $W^{\Gamma}_t(q,m) > W^{\varnothing}_t$ for $\Gamma \in \{T, N\}$. Second, because exit is exogenous, firm size does not matter for consumer surplus, i.e. $W^{\Gamma}_t(q, m_i) = W^{\Gamma}_t(q, m_j)$ for any i, j. Third, because customers can only be diverted to a firm which provides identical surplus, a firm's trademark status does not affect consumer surplus directly, conditional on quality.[38] Therefore, $W^T_t(q,m) = W^N_t(q,m)$. Given these properties, perceived quality is the only surplus-relevant attribute. The consumer surplus in (8) then simplifies to

$$W_t(q) = \widetilde{u}(q) + \beta \left\{ \begin{array}{c} \Lambda^0_{t+1}\left[(1-\chi)W_{t+1}(q) + \chi W^{\varnothing}_{t+1}\right] \\ + \sum_{i=1}^{\infty}\Lambda^k_{t+1}\left[(1-\chi)\int \max\{W, W_{t+1}(q)\}dZ^k_{t+1}(W) + \chi\int W dZ^k_{t+1}(W)\right] \end{array} \right\}.$$

Consider now a consumer informed of a firm with perceived quality q. Each such consumer can be of

[37] Because ads from firms arrive independently, one can write $Z^k_t(W) = [Z_t(W)]^k$, where $Z_t(W)$ is the probability that an ad is from a firm that offers a surplus of at most W.

[38] This is not to say that a firm's brand or reputation does not matter. Consuming a famous brand can enhance utility (not modelled here), but not simply because that brand has a registered trademark.

two types: one that purchased from the firm in the previous period, and the other that did not. Let the probability of purchase for these two types of consumers be $z_t(q)$ and $\omega_t(q)$, respectively. The function $z_t(q)$ can be written as

$$z_t(q) = \sum_{k=0}^{\infty} \Lambda_t^k Z_t^k(W_t(q)), \tag{11}$$

where the kth term is the probability of sale to a consumer who has received k ads from firms other than firm q. Furthermore,

$$\omega_t(q) = [\eta_t + (1-\eta_t)\psi_t(q)]z_t(q). \tag{12}$$

where η_t is the fraction of consumers with no information from other firms, and $\psi_t(q)$ is the probability that the firm a consumer purchased from in period $t-1$ provides more surplus than firm q.[39]

3.3 Firms

3.3.1 Product development and production technology

For an entrant with product development efficiency, e, the one-time cost of developing a product with quality q^o is

$$C(q^o; e) = B\left(\frac{q^o}{e}\right)^{\eta}, \quad B > 0, \eta > 1.$$

C is strictly convex in q^o, and strictly decreasing in e – a higher quality product costs more to develop, but more efficient firms incur lower cost at any quality.

After the product is developed, a firm's production technology in any period is linear in the only input, labor n

$$F(n; q^o) = \mathcal{M} n \left(q^o\right)^{-\sigma}, \quad \mathcal{M} > 0, \sigma > 0. \tag{13}$$

F is decreasing in q^o – a higher quality product has a higher marginal cost of production.

3.3.2 Marketing technology

Persuasive advertising During product development, a firm engages in a one-time persuasive advertising campaign or "product image" development, which, if successful, permanently alters consumers' perceptions of the firm's product quality for the rest of the firm's life. Persuasive advertising is successful with probability φ, in which case the firm's perceived quality is $q = q^o + \Delta$, $\Delta > 0$. Otherwise, the perceived quality is identical to the actual quality, $q = q^o$.

Informative advertising Firms disseminate information using an informative advertising (marketing) technology, $\Phi^{\Gamma}(n_a)$, $\Gamma \in \{N, T\}$, which gives the total number of new consumers informed by the firm (equivalently, the number of ads sent) as a function of marketing labor, n_a

$$\Phi^{\Gamma}(n_a) = A^{\Gamma} n_a^{\rho}, \quad A^{\Gamma} > 0, \rho \in [0, 1).$$

[39] Because $\eta_t, \psi_t^{\Gamma}(q, m) \leq 1$, it follows that $z_t^{\Gamma}(q, m) \geq \omega_t^{\Gamma}(q, m)$, i.e. a firm faces more demand from its customers in $t-1$ than consumers it reaches through advertising in t.

Φ^Γ is strictly concave, reflecting diminishing returns to marketing labor[40]. There is no fixed cost of advertising, $\Phi^\Gamma(0) = 0$. These assumptions imply that all firms advertise every period.

The relative productivity of trademark firms in marketing is given by $\varrho = A^T/A^N > 1$. For any given marketing labor, a trademark firm reaches more consumers. In other words, consumer awareness is higher for trademark firms, because it is easier for consumers to recognize a registered trademark that cannot be confused with others – consistent with the key role registered trademarks play in reducing the informational frictions between a firm and consumers.

3.3.3 The cost of a trademark

There is a one-time cost, c_T, of obtaining a registered trademark. This cost not only includes the registration fees, but also any outlays on designing a trademark that is distinguished, searching the existing trademarks for potential similarity, and hiring legal professionals for the registration process. In addition, firms incur costs to maintain and protect their trademarks.[41] For simplicity, c_T represents the present discounted value of all costs associated with owning a registered trademark.

3.3.4 Firm's dynamic problem

A firm's type in any period is given by (Γ, q, q^o, m) where Γ is firm's trademark status, m is the firm's size in terms of number of customers, q, and q^o denote the perceived and actual quality, respectively. The period profit of a firm as a function of its price p, production labor n, and marketing labor n_a used for informative advertising is

$$\Pi_t^\Gamma(p, n, n_a; q, q^o, m) = p m_t^\Gamma(n_a; q, q^o, m) x(p, q) - n - n_a, \tag{14}$$

where the wage rate is normalized to one. The production constraint is

$$F(n; q^o) = m_t^\Gamma(n_a; q, q^o, m) x(p, q), \tag{15}$$

where the right hand side is the total demand for a type Γ firm's product with perceived quality q and price p, if the firm starts the period with m customers and utilizes n_a marketing labor. A customer's demand for a product with perceived quality q and price p is $x(p, q)$, and m^Γ denotes the firm's end-of-period number of customers

$$m_t^\Gamma(n_a; q, q^o, m) = \underbrace{m[1 - (1 - I(\Gamma = T))\delta_N - I(\Gamma = T)\delta_T] z_t(q)}_{\text{continuing customers}} + \underbrace{\Phi^\Gamma(n_a)\omega_t(q)}_{\text{new customers}} + \underbrace{d_t(q)}_{\text{diverted customers}}, \tag{16}$$

where $I(\cdot)$ is the indicator function.

[40] From an individual firm's perspective, there is a very large pool of consumers and each ad reaches a distinct consumer. If the marketing technology exhibits constant or increasing returns to scale, some firms can reach an arbitrarily large number of consumers and grow without bound. Decreasing returns to advertising is documented in Sutton (1991). Grossman and Shapiro (1984) point out that decreasing marginal productivity of advertising can stem from media saturation, overlapping media, or different predispositions to view ads.

[41] For instance, trademarks have to be renewed periodically and be defended against infringement. As discussed earlier, the costs of protecting a trademark can be large, as firms need to be able to detect infringement, hire lawyers, and engage in litigation.

The first term in (16) represents customers from the previous period that continue to purchase from the firm. A firm starts the period with m customers. A non-trademark firm ($\Gamma = N$) loses a fraction δ_N of its customers to customer diversion, whereas a trademark firm ($\Gamma = T$) loses a fraction $\delta_T < \delta_N$. Of the remaining consumers, a fraction $z_t(q)$ continue to purchase from the firm. The second term gives the mass of new customers the firm gains through informative advertising. The firm reaches $\Phi^{\Gamma}(n_a)$ new consumers, but only a fraction $\omega_t(q)$ choose to purchase from the firm.

The third term is the mass of consumers, $d_t(q)$, the firm receives as a result of customer diversion.[42] Each firm with perceived quality q gets an equal share of the total mass of consumers diverted, i.e. $d_t(q)$ is the same across all firms with quality q. The customer depreciation rate δ_Γ can thus be viewed as a "tax", and $d_t(q)$ represents the equal redistribution of the gains from this tax across all firms at that quality. Because firms of quality q differ by customer size, m, the process of customer diversion implies that among firms for a given $\Gamma = T, N$ smaller ones will gain more on the net in terms of customers – consistent with casual evidence that less known, smaller, and younger firms obtain a higher benefit more from customer diversion due to imitating and infringing activity, compared to larger, more established firms.

The value of a type (T, q, q^o, m) firm is

$$V_t^T(q, q^o, m) = \max_{p,n,n_a} \left[\Pi_t^T(p, n_a, n; q, q^o, m) + ((1-\chi)/(1+r))V_{t+1}^T(q, q^o, m^T)\right]. \tag{17}$$

In (17), the firm exits with probability χ in the beginning of the next period, and the firm value upon exit is normalized to zero.

Similarly, the value of a type (N, q, q^o, m) is given by

$$V_t^N(q, q^o, m) = \max \left\{ \begin{array}{c} \max_{p,n,n_a} \left[-c_T + \Pi_t^T(p, n, n_a; q, q^o, m) + ((1-\chi)/(1+r))V_{t+1}^T(q, q^o, m^T)\right], \\ \max_{p,n,n_a} \left[\Pi_t^N(p, n, n_a; q, q^o, m) + ((1-\chi)/(1+r))V_{t+1}^N(q, q^o, m^N)\right] \end{array} \right\}, \tag{18}$$

In (18), the firm chooses between registering a trademark by paying c_T, or continuing without a trademark.

3.3.5 Entry

Each period a mass of $M_t \geq 0$ new firms enter. If a firm decides to stay after entry, it starts with zero customers ($m = 0$) and no trademark, and has to first develop its product and undertake a persuasive advertising campaign. An entrant with product development efficiency e chooses its actual quality to maximize its expected value *before* the outcome of persuasive advertising is known

$$V_t^E(e) = \max_{q^o} \left[\varphi V_t^N(q^o + \Delta, q^o, 0) + (1-\varphi)V_t^N(q^o, q^o, 0) - C(q^o; e)\right]. \tag{19}$$

An entrant exits in its first period without developing a product if $V_t^E(e) < 0$. Free entry implies

$$\int \max\left\{0, V_t^E(e)\right\} h(e)de \leq \kappa, \tag{20}$$

[42] Note that $d_t(q)$ is a function of quality, since only the firms providing identical surplus can divert consumers from each other.

which holds with equality when $M_t > 0$.

3.3.6 Trademark decision

A firm without a trademark registers a trademark in period t if the net benefit from registering a trademark exceeds that from staying as a non-trademark firm

$$\begin{aligned} & -c_T + \max_{p,n,n_a} \left[\Pi_t^T(p,n,n_a;q,q^o,m) + ((1-\chi)/(1+r))V_{t+1}^T(q,q^o,m^T)\right] \\ \geq & \max_{p,n,n_a} \left[\Pi_t^N(p,n,n_a;q,q^o,m) + ((1-\chi)/(1+r))V_{t+1}^N(q,q^o,m^N)\right]. \end{aligned} \tag{21}$$

3.4 Stationary equilibrium

A stationary equilibrium for the model economy is defined as follows.

Definition (Stationary Equilibrium). *A stationary equilibrium consists of product quality, employment, pricing, informative advertising, and trademark policies, $q^o(e)$, $n(\Gamma,q,q^o,m)$, $p(q,q^o)$, $n_a(\Gamma,q,q^o)$, and $T(q,q^o,m)$, mass of customers gained from diversion, $d(q)$, a cumulative distribution of consumer surplus (W) across ads $Z(W)$, a measure $\mu(\Gamma,q,q^o,m)$ over firm types, and an entry mass $M \geq 0$, such that*

(i) Firms maximize their profits,
(ii) Consumers maximize their surplus,
(iii) M satisfies the free entry condition (20),
(iv) Markets clear,
(v) Z and μ are consistent with conditions (i) through (iv).

At any perceived quality q, the mass of customers gained by all firms from diversion equals the mass of customers lost by all firms to diversion

$$d(q)\left[\int_m d\mu(\cdot,q,\cdot,m)\right] = \delta_N\left[\int_m m d\mu(N,q,\cdot,m)\right] + \delta_T\left[\int_m m d\mu(T,q,\cdot,m)\right] \tag{22}$$

Finally, the interest rate is given by $r = (1/\beta) - 1$.

4 Characterization of stationary equilibrium

4.1 Pricing policy

Because firms cannot commit to prices and prices are not advertised, consumers decide which firm to purchase from by correctly anticipating that once they decide to purchase from a firm they will face monopoly pricing. Firms set prices taking this consumer behavior into account, and the resulting pricing policy follows a constant mark-up rule. Consumer's optimal consumption decision is based on perceived

quality. The efficiency condition for optimal consumption of each variety yields[43]

$$\frac{x(p(q_i),q_i)}{x(p(q_j),q_j))}=\left[\frac{p(q_j)q_i^{\xi}}{p(q_i)q_j^{\xi}}\right]^{1/(1-\theta)},$$

for all i,j. Without loss of generality, designate the lowest quality $\underline{q}$ as the reference, and express the consumer demand for quality q as

$$x(p,q)=\left[\frac{q}{\underline{q}}\right]^{\frac{\xi}{1-\theta}}\left[\frac{p(\underline{q})}{p}\right]^{\frac{1}{1-\theta}}x(p(\underline{q}),\underline{q}). \tag{23}$$

The consumer's total expenditure, y, is given by

$$y=(1-\omega_o)\int_{\underline{q}}^{\overline{q}^o}p(\mathfrak{q})x(p(\mathfrak{q}),\mathfrak{q})d\omega(\mathfrak{q}) \tag{24}$$

where ω_o is the fraction of varieties the consumer has no information of, and $\omega(\mathfrak{q})$ is the probability that the consumer purchases quality $\mathfrak{q}$ of a given variety.[44]

Using (23) and (24), the demand function can be written as

$$x(p,q)=\frac{\left(\underline{q}^{\xi}/p\right)^{\frac{1}{1-\theta}}y}{(1-\omega_o)\int_{\underline{q}}^{\overline{q}^o}p(\mathfrak{q})^{\frac{-\theta}{1-\theta}}\mathfrak{q}^{\xi}d\omega(\mathfrak{q})}.$$

For a firm with no success in persuasive advertising, consumers' quality perception matches the actual quality. Note that the firm's demand in (15) is separable in customer size and individual consumer's demand. Therefore, the optimal price of this firm is the solution to

$$\pi(q,q^o)\equiv\max_p x(p,q^o)\left(p-(q^o)^{\sigma}/\mathcal{M}\right), \tag{25}$$

where $(q^o)^{\sigma}/\mathcal{M}$ is the marginal cost of production based on (13).

Proposition 1 *Price of a firm without persuasive advertising follows a constant mark-up rule,* $p(q)=p(q^o)=\frac{1}{\theta}\frac{(q^o)^{\sigma}}{\mathcal{M}}$.

Proof. See Appendix B for all proofs. ■

The pricing policy of a firm with persuasion is more subtle. For these firms, the perceived quality exceeds the actual quality. Thus, consumers expect these firms to price their products consistent with their (higher) perceived quality, $p(q)=\frac{1}{\theta}\frac{(q^o+\Delta)^{\sigma}}{\mathcal{M}}$. The firm will need to charge this price to be consistent with consumers' perceptions. Thus, firms with persuasive advertising have higher markups for any q^o.

Higher quality products provide consumers higher utility, but they are also more expensive. Consumers prefer a higher quality product to a lower quality one under the condition $\xi>\sigma$, which is assumed hereafter. Consumers then purchase from the firm that offers the highest perceived quality

[43] Consumer optimization yields $\frac{\frac{\partial u(q_i,x_i)}{\partial x_i}}{\frac{\partial u(q_j,x_j)}{\partial x_j}}=\frac{\theta q_i^{\xi}x_i^{\theta-1}}{\theta q_j^{\xi}x_j^{\theta-1}}=\frac{p(q_i)}{p(q_j)}$.

[44] This is equivalent to the (unconditional) probability in (12) that the highest quality a consumer is informed of is $\mathfrak{q}$.

(or surplus) among the firms he is informed of. Furthermore, assume that customers gained through diversion, $d(q)$, is non-decreasing.[45] A firm's value is then an increasing function of its size and actual quality.

Proposition 2 *(i)* $W(q)$, $z(q)$, *and* $\omega(q)$ *are increasing in* q, *(ii)* $V^{\Gamma}(q, q^o, m)$ *is increasing in* q^o *and* m, *(iii)* $\pi(q^o + \Delta, q^o) > \pi(q^o, q^o)$ *and* $V^{\Gamma}(q^o + \Delta, q^o, m) > V^{\Gamma}(q^o, q^o, m)$.

4.2 Quality policy

In stationary equilibrium, the optimal quality choice for an entrant with product development efficiency e solves (19). Assume a unique solution, $q^o(e) \in (0, \infty)$. Firms with higher efficiency e choose higher quality levels.

Proposition 3 *Quality,* $q^o(e)$, *is increasing in* e.

4.3 Informative advertising policy

Consider a firm that is $\tau \in \mathbb{Z}$ periods away from its trademark registration, where $\tau = 0$ corresponds to the period of trademark registration, and $\tau < 0$ (> 0) to the periods before (after) registration. For a that firm never registers a trademark, $\tau = -\infty$. The optimal informative advertising policy (or marketing labor), $n_a^{\Gamma(\tau)}(q, q^o, \tau)$, satisfies the first order condition

$$\rho A^{\Gamma(\tau)} [n_a^{\Gamma(\tau)}(q, q^o, \tau)]^{\rho-1} \omega(q) \pi(q, q^o) L(q, \tau) = 1, \tag{26}$$

where $\Gamma(\tau) = N$ if $\tau < 0$ and $\Gamma(\tau) = T$, otherwise, and

$$L(q, \tau) = \sum_{t=0}^{-\tau} \left(\tfrac{(1-\chi) l^N(q)}{1+r} \right)^t + \left[\left(\tfrac{(1-\chi) l^N(q)}{1+r} \right)^{-\tau I(\tau<0)} \sum_{t=1}^{\infty} \left(\tfrac{(1-\chi) l^T(q)}{1+r} \right)^t \right], \tag{27}$$

where $I(\cdot)$ is an indicator function.

The left hand side of (26) is the marginal benefit of increasing the marketing labor, and the right hand side is the marginal cost (normalized wage rate). The marginal benefit depends on the discounted flow of profit from a customer acquired through advertising. This customer continues to patronize the firm from one period to another with probability $l^{\Gamma(\tau)}(q)$, which is the effective customer loyalty rate

$$l^{\Gamma(\tau)}(q) = \begin{cases} z(q)(1 - \delta_T) & \text{if } \Gamma(\tau) = T, \\ z(q)(1 - \delta_N) & \text{if } \Gamma(\tau) = N. \end{cases}$$

Hence, $L(q, \tau)$ in (27) is the (discounted) expected number of times a customer purchases from the firm.

The solution to (26) is

$$n_a^{\Gamma(\tau)}(q, q^o, \tau) = [\rho A^{\Gamma(\tau)} \omega(q) \pi(q, q^o) L(q, \tau)]^{1/1-\rho}. \tag{28}$$

[45] Note that $d(q)$ is endogenously determined as part of the equilibrium. As shown in the proof of Proposition 2, a sufficient condition for this monotonicity to hold is δ_T is not too small relative to δ_N.

Two special cases are noteworthy. One is the case when the firm has a trademark ($\tau \geq 0$), and the other is when the firm never registers a trademark ($\tau = -\infty$). For these two cases, optimal advertising policy is given by

$$n_a^{\Gamma(\tau)}(q, q^o, \tau) = \left[\rho A^{\Gamma(\tau)} \omega(q) \pi(q, q^o) \frac{1+r}{1+r-(1-\chi) l^{\Gamma(\tau)}(q)}\right]^{\frac{1}{1-\rho}}, \text{ for } \tau = -\infty \text{ or } \tau \geq 0. \tag{29}$$

What is the effect of a trademark on advertising? For a firm with a trademark ($\tau \geq 0$), (29) implies

$$\frac{n_a^T(q, q^o, \tau)}{n_a^N(q, q^o, -\infty)} = \left[\varrho \left(\frac{1-\delta_T}{1-\delta_N}\right) \left(\frac{1+r-(1-\chi) l^N(q)}{1+r-(1-\chi) l^T(q)}\right)\right]^{\frac{1}{1-\rho}} > 1, \text{ for } \tau \geq 0, \tag{30}$$

because $\varrho > 1$, and $l^T(q) > l^N(q)$. In other words, the firm advertises more when it has a trademark.[46]

Higher quality firms advertise more, ceteris paribus, as they make higher profit per consumer and have higher probability of attracting and retaining consumers. Moreover, for a firm that eventually registers a trademark, advertising expenditure increases as the firm approaches its trademark date. This effect arises from the anticipated increase in the effective loyalty rate, $l^{\Gamma(\tau)}(q)$, that occurs at trademark registration. Note also that for any q^o, firms with persuasion make higher profit per consumer, and enjoy higher probability of sale and customer loyalty, and therefore engage in higher levels of informative advertising. The properties of the advertising policy is summarized below.

Proposition 4 *i)* $n_a^{\Gamma(\tau)}(q, q^o, \tau)$, *is increasing in* q^o,

ii) $n_a^{\Gamma(\tau)}(q, q^o, \tau) > n_a^{\Gamma(\tau)}(q, q^o, -\infty)$ *for* $\tau > -\infty$,

iii) For a firm that registers a trademark at some point in its life-cycle, $n_a^{\Gamma(\tau)}(q, q^o, \tau)$ *is increasing in* τ *before its trademark date* ($-\infty < \tau < 0$), *and constant after registration (for* $\tau \geq 0$*); for firms that never register a trademark,* $n_a^{\Gamma(\tau)}(q, q^o, \tau)$ *is time-invariant,*

iv) $n_a^{\Gamma(\tau)}(q^o + \Delta, q^o, \tau) > n_a^{\Gamma(\tau)}(q^o, q^o, \tau)$.

Proposition 4(iii) implies advertising expenditure is higher after a trademark registration compared to any period before the registration. As discussed in Section 4.5, firms that register trademarks have higher quality than those that do not register trademarks. Hence, by Proposition 4(i) and (ii) firms with a trademark advertise more on average, both before and after registration, compared to those without trademarks. These implications are consistent with the empirical findings in Section (2.6) that suggest higher average advertising spending by trademark firms both pre and post-registration compared to firms with no trademarks, and an increase in advertising expenditure after registration for those firms that register trademarks.

The fraction of firm revenue a firm allocates to advertising, or advertising intensity, is given by

$$\mathfrak{a}^{\Gamma(\tau)}(q, q^o; \tau) = \frac{n_a^{\Gamma(\tau)}(q, q^o; \tau)}{m^{\Gamma(\tau)}(n_a^{\Gamma(\tau)}; q, q^o, m) p(q) \widetilde{x}(q)}, \tag{31}$$

where $\widetilde{x}(q) = x(p(q), q)$ and m is the size at the beginning of the period. For firms with a trademark and those that never register one, advertising expenditures do not vary over time (Proposition 4) but

[46] Note that in stationary equilibrium a given firm type (q, m) either registers a trademark or not – hence the comparison of advertising expenditures is a counterfactual exercise.

revenue increases, implying that advertising intensity declines as the firm ages (or equivalently, as its size increases). For firms that register a trademark, advertising expenditures increase until the trademark date (Proposition 4), along with revenue. However, the change in advertising intensity during this period is ambiguous and depends on the parameters.

Overall, in a panel of firms, advertising intensity tends to decline over time (as a firm grows or ages) if the composition of firms is dominated by firms that never register a trademark or firms already in their post-trademark period. Consistent with this scenario, in the sample of firms studied in Section 2.6, advertising intensity declines over time as firms grow.[47]

4.4 Firm size dynamics and heterogeneity

Using the law of motion in (16) and (29), the customer size of an age-A firm which registers its trademark at birth is

$$m_{\text{A}}^T(q,q^o) = \sum_{j=0}^{\text{A}} \left(\Phi^T(n_a^T(q,q^o,0))\omega(q) + d(q)\right)[l^T(q)]^j = [\Phi^T(n_a^T(q,q^o,0))\omega(q)+d(q)]\frac{1-[l^T(q)]^{\text{A}+1}}{1-l^T(q)}. \quad (32)$$

Similarly, the size of an age-A firm that never registers a trademark is

$$m_{\text{A}}^N(q,q^o) = \left[\Phi^N(n_a^N(q,q^o,-\infty))\omega(q) + d(q)\right]\frac{1-[l^N(q)]^{\text{A}+1}}{1-l^N(q)}. \quad (33)$$

The long-run size of a firm that registers a trademark at birth is related to its initial size (at age zero) as follows

$$\frac{m_\infty^T(q,q^o)}{m_0^T(q,q^o)} = \frac{\lim_{\text{A}\to\infty} m_{\text{A}}^T(q,q^o)}{m_0^T(q,q^o)} = \frac{1}{1-l^T(q)}.$$

Since $l^T(q)$ is increasing, this ratio is higher for higher quality firms. For the firm with the highest perceived quality, $l^T(\overline{q}^o+\Delta) = (1-\delta_T)$, and this ratio becomes $1/\delta_T$.

Similarly, for a firm which never registers a trademark

$$\frac{m_\infty^N(q,q^o)}{m_0^N(q,q^o)} = \frac{1}{1-l^N(q)}.$$

Since $l^N(q) = z(q)(1-\delta_N)$, the growth potential (the ratio of the highest possible size to initial size) for firms without a trademark depends critically on δ_N. For the lowest quality firm that never trademarks, the growth potential is the lowest, since $z(\underline{q}^o)$ is the lowest. Because the lowest quality firm can only sell to consumers that do not receive any other ads, $z(\underline{q}^o)$ is lower in an economy with higher number of ads per consumer, implying a lower growth potential for this type of firm when there is more aggregate advertising activity.

The range of firm size depends on the range of quality. In an economy where a positive mass of firms register trademarks, the support of the firm size distribution is $[m_0^N(\underline{q}^o,\underline{q}^o), m_\infty^T(\overline{q}^o+\Delta,\overline{q}^o)]$, such that

[47] In general, the average advertising intensity of a trademark firm in its post-trademark period can be higher or lower than that in the pre-trademark period, depending on the parameters. The analysis in Section 2.5 indicates that advertising intensity tends to be higher in the post-trademark period in the sample studied.

$$\frac{m_\infty^T(\overline{q}^o+\Delta,\overline{q}^o)}{m_0^N(\underline{q}^o,\underline{q}^o)} = \frac{\Phi^T(n_a^T(\overline{q}^o+\Delta,\overline{q}^o,0))\omega(\overline{q}^o+\Delta)+d(\overline{q}^o+\Delta)}{\Phi^N(n_a^N(\underline{q}^o,\underline{q}^o,-\infty))\omega(\underline{q}^o)+d(\underline{q}^o)}\frac{1}{1-l^T(\overline{q}^o)},$$
$$= \frac{\Phi^T(n_a^T(\overline{q}^o+\Delta,\overline{q}^o,0))+d(\overline{q}^o+\Delta)}{\Phi^N(n_a^N(\underline{q}^o,\underline{q}^o,-\infty))\omega(\underline{q}^o)+d(\underline{q}^o)}\frac{1}{\delta_T},$$

where the last equality follows because $\omega(\overline{q}^o+\Delta)=1$, and $l^T(\overline{q}^o)=(1-\delta_T)$.

4.5 Trademark policy

Consider a firm without a trademark. Based on (18), the value of this firm is given by

$$V^N(q,q^o,m)=\max\{V^{NN}(q,q^o,m),V^T(q,q^o,m)-c_T\}.$$

Here, V^{NN} is the value from not registering a trademark in the current period and continuing as a non-trademark firm into the next period

$$V^{NN}(q,q^o,m)=\widetilde{\Pi}^N(q,q^o,m)+((1-\chi)/(1+r))V^N(q,q^o,m^N),$$

where $\widetilde{\Pi}^N(q,q^o,m)$ is the firm's period profit. The difference between the value from registering a trademark versus not registering in the current period is

$$V^D(q,q^o,m)=V^T(q,q^o,m)-V^{NN}(q,q^o,m). \tag{34}$$

The firm registers a trademark if $V^D(q,q^o,m)>c_T$. Now, denote the value of the firm in terms of its perceived and actual quality (q,q^o) and age A by $V_{\text{A}}^\Gamma(q,q^o)\equiv V^\Gamma(q,q^o,m)$, $\Gamma=N,T$, or NN, and $m\equiv m(\text{A},q,q^o)$. Let $V_{\text{A}}^D(q,q^o)=V_{\text{A}}^T(q,q^o)-V_{\text{A}}^{NN}(q,q^o)$.

Proposition 5 *i)* $V_{\text{A}}^D(q,q^o)$ *is increasing in* q^o *and* A*, ii)* $V_{\text{A}}^D(q^o+\Delta,q^o)>V_{\text{A}}^D(q^o,q^o)$ *for all* q^o *and* A*.*

Proposition (5) implies that at any age higher quality firms have trademarks, and given a quality level, the difference between the value of a trademark firm and a non-trademark one grows over time. Having a trademark is more valuable for a firm with a successful persuasive advertising campaign. Such firms tend to register a trademark at a lower quality level at any age, and at a lower age given a quality level.

When does stationary equilibrium feature both trademark firms and those that never trademark? The answer depends on how high the cost of registration, c_T, is relative to the benefits from registration. Suppose in equilibrium some firm with perceived quality $\widetilde{q}\in[\underline{q}^o,\overline{q}^o+\Delta]$ registers a trademark at some age A. This outcome is obtained if c_T is sufficiently low. Then, by Proposition (5), all firms with quality $q>\widetilde{q}$ also register a trademark at most by age A. Depending on the range of quality in equilibrium, there can also be a quality level $\widehat{q}>\tilde{q}$ such that all firms with $q>\widehat{q}$ register a trademark at birth (A$=0$). If c_T is high enough, no firm will register a trademark.

Figure 7 shows the regions of trademark policy in the firm (steady-state) size-quality plane – some of these regions may not exist depending on the parameters. In region I, only the firms with no persuasive advertising exist and the quality is so low that no firm will trademark. In region II, both types (persuasive

and non-persuasive) exist but do not trademark. In region III, firms with persuasive advertising will trademark at some point, but others do not trademark. Both types of firms trademark at some point (that differs by type) in region IV. In region V, firms with persuasive advertising trademark at birth, while others trademark at a different age. Both types of firms trademark at birth in region VI. The last region (VII) consists only of persuasive firms with very high perceived quality which trademark at birth (upon entry).

5 Calibration

The model is now taken to data. The empirical moments used for calibration are shown in Table 9. The 19 parameters in Table 10 are calibrated so that the key features of the model economy closely match those of the U.S. economy.[48]

The efficiency distribution, $H(e)$, is assumed to be uniform over $[\underline{e}, \overline{e}]$. Without loss of any economic representation, three parameters are normalized: $\mathcal{M}, B, \underline{e} = 1$. The time preference parameter is set at $\beta = 0.99$. For the utility function parameters, we use the estimates in Bils and Klenow (2001). Based on the averages of the Engel curve slopes for quality and quantity reported in their Table 4, we set the elasticity parameters to $\xi = 0.57$ and $\theta = 0.76$. The exogenous exit rate δ is set to 0.1 which is taken from the Business Dynamics Statistics.

The advertising intensities in Table 9 are obtained from our estimates in Table 8: 0.57% for non-trademark firms and 0.79% for trademark firms. For the depreciation rate of advertising stock, we use 55% per year in Hall (2014) that is based on various estimates in the literature. For instance, in an early study, Lambin (1976) estimates that advertising depreciation rates differ across product groups, averaging around 50% per year. Similarly, Brown (1978) reports depreciation rates of approximately 60% in his study of the cigarette industry. In a more recent macroeconomic analysis, Corrado et al. (2009) examine the role of intangible capital, including advertising, in U.S. economic growth and use a 60% annual depreciation rate for advertising capital – see also Bagwell (2007) and Tellis (2004) for further discussion.

For the two parameters related to persuasive advertising, φ and Δ, there is limited information available in the literature. However, a recent study on the effects of persuasive advertising in the automobile industry is quite useful for our purposes. Guitart et al. (2018) examine the effects of persuasive advertising in the form of promoting non-premium products as if they were premium ("advertising up"). Using data on automobile ads, they estimate the impact of advertising up on advertising elasticity and brand value. As part of their analysis, Guitart et al. (2018) construct an advertising up index, defined over $(0, 1)$, that measures how aggressively firms market their products as premium. By using the mean of this index and the estimated impact of this index on advertising elasticity, we calibrate φ and Δ. Specifically, we set φ so that the expected success rate in persuasion equals the mean of the advertising-up index (Table 2 in Guitart et al. (2018)). We calibrate Δ by running a regression on our simulated data to match the coefficient of the interaction term "Advertising Stock $\times$ Advertising-up index" in Table 3 of Guitart et al. (2018), which equals 0.037.

[48] To calibrate the model, the state space is discretized into a grid. For each of the state variables, q and m, 100 grid points are used.

To discipline trademark costs, c_T, we rely on the estimates of trademark litigation costs. The largest portion of trademark-related costs pertain to litigation expenses in trademark infringement cases. According to the American Intellectual Property Law Association (AIPLA), these costs can vary widely depending on the stage and complexity of the case. For instance, litigating a trademark infringement case through the end of discovery can vary in the range \$375,000 - \$500,000, and \$750,000 - \$1.5 million or more if the case goes to trial.[49] Additionally, the number of trademark litigation cases in the U.S. has been trending up in recent years. In 2023, around 3,500 trademark litigation cases were filed in U.S. district courts. Using this information, we calculate a conservative estimate of the average trademark litigation cost-to-revenue ratio of 0.02% for large U.S. firms that tend to have registered trademarks.[50]

The 12 parameter values whose values are not directly assigned are jointly calibrated by using the data targets shown in Table 9. Let d_j represent the j-th data target and $m_j(p)$ be its analogue for the model as a function of the parameter vector $p \equiv (A^T, A^N, \delta_T, \delta_N, \eta, \rho, \sigma, \overline{e}, \kappa, c_T, \varphi, \Delta)$. The calibrated parameter values p solve

$$\min_p \sum_j \left[\frac{d_j - m_j(p)}{d_j}\right]^2.$$

The resulting parameter values are shown in Table 10.

5.1 Benchmark economy

The key features of the benchmark economy resulting from the calibration exercise are summarized in Figure 8. Most graphs are plotted in relation to firm efficiency levels in the calibrated model, which range from 1 (normalization) to 3.01. In some graphs, efficiency levels are indexed to 1-100.

In the benchmark economy, 17% of firms engage in persuasive advertising (persuasive firms). 4.2% of firms register a trademark, close to the 4% observed in the data. Among persuasive firms, 6% register for a trademark. For others this figure is 4%. This difference occurs because registering a trademark is relatively more profitable for persuasive firms, which enjoy higher perceived quality and higher markups.

Figure 8(a) shows trademark policy. The highest three efficiency levels among persuasive firms trademark their products, while only the highest two efficiency levels do so among other firms. The quality policy depicted in Figure 8(b) is monotonic, and quality increases more rapidly for efficiency levels near and beyond the trademark threshold. The likelihood of sale from an ad is shown in Figure 8(c). All ads sent by the highest quality firms generate a sale, whereas the lowest quality firms' ads turn into a sale with probability 60%. Since the lowest quality firms are the least favored by consumers, this probability is equal to the mass of consumers with no information about firms.

Higher quality firms with trademarks reach consumers more easily. They are also preferred more by consumers and their ads are more likely to turn into sales. As a result, they grow more quickly. Figure 8(d) shows the life-cycle firm size profile for the lowest quality firm that registers a trademark and the counterfactual profile if that firm did not register a trademark. This firm registers a trademark at birth. It is 150% larger at birth and grows much faster compared to its counterfactual, achieving nearly 220% larger size than the counterfactual by the end of 30 periods.

[49] See 2019 AIPLA Report of the Economic Survey, available at https://www.aipla.org/.

[50] We obtain this ratio by dividing the total cost of litigation for cases that went to trial (based on average trial cost) by the total estimated revenue of firms with trademarks: $(3,500 \times (\$750,000 + \$1,500,000)/2)/\$18.8T = 0.0002$.

Despite the uniform distribution of quality development efficiency among firms, firm size distribution is positively skewed and has a relatively long tail populated by high-quality firms with trademarks (Figure 8(e)). This skewness results from the higher advertising activity and higher likelihood of sale for higher quality firms, which allow them to accumulate a large customer base. The flat portion on the left tail is attributable to small firms with low efficiency that mainly gain customers through customer diversion. High quality products constitute a considerable share of a consumer's expenditure. Figure 8(f) plots the expenditure share of goods as a function of firm efficiency. Around 18% of the expenditure is dedicated to firms with the highest 1% efficiency (quality). The concentration of expenditures in high quality firms with trademarks is an important source of welfare, as highlighted below in model simulations.

Advertising intensity increases with efficiency, and firms with trademarks have significantly higher advertising intensity (Figure 8(g)). The saw-tooth appearance in this figure is due to the co-existence of persuasive and non-persuasive firms at the same actual quality level. The firms with the highest quality have an average advertising intensity of 0.85%, whereas the lowest quality firms' average advertising intensity is 0.5%. As firms age, their advertising intensity decreases (Figure 8(h)). Persuasive firms have higher advertising intensity early on. At birth, their advertising intensity is as high as 3%, but drops below 0.3% as they age. On the other hand, non-persuasive firms have an advertising intensity of 1.3% at birth, which declines to below 0.4% as they grow. These life-cycle profiles for advertising intensity generally line up with the empirical evidence in Table 7.

How important is persuasive advertising in increasing perceived quality? Figure 8(i) shows the percentage gain in the perceived quality for successful firms in relation to their efficiency level. For the lowest efficiency level, successful persuasive advertising results in a 3% increase in perceived quality. As efficiency increases, the importance of persuasive advertising in enhancing perceived quality diminishes. For highly efficient firms, the percentage increase in quality is below 0.5%.

Persuasive firms also enjoy higher markups. In Figure 8(i), the percentage markup gain of persuasive firms for each *actual* quality level is shown. The gain is higher for low-quality firms. The firm with the lowest *actual* quality experiences a 7.5% increase in its markup if it succeeds in persuasive advertising. In contrast, the corresponding increase is below 1% for the highest-quality persuasive firm. Overall, the average markup of persuasive firms is 2.9% higher than that of non-persuasive firms.

6 Counterfactual simulations

The model is now simulated to conduct some thought experiments. First, to evaluate the effects of trademarks at large, an economy with no trademarks is analyzed. This is done by setting the one-time trademark cost c_T prohibitively high – all other parameters fixed at their benchmark values. This economy is called the *no-trademark economy*.

The second experiment examines the effects of an increase in imitative and infringing activity which may result, for instance, from a foreign trade shock or advances in imitation and infringement technologies. Such higher activity has two effects. First, firms without trademarks are subject to a higher customer diversion rate, δ_N. Second, as a result of higher diversion, their ability to differentiate themselves and generate customer awareness is also diminished – represented by a lower A^N. This economy is referred to as the *high-diversion economy*. We also explore the effects of high diversion if there were no trademark protection in the economy to start with.

In the third experiment, we consider an economy where trademarks are better protected, similar in spirit to the case of stronger trademark protection following the 1984 Trademark Counterfeiting Act examined empirically in section 2.4. Our final experiment studies an economy with no persuasive advertising.

6.1 No-trademark economy

The absence of a trademark system results in important changes. Figure 9(a) compares the quality policy for the benchmark and no-trademark economies. In the benchmark economy, firms with trademarks develop products with 12% higher quality on average, compared to the no-trademark economy. However, firms without trademarks offer lower quality products. In the benchmark economy, firms without trademarks are discouraged from investing in quality due to more intense competition from firms with trademarks. This is a result of the more efficient marketing technology and higher incentives to advertise induced by trademark protection.

Figure 9(b) plots the probability of sale. Note the distinct increase in probability at the efficiency level corresponding to the trademark threshold in the benchmark economy. The probability of sale for trademark firms in the benchmark economy is much higher than that in the no-trademark economy. Increased consumer information about firms leads to more intense competition in the benchmark economy. However, the probability of sale is not uniformly lower in the benchmark economy compared to the no-trademark economy; in fact, the opposite occurs for lower quality levels.

Despite the negative externality on less efficient (lower quality) firms, consumers in the benchmark economy purchase higher quality products, and more consumers have access to them. 18% of consumers purchase the highest 1% quality products, compared to 7% in the no-trademark economy (Figure 9(c)). These differences highlight the critical role of trademarks in ensuring quality provision.

Trademark protection also leads to a significant reallocation of output and employment across firms. Figure 9(d) indicates substantial job reallocation towards high-quality firms with trademarks from firms without trademarks. The highest quality firms experience nearly a 60% increase in employment when trademark protection is in place. The biggest losers in a trademark economy are the firms just below the trademark threshold – they would gain around 150% in employment if there were no trademarks.

As shown in Figure 9(e), trademarks generate a high degree of skewness in firm size in the benchmark economy, whereas their absence induces lower inequality in firm size. Figure 9(f) focuses on the right tail of the distribution, plotting the largest 2% of the firms in both economies. The firms with trademarks are much larger than the others in the benchmark. Overall, the benchmark economy behaves more like an economy with superstar firms, where trademarks reinforce the dominance of of high quality firms.

The superstar nature of the benchmark economy is more evident in Figure 9(g). In the no-trademark economy, young and small firms gain diverted customers at a higher rate, benefiting them initially. In contrast, in the benchmark economy, a young high-quality firm grows rapidly after registering a trademark, reaching a size more than 2.5 times that of the corresponding firm in the no-trademark case. The life-cycle size profile differs significantly for the lowest quality firms, which have an eventual size that is 9% smaller in the benchmark economy compared to the no-trademark one.[51]

[51] Notice also that the lowest quality firms approach their steady-state size much sooner compared to the highest quality firms. The more gradual convergence to steady-state size for high quality firms is driven by the fact that marketing technology

Concentration of customers in firms is plotted in Figure 9(h). Concentration is measured by the cumulative customer share of firms below a given efficiency (quality) level. In the benchmark economy, the highest 2% quality firms have a share of 35%, but only around 10% in the no-trademark economy. Note that customer concentration is uniformly higher for all quality levels in the benchmark economy.

Advertising intensity is higher in the benchmark economy at all efficiency levels (Figure 9(i)). The biggest difference occurs for high quality firms with trademarks. In particular, the highest-quality firms with trademarks have more than 70% higher advertising intensity in the benchmark economy.

Overall, the availability of trademarks improves consumer welfare. However, the mechanism by which it does so is not uniform across different dimensions of welfare, which include product quality, variety, and consumer information. In the benchmark economy, low-quality firms are at such a disadvantage in competition that they do not market their products as aggressively. High-quality firms with trademarks market their products heavily, but this is not enough to offset the lower advertising from low-quality firms. Consequently, the total number of ads is lower in the benchmark economy, leading to a decrease in the variety of products consumed. However, the average product quality is significantly higher, which dominates the overall change in welfare.

As shown in Table 11, the product information available to consumers is 7.5% lower, and the product variety consumed is 5.8% lower in the benchmark. Nevertheless, the average product quality is 12.4% higher. This increase in average product quality results from two sources: first, firms protected by trademarks invest more in quality; second, there is a reallocation of consumption towards higher quality products due to greater consumer awareness of these firms. As a result, consumer welfare, measured by compensating variation (C.V.), is 3.6% higher. At the same time, the benchmark economy exhibits much higher concentration in revenue, as measured by the Hirschman-Herfindahl Index (HHI).[52] The asymmetric effects of trademarks on different dimensions of welfare is due to the calibrated advertising elasticity being very low. With this calibration, the cost of advertising is highly convex. Consequently, the increase in advertising by trademark firms does not compensate for the decline in advertising by lower-quality firms.

6.2 High-diversion economy

Customer diversion rates can vary across industries, regions, or countries depending on the costs and benefits of imitation and infringement. For instance, customer diversion may increase when firms face foreign competition, some of which may imitate domestic products. The strength of institutions and regulations designed to protect brands also plays a crucial role. Loosening regulations on product quality in an industry can reduce entry barriers, inviting infringing competitors and resulting in higher customer diversion. Imitation technologies can also improve over time, leading to higher infringement.

Our analysis here is primarily motivated by the China trade shock examined in section 2.5. Our premise, as in the empirical analysis, is that this trade shock generated, in part, an increase in imitative and infringing foreign competition. We assume that this increase affects both the customer diversion rate

has diminishing returns, limiting the number of new consumers a firm can acquire in any period.

[52] Since there is a continuum of firms in each industry, conventionally computed HHI would inevitably be always zero. Instead, a converted (discretized) HHI is considered, where a constant measure of firms is set to be one firm and that measure is calibrated so that the benchmark HHI matches the average 6-digit NAICS HHI in the US economy (1280). This constant measure of firms is used across all equilibria pertaining to the different thought experiments to be consistent with the benchmark measurement of HHI.

and the marketing effectiveness for non-trademark firms.[53] Our empirical analysis indicates that after the shock employment in non-trademark firms decreases by around 1.72% relative to trademark firms. We use this information to discipline the implied effects of the shock in the model. We assume that both the customer diversion rate and the marketing efficiency of non-trademark firms are scaled by the same proportional rate due to the shock. This implies a 7.2% increase in the customer diversion rate and the same percent decrease in the marketing productivity of non-trademark firms from their benchmark values.

The quality policies shown in Figure 10(a) indicate that high-efficiency firms have slightly higher quality levels compared to the benchmark, with the highest quality firms seeing an increase of 0.2%. However, at lower efficiency levels, firms reduce quality – particularly those just below the trademark threshold in the high-diversion economy, which lower their quality by 1.4%. Additionally, the probability of sale decreases for firms near the benchmark trademark threshold but rises for low-quality firms (Figure 10(b)). Overall, high-quality trademark firms account for a larger share of consumer spending in the high-diversion economy (Figure 10(c)). The change is more pronounced for persuasive firms with high quality and trademarks.

Higher customer diversion also has large employment reallocation effects (Figure 10(d)). Firms with the highest efficiency experience a 3% increase in employment in the high-diversion economy. In contrast, high-quality firms without trademarks suffer the most, losing 7% of their employment.

Figure 10(e) indicates that both economies exhibit significant skewness in size distribution, but the higher diversion economy shows slightly greater skewness. The highest quality firm type has a slightly higher life-cycle size profile in the high-diversion economy (Figure 10(g)). Conversely, a high diversion rate adversely affects low-quality firms, making them smaller compared to their counterparts in the benchmark economy. The steady-state size of the lowest quality firm type is about 1% smaller in the high-diversion economy (Figure 10(f)). Industry concentration increases at the efficiency levels for firms around the trademark threshold (Figure 10(h)).

The advertising intensity of trademark firms increases very slightly, while it decreases more for non-trademark firms (Figure 10(f)). In particular, the lowest quality firms experience a decrease of around 2% from their benchmark level of advertising intensity.

Both product information and the number of varieties consumed are lower, while the average product quality is higher in the high-diversion economy. As seen in Table 11, product information declines by 3.4%, and the number of varieties consumed decreases by 2.5%. On the other hand, average product quality increases by 0.6%. Despite this improvement in average quality, the negative impacts on product information and variety dominate, leading to decline of 1.0% in welfare. Additionally, higher diversion rate increases industry concentration by 7.2%, driven by the increased customer share of high-quality firms with trademarks.

6.2.1 High diversion in a no-trademark economy

To further assess the role of trademarks in a high-diversion economy, consider the following thought experiment: What would be the effects of higher customer diversion if there were no trademark protection

[53] Trademark firms' diversion rate and marketing effectiveness can also be affected to some extent. However, trademark protection would limit the extent of such effects. For simplicity, we model the effects of a high diversion shock through its effects on non-trademark firms that are relatively more vulnerable to such a shock.

to start with? In an economy with a similar customer diversion shock but without trademark protection, high-quality firms would not invest as much in quality and customer accumulation. In contrast to the earlier exercise, the largest effect would be on high quality firms. As depicted in Appendix Figure A2(a), in high diversion environment without trademark protection the highest quality firms' investment in quality are hit the hardest – they reduce their quality by 0.6%. In contrast, lowest quality firms' quality increases by around 0.2%.

Low-quality firms face less competition from high-quality firms, and their likelihood of sale from an ad goes up by 0.8% as shown in Figure A2(b). Firms' incentives for marketing are also adversely affected. The average advertising intensity in the economy goes down by 4% (Figure A2(i)). Highest quality firms experience a larger decline (6%) in advertising intensity. Consequently, consumption shifts from high-quality to low-quality firms. As a result, the share of highest quality firms in consumers' spending decreases by 2.5% whereas the lowest quality firms see an increase around 3.6% (Figure A2(c)).

The life-cycle firm size evolution of the highest and lowest quality firms are shown in Figures A2(f)-(g). The lowest quality firms experience a 0.5% increase in their steady-state size, whereas the highest quality firms' size falls by 4.4%. This is partly a result of declining marketing activity of firms. Advertising intensity falls by around 4% at all efficiency levels – Figure A2(i). As a result, industry concentration decreases – Figure A2(h).

Overall, trademarks mitigate the welfare-reducing effects of a shock that increases customer diversion. Without trademarks, higher customer diversion has a larger impact on all dimensions critical for consumer welfare. As shown in Table 11, the availability of product information would go down by 3.7%, and the consumed product variety by 2.9%. Average product quality would also be lower by 0.8%. All of these effects would add up to a 1.9% lower welfare. At the same time, industry concentration would go down by around 3.4%.

6.3 Stronger trademark protection

Motivated by the Trademark Counterfeiting Act of 1984 analyzed empirically in section 2, we next consider the effects of stronger trademark protection. Our quantitative exercise will be disciplined by the estimated effects in section 2. As depicted in Figure 11(a), following the stronger trademark protection enacted by the Trademark Counterfeiting Act of 1984, employment in trademark firms increased by 10% relative to the matched non-trademark firms over the next five years. Assuming that the full impact of the act materialized within the first five years, we will calibrate the parameters of stronger trademark protection so that our model yields a similar relative employment change.[54]

We assume that stronger protection decreases the customer diversion rate and increases the marketing effectiveness of trademark firms. Specifically, assume that both the customer diversion rate, (δ_T), and the marketing productivity of trademark firms, (A^T), are scaled by the same magnitude to match the observed employment effects of stronger trademark protection provided by the Act. This implies a 10.2% decrease in the customer diversion rate and a similar increase in the marketing productivity of trademark firms from their benchmark values. The results of the simulation for this scenario are in Figure 11.

[54]Note that our benchmark model is calibrated using data moments that pertain to a time period that is much more recent than that for the Trademark Counterfeiting Act. Thus, our counterfactual exercise should be viewed as assessing the effects of a legislation that would have created an effect of similar magnitude on employment in recent years as in the case of the Trademark Counterfeiting Act.

Higher quality firms, especially those with trademarks, benefit from stronger trademark protection. The highest quality firms invest even more in quality development, increasing their quality by 1%, while non-trademark firms' quality levels fall by an average of 1.7% (Figure 11(a)). Moreover, non-trademark firms' likelihood of sale from an ad decreases by 0.6% (Figure 11(b)).

Trademark firms see a large rise in their share of consumer spending, with an average increase of approximately 6%, reducing the share for non-trademark firms (Figure 11(c)). As a result, there is a large reallocation across firms. Trademark firms gain 6% in employment on average, while non-trademark firms, especially those near the trademark threshold, experience a decline in employment by 11% (Figure 11(d)).

High-quality trademark firms are now born larger and experience faster growth at the expense of low-quality non-trademark firms (Figures 11(f)-(g)). The highest quality firms are around 8% larger at birth and their steady-state size is more than 16% larger. In contrast, the lowest quality firms start out approximately 1.5% smaller and reach a steady-state size that is 2.2% smaller.

With stronger protection, trademark firms have now a higher incentive to advertise their products, leading to an increase in their advertising intensity by 3.5%, whereas non-trademark firms experience a slight decrease in their advertising intensity (Figure 11(i)). As a result of all the effects above, firm size inequality and industry concentration increase significantly due to the the stronger trademark protection (Figure 11(e)-(h)).

Overall, stronger protection leads to improvements across all three dimensions of welfare: product information, variety, and average quality increase by 1.2%, 1.0%, and 1.9%, respectively. Consequently, the overall welfare measure rises by 1.3%. However, a significant downside is the notable increase in industry concentration.

6.4 No persuasive advertising

To understand the effects of persuasive advertising, consider now an economy with no persuasive advertising. Eliminating persuasive advertising disproportionately benefits high-quality trademark firms. High-quality trademark firms invest more in quality. As depicted in Figures 12(a) and (b), the highest quality firms increase their quality by 0.4%. Their share in consumer spending and employment both increase by 5%. Conversely, low-quality non-trademark firms invest even less in quality, and their share of consumer spending and employment decrease.

Without persuasive advertising, the lowest quality firms start out about 1% smaller and grow relatively less compared to the highest quality firms. On the other hand, the highest quality firms see a more dramatic increases in size; at birth, they are 3% larger and grow more over their life-cycle (Figures 12(d)-(e)). The right tail of the firm size distribution extends further, leading to an increase in firm-size inequality (Figure 12(c)). High-quality firms also advertise more aggressively, increasing their advertising intensity by around 2% – Figure 12(f).

Overall, as reported in Table 11, eliminating persuasive advertising improves all three dimensions of welfare. Product information increases by 1%, the number of varieties expands by 1%, and average product quality improves by 0.8%. As a result, consumer welfare increases by 0.8%. However, firm-size inequality rises, leading to a 4% higher HHI. This occurs because persuasive advertising discourages high-efficiency firms from investing in quality and, consequently, from expanding to their full potential,

thereby reducing firm-size inequality. Furthermore, with the elimination of persuasive advertising, average markup goes down slightly by around 0.5%.

6.4.1 How much does persuasive advertising reduce the benefits from trademarks?

Our results indicate that persuasive advertising can have a significant negative impact on consumer welfare. On the other hand, trademarks overall enhance consumer welfare. However, in the calibrated economy, trademarks disproportionately protect persuasive firms, as these firms have a higher tendency to trademark their products—6% of persuasive firms register for a trademark, compared to 4% of non-persuasive firms. How much does this pattern reduce the potential benefits of trademarks? To answer this question, we conduct an exercise similar to the high-diversion experiment and compare two hypothetical economies without persuasive advertising: one with trademarks and the other one, without.

The results show that without persuasive advertising, trademarks improve overall welfare by 18%. In other words, persuasive advertising undermines the potential welfare benefits of trademarks by 18%. This result gives rise to the following question: Can sufficiently more effective persuasive advertising (i.e. Δ values higher than the calibrated value of 0.02) eliminate the benefits of trademarks entirely? Our experimentation with higher values of Δ indicates that this mitigation is highly concave in Δ. Even with a Δ value 10 times larger than the calibrated value, persuasive advertising mitigates 29% of the welfare benefits of trademarks. Therefore, it is very unlikely that trademarks would hurt overall consumer welfare due to excessive persuasive advertising. Nevertheless, in practice for some industries or products with higher Δ such large effects may occur.

7 Conclusion

What is the significance of the trademark system in the U.S. economy? This paper has provided empirical evidence on the types of firms that register trademarks, and how firms with registered trademarks perform compared to those without. The evidence suggests that firms select into trademark registration based on a variety of characteristics associated with high growth and innovation. Beyond this selection, firms that register trademarks tend to experience higher growth compared to their counterparts and are more likely to achieve outcomes in the right tail of employment and revenue distributions. Stronger trademark protection is associated with better outcomes for firms with trademarks. Trademarks also appear to protect firm value in the face of more intense competition – in the form of higher import competition. These findings suggest an important role for trademarks in the making of skewness in firm size and in the emergence of superstar firms.

Motivated by this evidence, a general equilibrium model of trademarks is developed. Firms invest in product quality and engage in marketing to gradually build a customer base subject to customer attrition due to imitation and infringement. Advertising by firms can be both informative and persuasive in nature. These two types of advertising matter distinctly in the welfare effects of trademarks. Firms can protect their customer base from erosion by registering trademarks. Trademarks have two basic functions in the model: they protect a firm's customer base from depreciation due to imitation and infringement of its brand, and they allow firms to spread information about themselves more effectively. The analysis of the model indicates that the existing trademark system in the United States has economically significant

effects. The absence of trademark protection would result in a significant loss of welfare, driven mainly by a decline in average product quality. Counterfactual experiments suggest that trademarks are even more critical for welfare and quality provision in environments where imitation and infringement are inherently more intense. In other words, an economy with a trademark system is more resilient in the face of higher customer diversion.

To focus on the fundamental roles of trademarks, the model has abstracted from a variety of other considerations. With some modifications, the model can accommodate many of them. For instance, the model assumes exogenous entry and exit. Consequently, we do not examine the potential adverse effects on entry arising from the dominance of trademark firms and the associated high concentration. The model also does not consider any prestige effects from consuming products of well-known firms, which could further reinforce reallocation towards firms with more established trademarks. Furthermore, product quality is perfectly revealed in ads, and future work can explore the effects of imperfect information about quality. Finally, the set of feasible product varieties is fixed, and the connection between new product innovations and trademarks can be studied in an extension.

References

Acemoglu, Daron, David Autor, David Dorn, Gordon H Hanson, and Brendan Price, "Import Competition and the Great US Employment Sag of the 2000s," *Journal of Labor Economics*, 2016, *34* (S1), S141–S198.

Alfaro, Laura, Maggie Chen, Ge Bao, Junjie Hong, and Claudia Steinwender, "Omnia Juncta in Uno: Foreign Powers and Trademark Protection in Shanghai's Concession Era," *Working Paper*, 2022.

Autor, David, David Dorn, Lawrence F Katz, Christina Patterson, and John Van Reenen, "The Fall of the Labor Share and the Rise of Superstar Firms," *The Quarterly Journal of Economics*, 2020, *135* (2), 645–709.

Autor, David H, David Dorn, and Gordon H Hanson, "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," *American Economic Review*, 2013, *103* (6), 2121–68.

_ , _ , and _ , "The China Shock: Learning from Labor-Market Adjustment to Large Changes in Trade," *Annual Review of Economics*, 2016, *8*, 205–240.

Bagwell, Kyle, "The Economic Analysis of Advertising," *Handbook of Industrial Organization*, 2007, pp. 1701–1844.

Bayard, Kimberly, Emin Dinlersoz, Timothy Dunne, John Haltiwanger, Javier Miranda, and John Stevens, "Early-stage Business Formation: An Analysis of Applications for Employer Identification Numbers," Technical Report, National Bureau of Economic Research 2018.

Bils, Mark and Peter J. Klenow, "Quantifying Quality Growth," *American Economic Review*, September 2001, *91* (4), 1006–1030.

Brown, S., "Estimating the Effect of Advertising on Sales Over Time," *Journal of Marketing Research*, 1978, *15* (3), 345–355.

Butters, Gerard R, "Equilibrium Distributions of Sales and Advertising Prices," *The Review of Economic Studies*, 1977, *44* (3), 465–491.

Cavenaile, Laurent, Murat Alp Celik, Pau Roldan-Blanco, and Xu Tian, "Style over substance? Advertising, innovation, and endogenous market structure," *Journal of Monetary Economics*, 2024, p. 103683.

Chow, Melissa C, Teresa C Fort, Christopher Goetz, Nathan Goldschlag, James Lawrence, Elisabeth Ruth Perlman, Martha Stinson, and T Kirk White, "Redesigning the Longitudinal Business Database," Technical Report, National Bureau of Economic Research 2021.

Corrado, C., C. Hulten, and D. Sichel, "Intangible Capital and U.S. Economic Growth," *Review of Income and Wealth*, 2009, *55* (3), 661–685.

Davis, Steven J, John C Haltiwanger, and Scott Schuh, "Job Creation and Destruction," *MIT Press Books*, 1998, *1*.

Dinlersoz, Emin M and Mehmet Yorukoglu, "Informative Advertising by Heterogeneous Firms," *Information Economics and Policy*, 2008, *20* (2), 168–191.

_ and _ , "Information and Industry Dynamics," *American Economic Review*, 2012, *102* (2), 884–913.

Dinlersoz, Emin, Nathan Goldschlag, Amanda Myers, and Nikolas Zolas, "An Anatomy of US Firms Seeking Trademark Registration," in "Measuring and Accounting for Innovation in the 21st Century," University of Chicago Press, 2018.

Economides, Nicholas, "Trademarks," *The New Palgrave Dictionary of Economics and the Law (Peter Newman, Ed.)*, 1998.

Economides, Nicholas S, "The Economics of Trademarks," *Trademark Rep.*, 1988, *78*, 523.

Foster, Lucia, Cheryl Grim, and Nikolas Zolas, "A Portrait of US Firms that Invest in R&D," *Economics of Innovation and New Technology*, 2020, *29* (1), 89–111.

Galbi, Douglas, "Communications Policy, Media Development, and Convergence," *Working Paper*, 2001.

_ , "Some Economics of Personal Activity and Implications for the Digital Economy," *Working Paper*, 2001.

Gourio, Francois and Leena Rudanko, "Customer Capital," *Review of Economic Studies*, 2014, *81* (3), 1102–1136.

Graham, Stuart JH, Alan C Marco, and Amanda F Myers, "Monetizing Marks: Insights from the USPTO Trademark Assignment Dataset," *Journal of Economics & Management Strategy*, 2018, *27* (3), 403–432.

_ , Galen Hancock, Alan C Marco, and Amanda Fila Myers, "The USPTO Trademark Case Files Dataset: Descriptions, Lessons, and Insights," *Journal of Economics & Management Strategy*, 2013, *22* (4), 669–705.

Greenwood, Jeremy, Yueyuan Ma, and Mehmet Yorukoglu, ""You Will:" A Macroeconomic Analysis of Digital Advertising," *The Review of Economic Studies*, 06 2024, p. rdae067.

Grossman, Gene M and Carl Shapiro, "Informative Advertising with Differentiated Products," *The Review of Economic Studies*, 1984, *51* (1), 63–81.

Guitart, Ivan A., Jorge Gonzalez, and Stefan Stremersch, "Advertising non-premium products as if they were premium: The impact of advertising up on advertising elasticity and brand equity," *International Journal of Research in Marketing*, 2018, *35* (3), 471–489.

Guzman, Jorge and Scott Stern, "The state of American entrepreneurship: New estimates of the quantity and quality of entrepreneurship for 32 US States, 1988–2014," *American Economic Journal: Economic Policy*, 2020, *12* (4), 212–43.

Hall, Robert, "What the Cyclical Response of Advertising Reveals about Markups and other Macroeconomic Wedges," *Working Paper*, 2014.

Heath, Davidson and Christopher Mace, "The Strategic Effects of Trademark Protection," *The Review of Financial Studies*, 2020, *33* (4), 1848–1877.

Ignaszak, Marek and Petr Sedláček, "Customer Acquisition, Business Dynamism and Aggregate Growth," *Working Paper*, 2022.

Jarmin, Ron S and Javier Miranda, "The longitudinal business database," Working Paper 02-17, U.S. Census Bureau, Center for Economic Studies 2002.

Kost, Kyle, Jeremy Pearce, and Liangjie Wu, "Market Power Through the Lens of Trademarks," Technical Report, Working Paper 2019.

Lambin, J. J., "Trademark Law: An Economic Perspective," *Advertising, Competition, and Market Conduct in Oligopoly Over Time.*, 1976.

Landes, William M and Richard A Posner, "Trademark Law: An Economic Perspective," *The Journal of Law and Economics*, 1987, *30* (2), 265–309.

_ **and** _ , "The Economics of Trademark Law," *Trademark Rep.*, 1988, *78*, 267.

_ **and** _ , *The Economic Structure of Intellectual Property Law*, Harvard university press, 2003.

Pearce, Jeremy and Liangjie Wu, "Brand Reallocation, Concentration, and Growth," *Working Paper*, 2022.

Roldan, Pau and Sonia Gilbukh, "Firm Dynamics and Pricing Under Customer Capital Accumulation," *Documentos de trabajo/Banco de España, 1838*, 2018.

Schautschick, Philipp and Christine Greenhalgh, "Empirical Studies of Trade Marks–The Existing Economic Literature," *Economics of Innovation and New Technology*, 2016, *25* (4), 358–390.

Sedláček, Petr and Vincent Sterk, "The Growth Potential of Startups Over the Business Cycle," *American Economic Review*, 2017, *107* (10), 3182–3210.

Stegeman, Mark, "Advertising in Competitive Markets," *American Economic Review*, 1991, pp. 210–223.

Sterk, Vincent, Petr Sedláček, and Benjamin Pugsley, "The nature of firm growth," *American Economic Review*, 2021, *111* (2), 547–79.

Stokey, Nancy and Robert Lucas, *Recursive Methods in Economic Dynamics*, Harvard University Press, 1989.

Sutton, John, *Sunk costs and market structure: Price competition, advertising, and the evolution of concentration*, MIT press, 1991.

Tellis, G. J., "Effective Advertising: Understanding When, How, and Why Advertising Works," 2004.

Törnqvist, Leo, Pentti Vartia, and Yrjö O Vartia, "How should relative changes be measured?," *The American Statistician*, 1985, *39* (1), 43–46.

Table 1: Some characteristics of firms with and without trademarks in 2014

Fraction of firms with trademark	4.01%
Fraction of new-born firms with trademark	0.90%
Avg. age at first trademark registration (years)	6.83
	(0.02)

	Firms with	
	TM	no TM
Avg. employment	285	12
	(10.37)	(0.06)
Avg. employment (largest 1% of firms)	19,360	427
	(939.20)	(5.68)
Employment share	49%	51%
Employment share (newborn (age zero) firms)	1.5%	98.5%

Notes: Standard errors in parentheses. Largest 1% of firms is defined separately for firms with and without TMs.

Table 2: Early firm characteristics and the first trademark registration

	TM Indicator within 5 years of birth (0/1)	Duration to first TM (years)
	LPM	OLS
Birth Size Percentile (50-75)	0.006^{***}	-0.164^{***}
	(0.0007)	(0.0194)
Birth Size Percentile (75-95)	0.012^{***}	-0.442^{***}
	(0.0015)	(0.0230)
Birth Size Percentile (95+)	0.022^{***}	-0.963^{***}
	(0.0021)	(0.0353)
First Year Growth Percentile (50-75)	0.006^{***}	0.214^{***}
	(0.0007)	(0.0224)
First Year Growth Percentile (75-95)	0.013^{***}	0.0005
	(0.0013)	(0.0185)
First Year Growth Percentile (95+)	0.023^{***}	-0.343^{***}
	(0.0023)	(0.0325)
Born Multi-State	0.007^{**}	-0.598^{***}
	(0.0029)	(0.0358)
Born with Patent	0.168^{***}	-1.429^{***}
	(0.0063)	(0.0688)
Born Exporter	0.030^{***}	-0.238^{***}
	(0.0017)	(0.0298)
N	$8,600,000$	$231,000$
R^2	0.052	0.205

Notes: Standard errors clustered by industry in parentheses. *, **, and *** indicate significance at 10, 5, and 1%, respectively. Models include birth cohort and 6-digit NAICS industry fixed effects. N is rounded to prevent disclosure of confidential information.

Table 3: Size-age profiles for firms with and without trademarks

Age (years)	Avg. Size (employment)		Relative Avg.
	TM	No TM	Size (TM/No-TM)
0	15.32	5.63	2.72
	(0.05)	(0.01)	
5	56.44	9.43	5.98
	(0.15)	(0.03)	
10	99.56	11.40	8.73
	(0.30)	(0.04)	
15	150.50	12.84	11.72
	(0.53)	(0.06)	
20	203.50	14.41	14.12
	(0.86)	(0.12)	
25	256.30	15.91	16.11
	(1.35)	(0.18)	
30	319.70	16.30	19.61
	(2.38)	(0.18)	
35	497.00	18.57	26.76
	(6.42)	(0.34)	

Notes: Standard errors in parentheses. Relative avg. size is the ratio of the average size of TM firms to that of no-TM firms.

Table 4: The effects of the 1984 Trademark Counterfeiting Act on firms with and without trademarks in 1984

	Employment	Payroll	Establishments	Employment	Payroll	Establishments
	(1)	(2)	(3)	(4)	(5)	(6)
$Post \times TM$	0.045***	0.055***	0.016***	0.041***	0.052***	0.026***
	(0.014)	(0.017)	(0.004)	(0.015)	(0.018)	(0.005)
$Post \times Large$	-0.038	-0.041	0.110**			
	(0.081)	(0.085)	(0.047)			
$Post \times High$				-0.087	-0.179	-0.037
				(0.115)	(0.126)	(0.063)
$Post \times TM \times Large$	0.235**	0.250**	0.184***			
	(0.101)	(0.104)	(0.067)			
$Post \times TM \times High$				0.133***	0.131**	0.010
				(0.046)	(0.058)	(0.020)
N	56,000	56,000	56,000	56,000	56,000	56,000
R^2	0.935	0.929	0.964	0.935	0.929	0.963

Notes: Dependent variables are in logarithms. All specifications include firm and industry-year fixed effects. Standard errors clustered by firm in parentheses. *, **, and *** indicate significance at 10, 5, and 1%, respectively. N is rounded to prevent disclosure of confidential information. The estimated coefficients correspond to the RHS variables in equation 3.

Table 5: The effects of China trade shock on firms with and without trademarks - second stage results from 2SLS IV estimation

	Employment		Revenue		Labor Productivity	
	(1)	(2)	(3)	(4)	(5)	(6)
$Post \times TM$	-0.0160*	-0.0346***	0.0097	-0.0349**	0.0044	-0.0155
	(0.0097)	(0.0103)	(0.0132)	(0.0141)	(0.0114)	(0.0120)
IC	0.0105***	0.0132***	0.0061***	0.0069***	-0.0027*	-0.0048***
	(0.0012)	(0.0013)	(0.0016)	(0.0017)	(0.0014)	(0.0014)
$Post \times IC$	-0.0137***	-0.0156***	-0.0115***	-0.0123***	-0.0005	0.0008
	(0.0010)	(0.0010)	(0.0013)	(0.0014)	(0.0011)	(0.0012)
$Post \times TM \times IC$	0.0030***	0.0031***	0.0056***	0.0065***	0.0033***	0.0043***
	(0.0009)	(0.0010)	(0.0012)	(0.0013)	(0.0011)	(0.0011)
N	63,500	53,000	48,500	40,500	48,500	40,500
R^2	0.0127	0.0111	0.0092	0.0049	0.0005	0.0003

Notes: Dependent variables are in logarithms. All specifications include firm and year fixed effects. Standard errors clustered by firm in parentheses. *, **, and *** indicate significance at 10, 5, and 1%, respectively. N is rounded to prevent disclosure of confidential information. Estimated coefficients correspond to the RHS variables in equation (4). Columns 1, 3, and 5 measure existing registered trademark ownership as of 2001, and columns 2, 4, and 6 measure trademark ownership as of 1999.

Table 6: Mean advertising expenditures and advertising intensity

	All firms			
	Expenditures ($K)		Intensity (%)	
	No TM	TM	No TM	TM
	12.4	147.2	0.48	0.65
	(0.3)	(9.6)	(0.001)	(0.005)
N	317,000	43,000	317,000	43,000

	Firms with positive advertising			
	Expenditures ($K)		Intensity (%)	
	No TM	TM	No TM	TM
	14.5	156.7	0.56	0.70
	(0.3)	(10.2)	(0.001)	(0.005)
N	271,000	40,500	271,000	40,500

Notes: Standard errors in parentheses. Advertising intensity is the ratio of advertising expenditures to revenue, (in %). N is rounded to prevent disclosure of confidential information.

Table 7: Advertising expenditures and advertising intensity over the firm life-cycle

	Pooled Cross Section			Fixed Effects		
	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	2SLS	OLS	OLS	2SLS	OLS
$ln(Revenue)$	-0.027^{***}	-0.021^{***}		-0.063^{***}	-0.023^{**}	
	(0.002)	(0.002)		(0.007)	(0.012)	
$ln(Employment)$			-0.002^{***}			-0.010^{**}
			(0.0006)			(0.051)
TM registration	0.091^{***}	0.085^{***}	0.079^{***}			
	(0.019)	(0.019)	(0.019)			
post-TM registration	0.047^{***}	0.046^{***}	0.047^{***}	0.066^{***}	0.060^{***}	0.058^{***}
	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)
N	$395,000$	$395,000$	$395,000$	$395,000$	$395,000$	$395,000$
R^2	0.350	0.350	0.349	0.010	0.027	0.009

Notes: Standard errors clustered by firm in parentheses. *, **, and *** indicate significance at 10, 5, and 1%, respectively. Pooled specification includes 6-digit NAICS industry, firm age, and year dummies. Fixed effects specifications include firm age dummies, and firm and year fixed effects. N is rounded to prevent disclosure of confidential information.

Table 8: Mean advertising intensity for the sample used in the analysis in Table 7

	Mean advertising intensity (%)	
Firms with no trademark	0.56	
	(0.001)	
	Pre-registration	Post-registration
Firms with trademark	0.70	0.79
	(0.002)	(0.006)

Notes: Standard errors corrected for firm-level clustering in parentheses. Advertising intensity is the ratio of advertising costs to revenue expressed as a percentage. Registration refers to the event of first trademark registration.

Table 9: Calibration targets: model and data

Target	Model	Data	Source
Avg. annual firm exit rate	0.1	0.1	Business Dynamics Statistics
% of firms with trademarks	4.17	3.99	Table 1
Ratio of avg. employment (TM firms vs. non-TM firms)	23.59	23.75	Table 1
Depreciation of advertising capital	0.54	0.55	Hall (2014)
Advertising cost to consumption ratio	1.96%	2%	Galbi (2001a,b)
Share of R&D spending to consumption	5.06%	5.38%	U.S. data for 2020 from OECD
Slope of Engel curve for quantity	0.76	0.76	Bils and Klenow (2001)
Slope of Engel curve for quality	0.57	0.57	Bils and Klenow (2001)
Mean rate of success in persuasive advertising	0.17	0.15	Guitart et al. (2018)
Increase in advertising intensity due to persuasive adv.	0.042	0.037	Guitart et al. (2018)
Advertising intensity of TM firms	0.82%	0.79%	Table 8
Advertising intensity of nonTM firms	0.53%	0.56%	Table 8
Relative size of TM firms at age 1 vs. age 20	13.95	13.28	Table 3
Average TM litigation cost to Revenue	0.022%	0.02%	Lex Machina
Relative employment growth in the next 5 years after TM	22.1%	22.6%	Figure 4a

Table 10: Calibrated parameter values

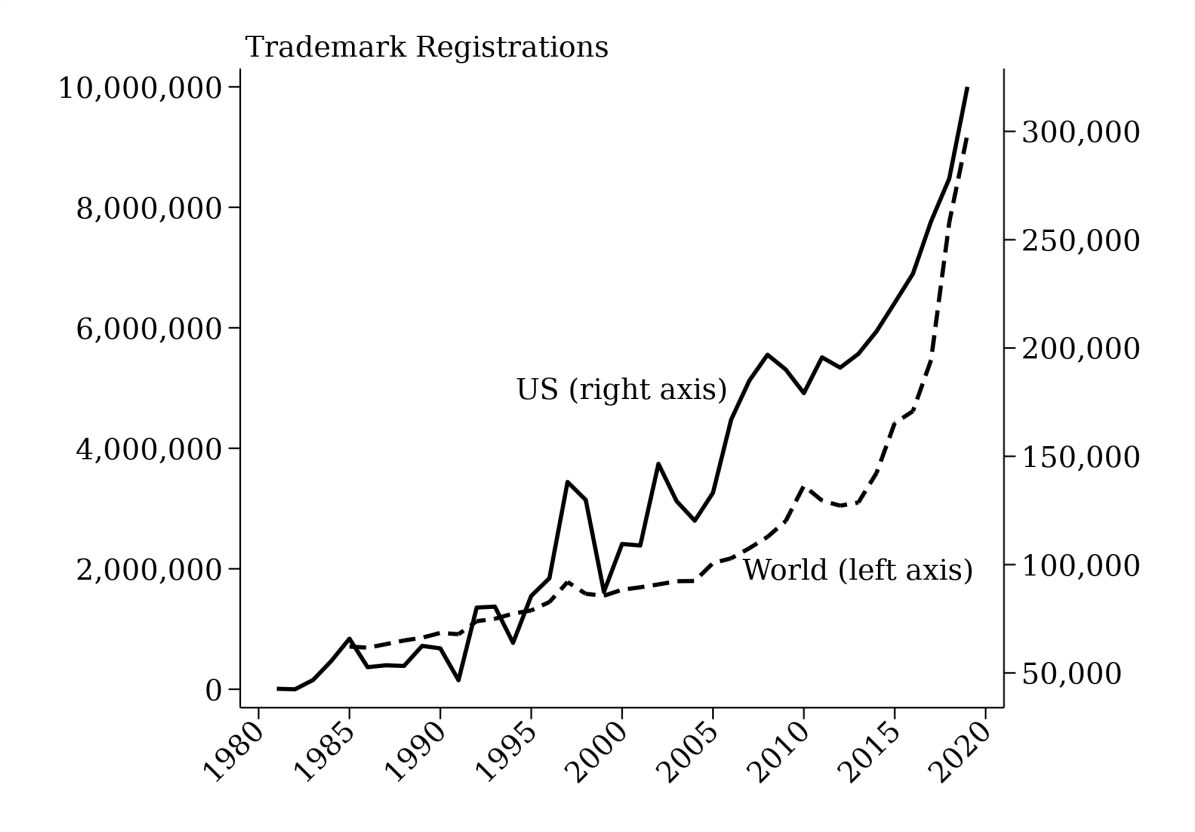
Parameter	Description	Value
Preference		
θ	Utility – exponent of quality	0.76
ξ	Utility – exponent of quality	0.57
β	Preference - time	0.99
Technology		
M	Production function – Hicks-neutral productivity (normalized)	1
A^N	Marketing productivity – non-TM firms	1.7
B	Quality development cost – scalar (normalized)	1
δ_T	Customer diversion rate – TM firms	2.5%
δ_N	Customer diversion rate – non-TM firms	5.2%
η	Quality development cost – exponent of quality	6.9
ρ	Advertising technology – exponent (elasticity) of marketing labor	0.022
χ	Exit rate	0.1
σ	Production function – exponent of quality	0.13
ϱ	Relative marketing productivity of TM firms – A^T/A^N	1.62
φ	Persuasive advertising success rate	0.15
Δ	Increase in perceived quality	0.02
Cost		
κ	Entry cost	6.2
c_T	Trademark cost	0.67
Distribution		
$\underline{e}$	Lower bound of quality development efficiency (normalized)	1
$\overline{e}$	Upper bound of quality development efficiency	3.01

Table 11: Key aggregates for the benchmark economy and the simulated economies

	Benchmark	No Trademark	High Diversion	*High Diversion (w/o TM)	Stronger TM Protection	No Persuasion
Ad per consumer	0.505	0.543 (7.5%)	0.488 (-3.4%)	0.523 (-3.7%)	0.511 (1.2%)	0.510 (1.0%)
No. of varieties	0.396	0.419 (5.8%)	0.386 (-2.5%)	0.407 (-2.9%)	0.4 (1.0%)	0.4 (1.0%)
Average quality	5.095	4.465 (-12.4%)	5.123 (0.6%)	4.43 (-0.8%)	5.195 (1.9%)	5.137 (0.8%)
Welfare (C.V.)	1.438	1.399 (-3.6%)	1.427 (-1.0%)	1.379 (-1.9%)	1.452 (1.3%)	1.446 (0.8%)
HHI	1279	583 (-54%)	1371 (7.2%)	563 (-3.4%)	1593 (25%)	1330 (4%)

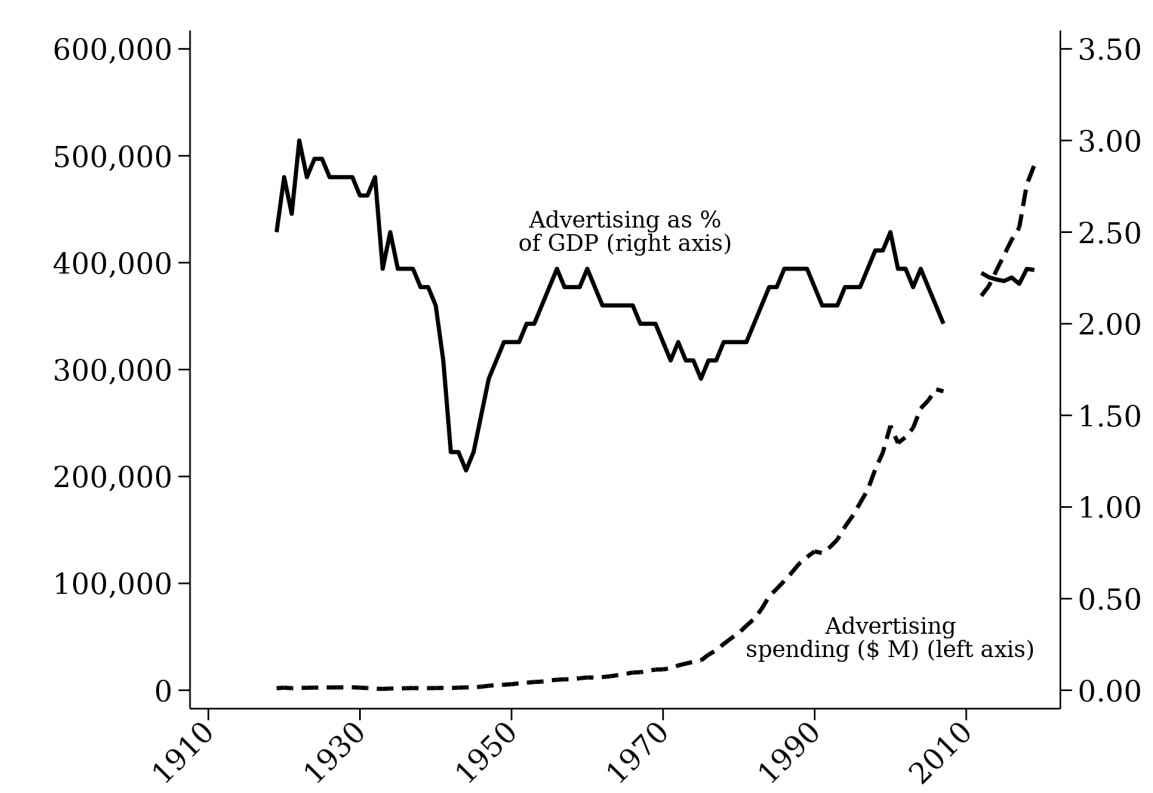
Notes: The percentages in parentheses indicate the percent change from the benchmark economy for the corresponding aggregate, except for the "*High Diversion (w/o Trademark)" case where the changes are with respect to the "No Trademark" economy. Welfare is based on compensating variation (C.V.).

Figure 1: Trademark registrations, direct and via the Madrid system, 1980-2019



Source: World Intellectual Property Organization (WIPO) Statistics Database.
Notes: The Madrid System is a platform for registering and managing trademarks in several countries at once (https://www.wipo.int/madrid/en/).

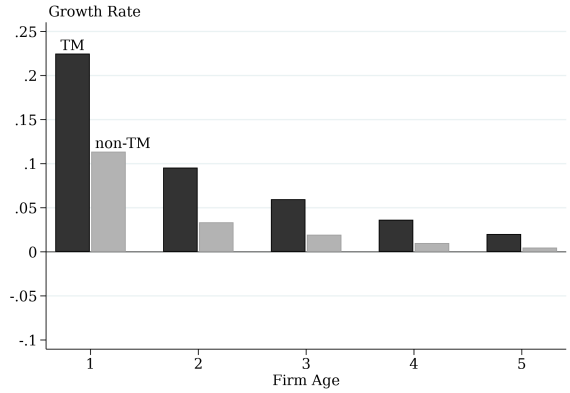
Figure 2: Advertising spending in the United States, 1919-2019



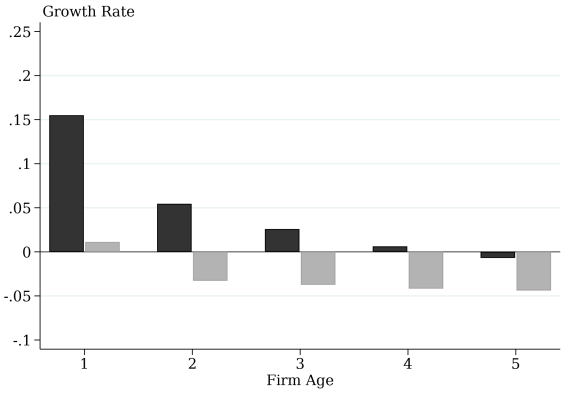
Source: The data are based on a revision of Galbi (2001a, 2001b). Advertising figures are from Coen Structured Advertising Expenditures Dataset. The GDP figures are from Johnston and Williamson (2005), Kendrick (1969), and the U.S. Bureau of Economic Analysis. The remaining GDP figures are from standard national accounts.
Notes: Total advertising spending includes spending for advertising in newspapers, magazines, radio, broadcast television, cable television, direct mail, billboards and displays, Internet, and other forms. The ad spending figures are in millions of current U.S. dollars.

Figure 3: Annual DHS growth rates for firms with and without trademarks: first five years (ages 1 through 5)

(a) Conditional on survival



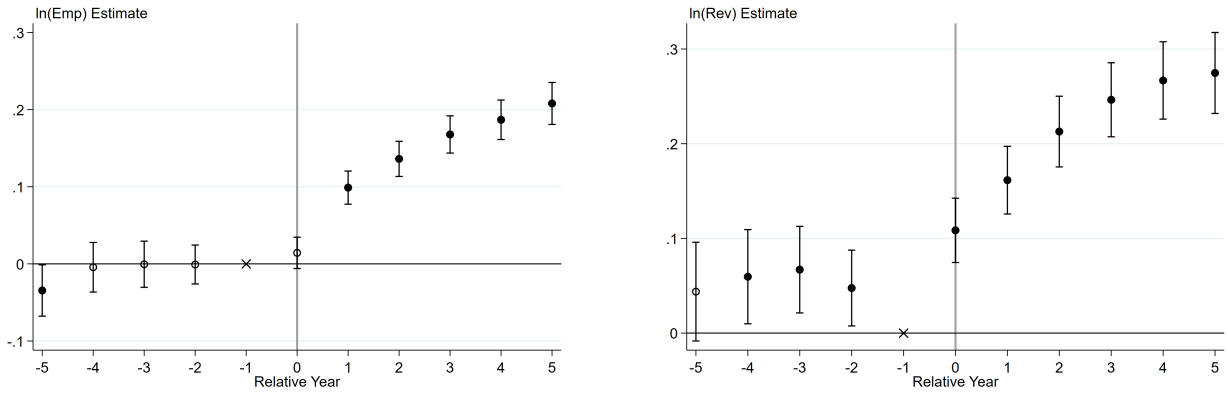
(b) Unconditional



Source: LBD, USPTO Trademark Case File Database
Notes: Davis-Haltiwanger-Schuh (DHS) growth rates are used. Panel (a) shows growth rates conditional on survival; Panel (b) shows growth rates inclusive of exits.

Figure 4: The evolution of firm outcomes before and after first trademark registration

(a) $ln(Employment)$ (b) $ln(Revenue)$

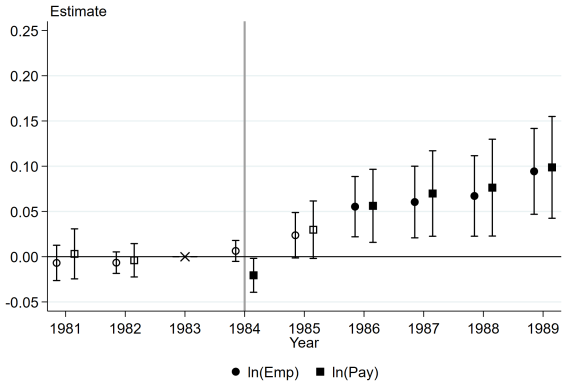


Source: LBD, USPTO Trademark Database

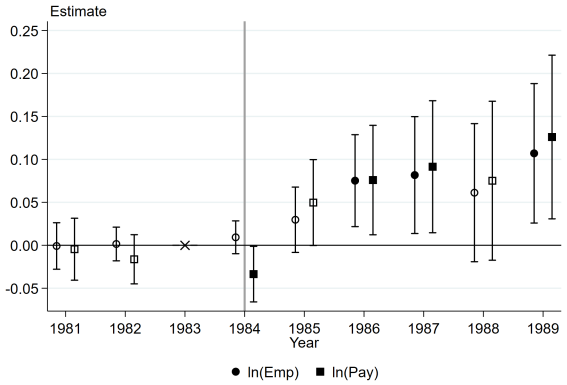
Notes: Panel (a) and (b) show estimates for employment and revenue, respectively. Estimated γ_j's from equation (1) plotted. 95% confidence bands are shown in vertical line segments. The event of trademark registration occurs at year t = 0 in x-axis.

Figure 5: The evolution of firm outcomes before and after the 1984 Trademark Counterfeiting Act

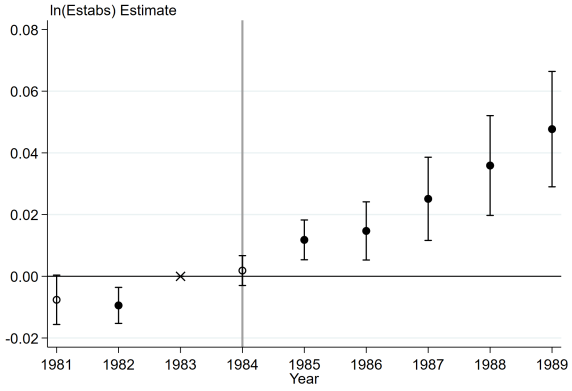
(a) $ln(Employment)$ and $ln(Payroll)$, TM as of 1984



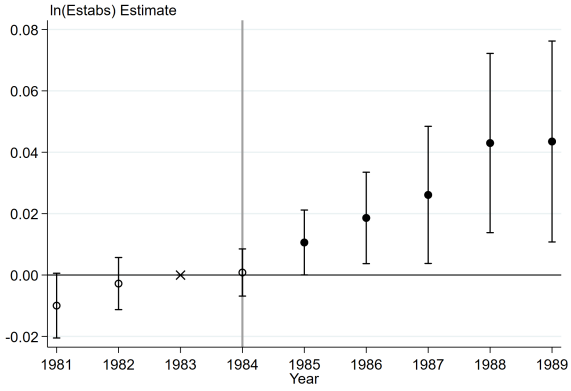
(b) $ln(Employment)$ and $ln(Payroll)$, TM as of 1981



(c) $ln(Establishments)$, TM as of 1984



(d) $ln(Establishments)$, TM as of 1981



Source: LBD, USPTO Trademark Database

Notes: Panels (a) and (b) show estimates for employment and payroll using treated firms as of 1984 versus 1981, respectively. Panels (c) and (d) show estimates for the number of establishments using treated firms as of 1984 versus 1981, respectively. Estimated γ_j's from equation (2) are plotted. 95% confidence bands are shown in vertical line segments. The timing of the Trademark Counterfeiting Act is shown as a vertical bar at year t = 1984 in x-axis.

Figure 6: Timing of decisions by firms within a period

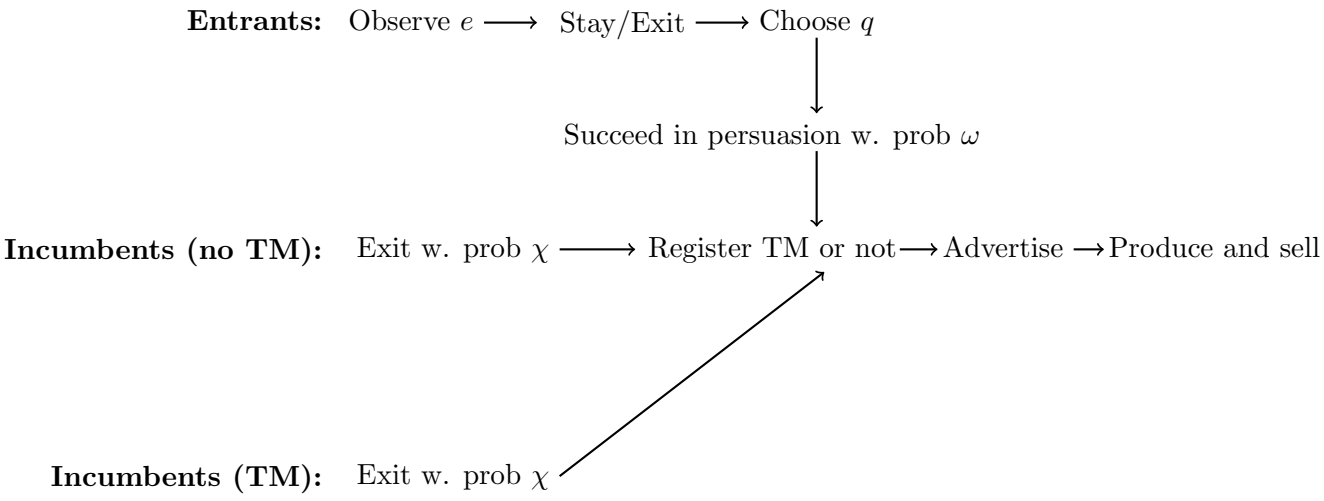


Figure 7: Regions of trademark policy in the firm size-quality plane

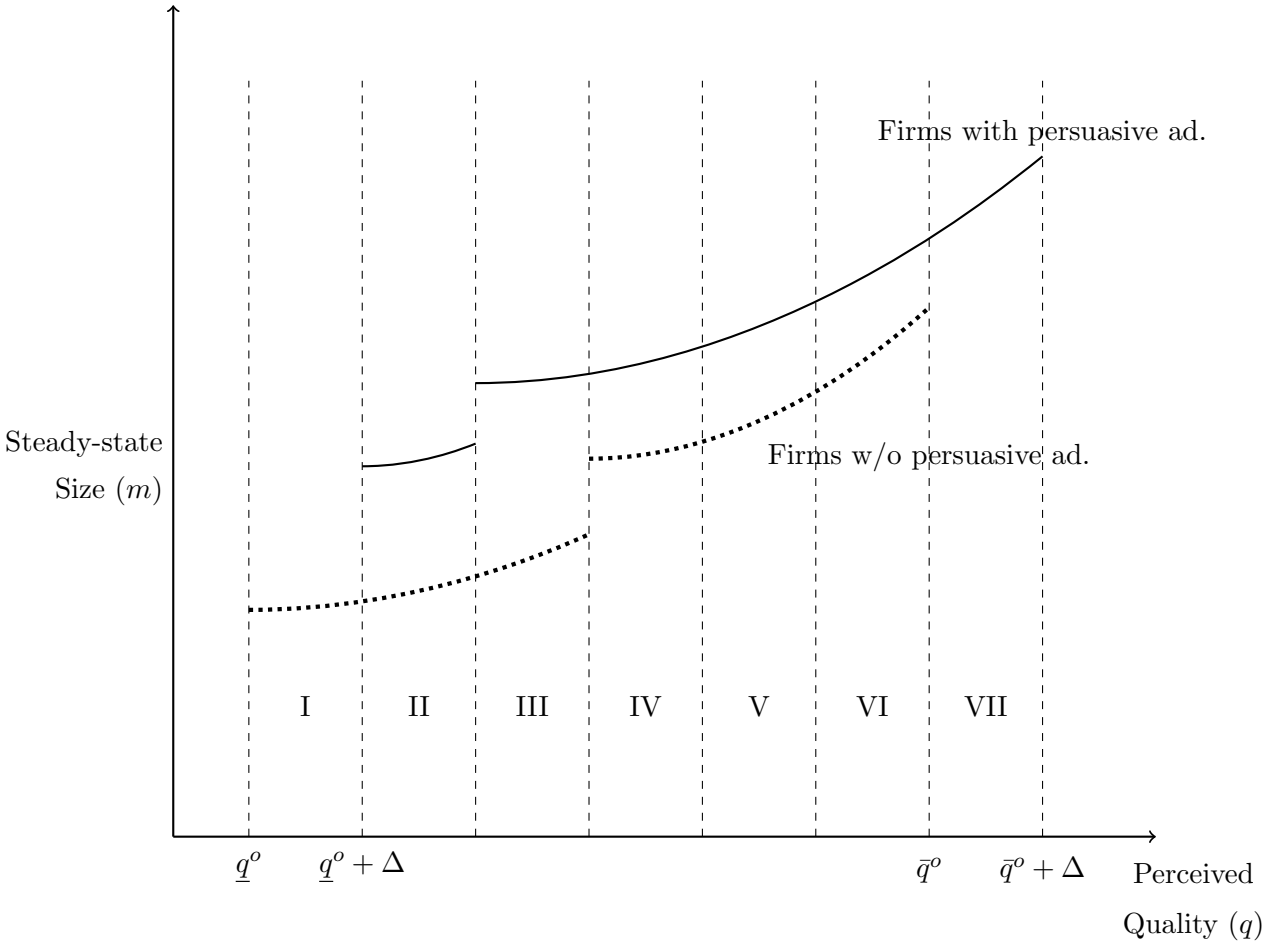
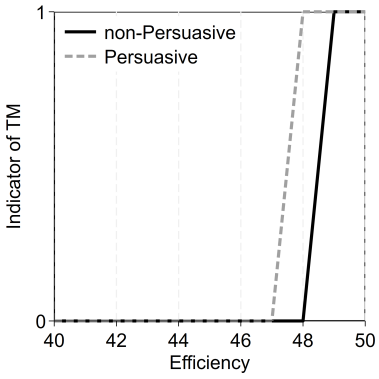
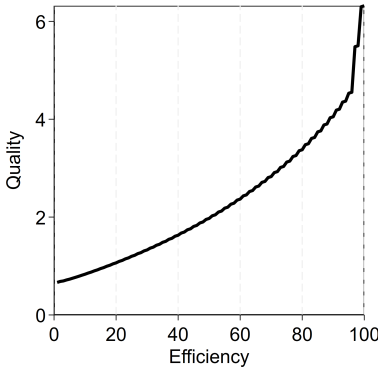


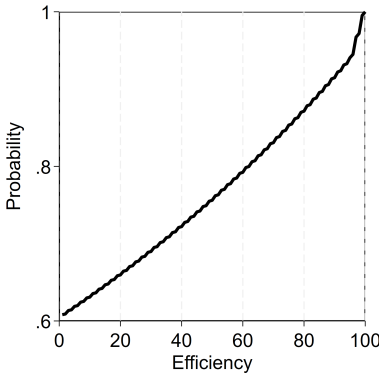
Figure 8: Benchmark economy



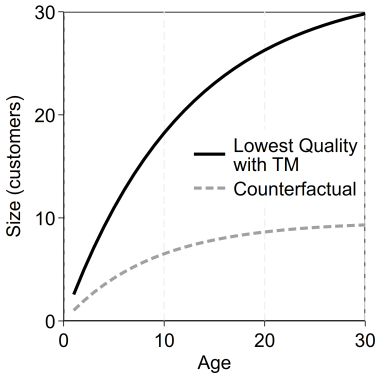
(a) Trademark policy



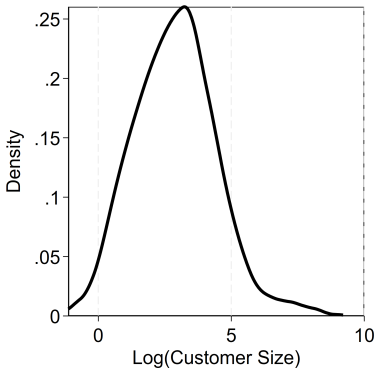
(b) Quality policy



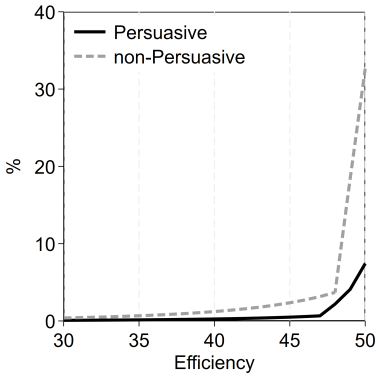
(c) Probability of sale



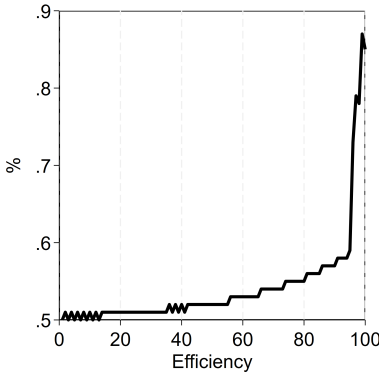
(d) Evolution of firm size



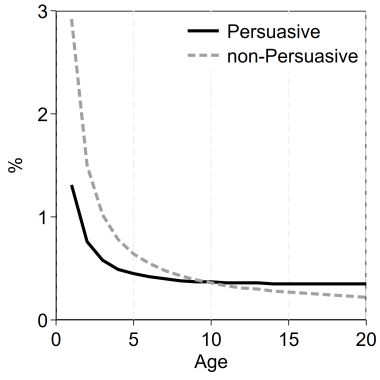
(e) Firm size distribution



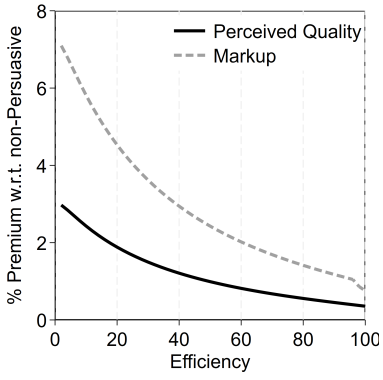
(f) Share of expenditures



(g) Advertising intensity

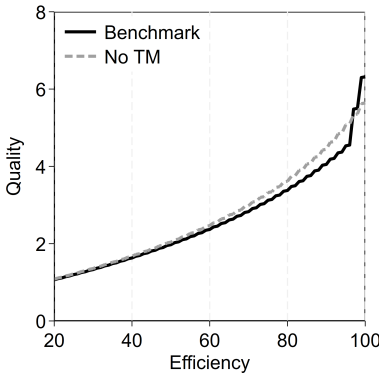


(h) Advertising intensity

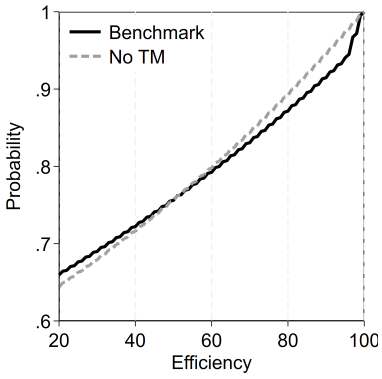


(i) Perceived quality and markups

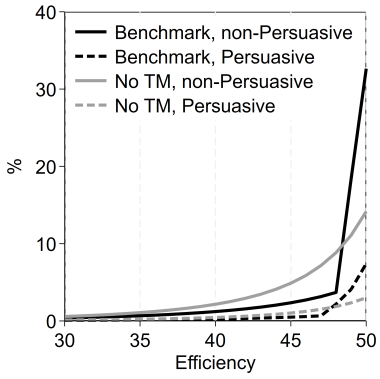
Figure 9: No-trademark economy



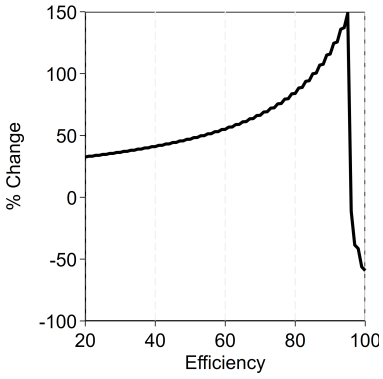
(a) Quality policy



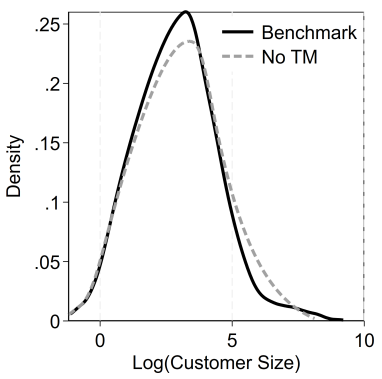
(b) Probability of sale



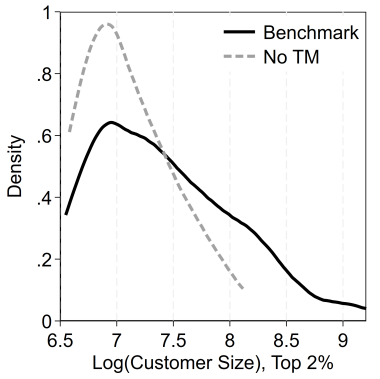
(c) Share of expenditures



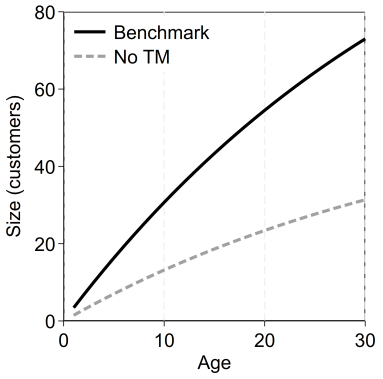
(d) Reallocation of employment



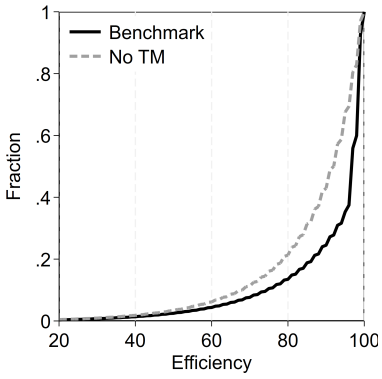
(e) Firm size distribution



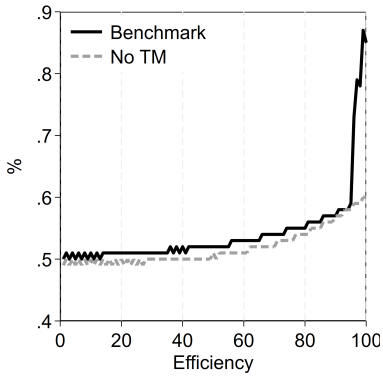
(f) Firm size distribution–right tail



(g) Size (customers)–Highest q

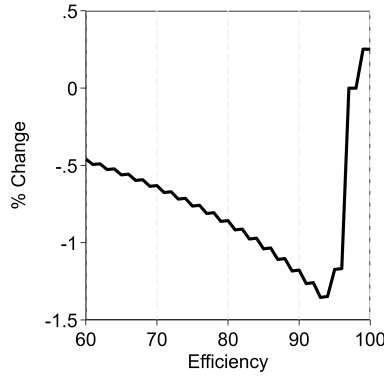


(h) Cumulative share of customers

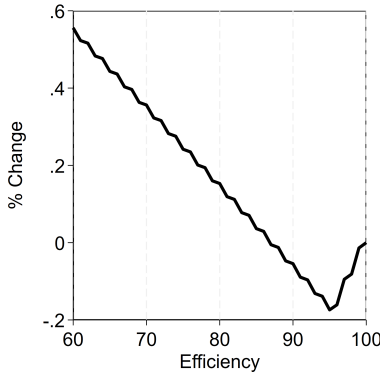


(i) Advertising intensity

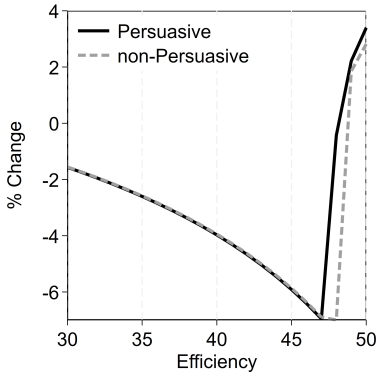
Figure 10: High-diversion economy



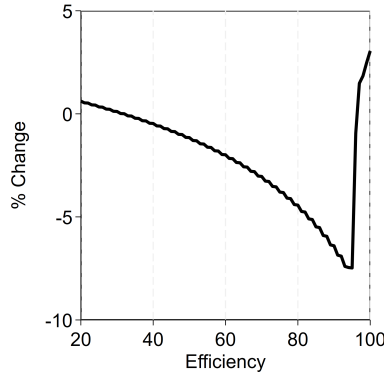
(a) Quality



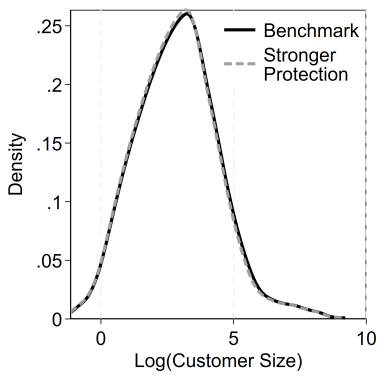
(b) Probability of sale



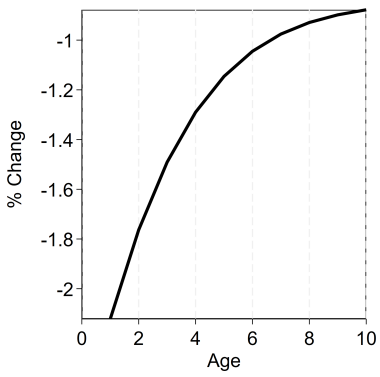
(c) Share of expenditures



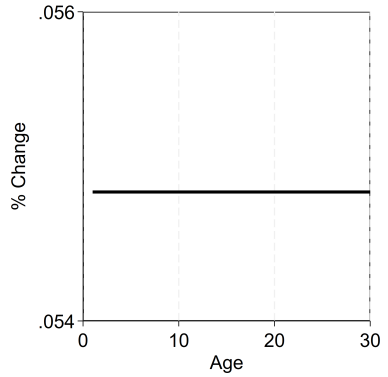
(d) Reallocation of employment



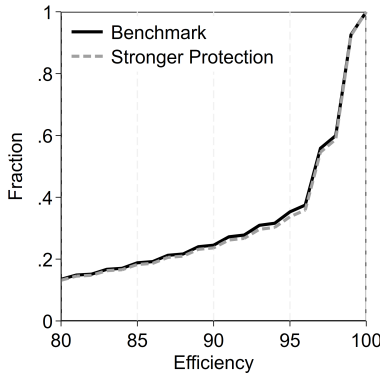
(e) Firm size distribution



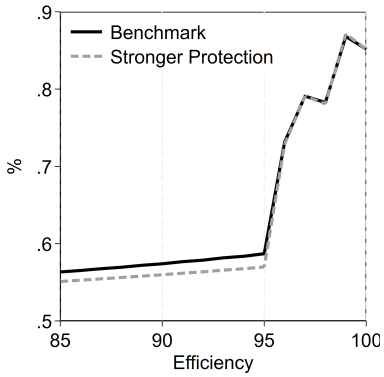
(f) Size (customers)–Lowest q



(g) Size (customers)–Highest q

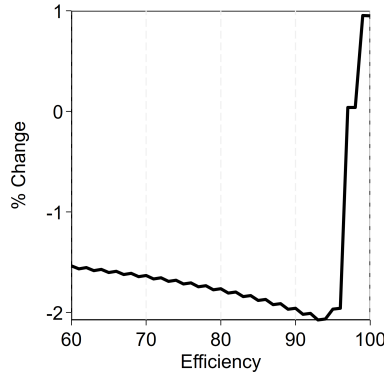


(h) Cumulative share of customers

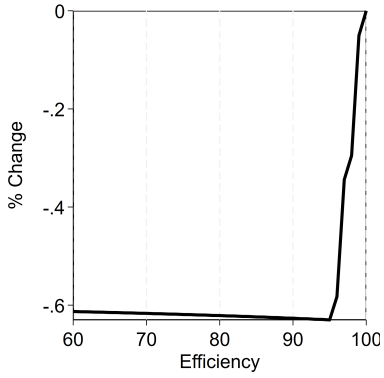


(i) Advertising intensity

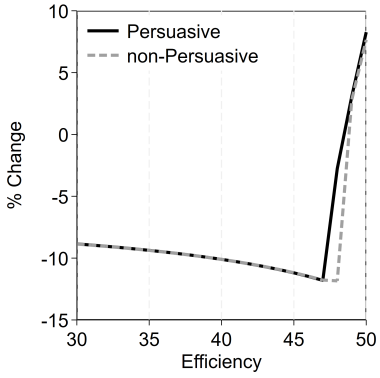
Figure 11: Stronger trademark protection



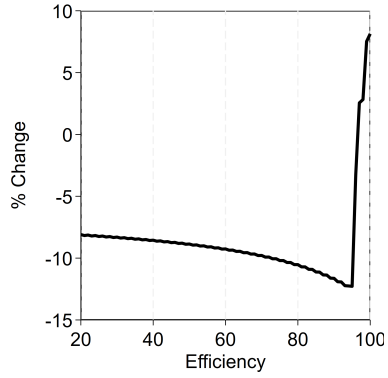
(a) Quality



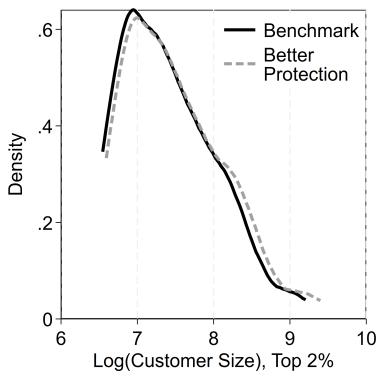
(b) Probability of sale



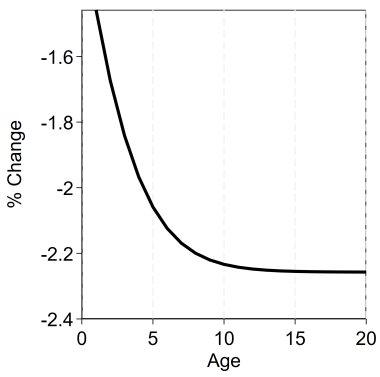
(c) Share of expenditures



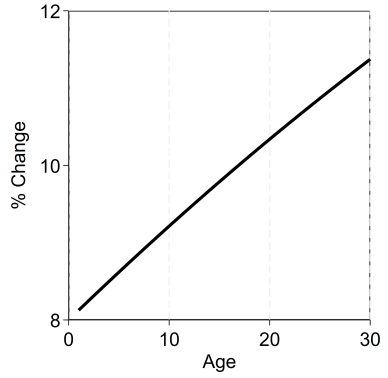
(d) Reallocation of employment



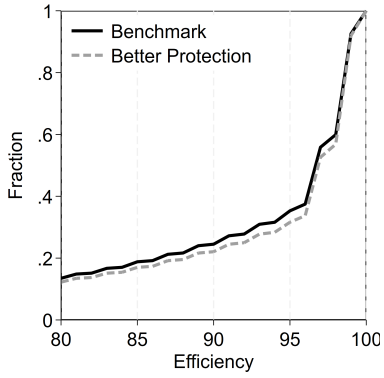
(e) Firm size dist.–right tail



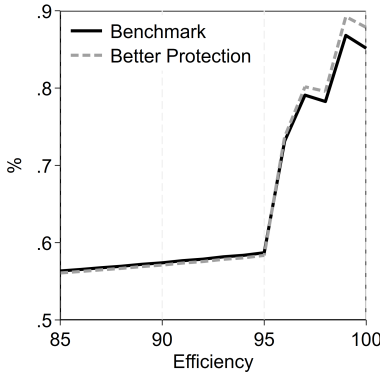
(f) Size (customers)–Lowest q



(g) Size (customers)–Highest q

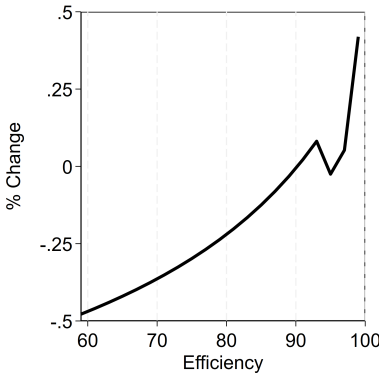


(h) Cumulative share of customers

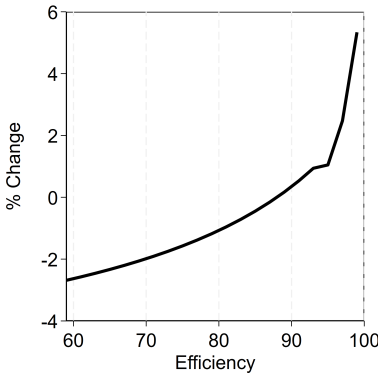


(i) Advertising intensity

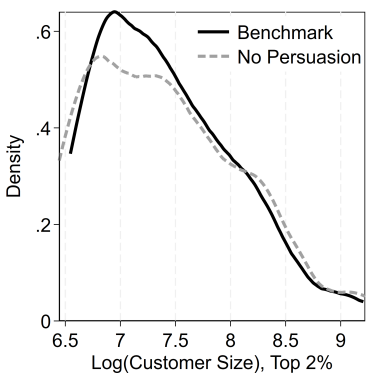
Figure 12: No persuasive advertising



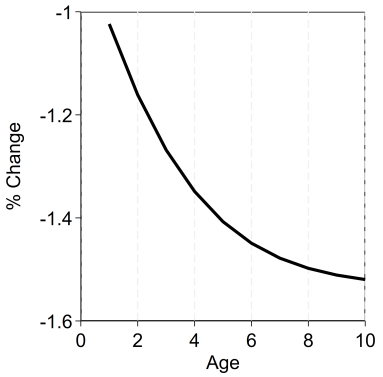
(a) Quality



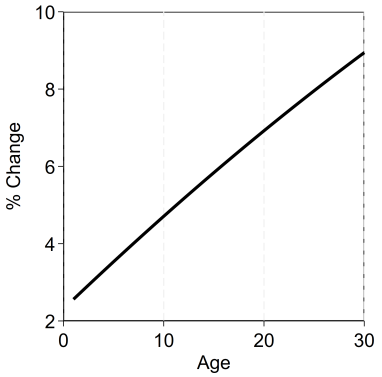
(b) Reallocation of employment



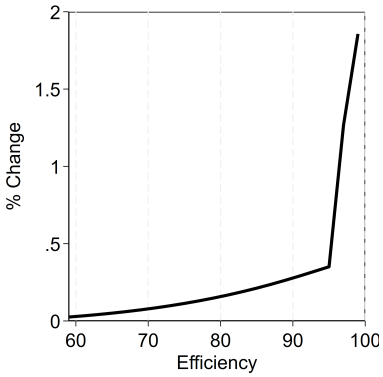
(c) Firm size dist.–right tail



(d) Size (customers)–Lowest q



(e) Size (customers)–Highest q



(f) Advertising intensity

Appendix

A Additional empirical results

Table A1: Estimates from regression (1)

	Employment (ln)	Revenue (ln)
γ_{-5}	-.0345**	.0438*
	(.017)	(.027)
γ_{-4}	-.0043	.0596**
	(.016)	(.025)
γ_{-3}	-.0005	.0670***
	(.015)	(.023)
γ_{-2}	-.0008	.0476**
	(.013)	(.020)
γ_{0}	.0143	.1085***
	(.010)	(.017)
γ_{+1}	.0988***	.1615***
	(.011)	(.018)
γ_{+2}	.1361***	.2128***
	(.012)	(.019)
γ_{+3}	.1678***	.2464***
	(.012)	(.020)
γ_{+4}	.1868***	.2668***
	(.013)	(.021)
γ_{+5}	.2080***	.2747***
	(.014)	(.022)
R^2	.017	.021
N	723,000	457,000

Notes: Standard errors clustered by firm in parentheses. *, **, and *** indicate significance at 10, 5, and 1%, respectively. N is rounded to prevent disclosure of confidential information.

Table A2: The effects of China trade shock on firms with and without trademarks - first stage statistics from 2SLS IV estimation

A. Employment

Table 5 Specification:	(1)	(1)	(1)	(2)	(2)	(2)
Instrumented Variable:	IC	$Post \times IC$	$Post \times TM \times IC$	IC	$Post \times IC$	$Post \times TM \times IC$
IV	0.736***			0.707***		
	(0.0027)			(0.0027)		
$Post \times IV$		0.732***			0.700***	
		(0.0024)			(0.0025)	
$Post \times TM \times IV$			0.765***			0.723***
			(0.0022)			(0.0023)
Constant	2.569***	1.612***	0.865***	2.595***	1.626***	0.873***
	(0.0252)	(0.0213)	(0.0158)	(0.0272)	(0.0232)	(0.0174)
N	63,500	63,500	63,500	53,000	53,000	53,000
R^2	0.548	0.599	0.656	0.557	0.605	0.655

B. Revenue

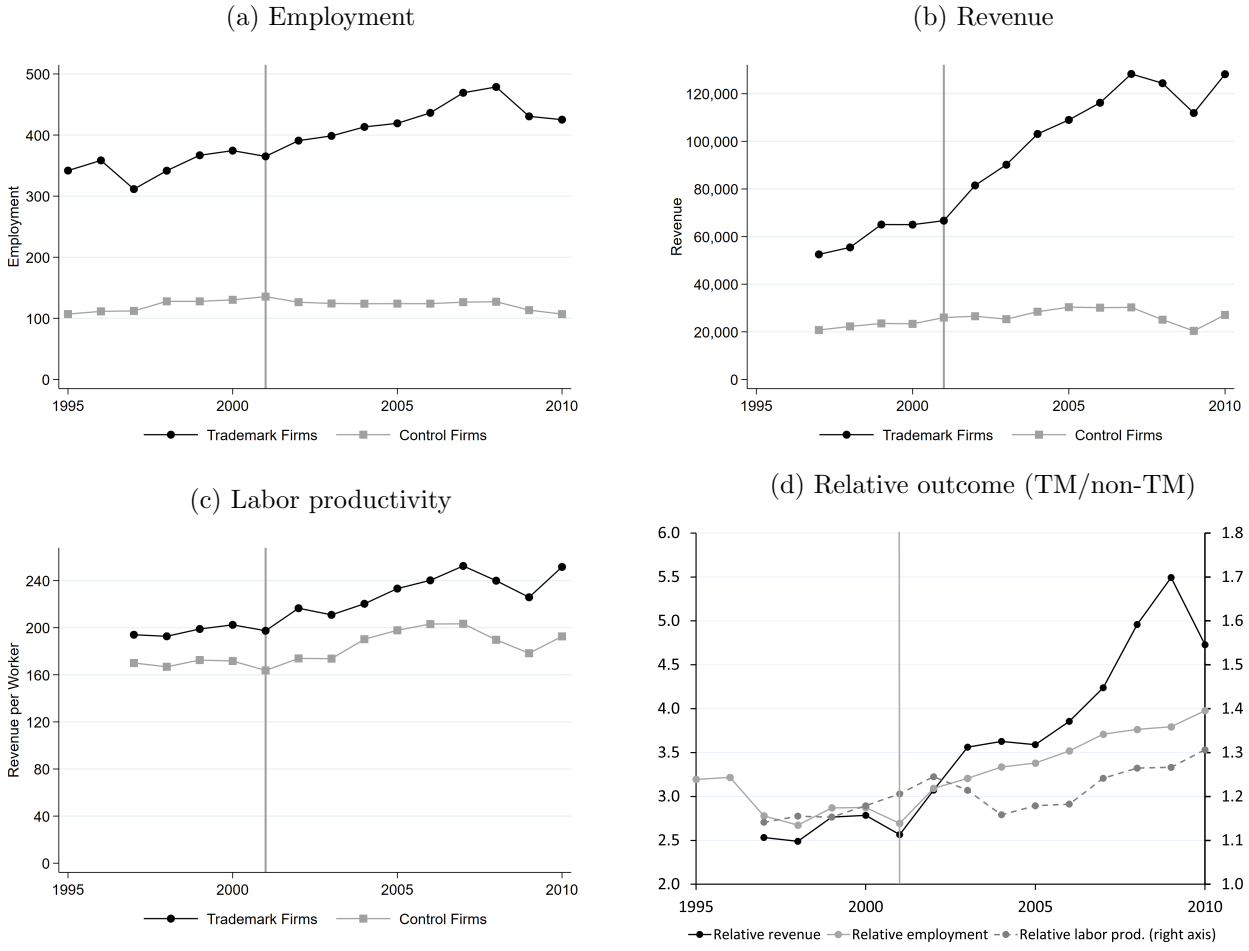
Table 5 Specification:	(3)	(3)	(3)	(4)	(4)	(4)
Instrumented Variable:	IC	$Post \times IC$	$Post \times TM \times IC$	IC	$Post \times IC$	$Post \times TM \times IC$
IV	0.736***			0.707***		
	(0.0027)			(0.0027)		
$Post \times IV$		0.732***			0.700***	
		(0.0024)			(0.0025)	
$Post \times TM \times IV$			0.765***			0.723***
			(0.0022)			(0.0023)
Constant	2.569***	1.612***	0.865***	2.595***	1.626***	0.873***
	(0.0252)	(0.0213)	(0.0158)	(0.0272)	(0.0232)	(0.0174)
N	63,500	63,500	63,500	53,000	53,000	53,000
R^2	0.548	0.599	0.656	0.557	0.605	0.655

C. Labor Productivity

Table 5 Specification:	(5)	(5)	(5)	(6)	(6)	(6)
Instrumented Variable:	IC	$Post \times IC$	$Post \times TM \times IC$	IC	$Post \times IC$	$Post \times TM \times IC$
IV	0.736***			0.707***		
	(0.0027)			(0.0027)		
$Post \times IV$		0.732***			0.700***	
		(0.0024)			(0.0025)	
$Post \times TM \times IV$			0.765***			0.723***
			(0.0022)			(0.0023)
Constant	2.569***	1.612***	0.865***	2.595***	1.626***	0.873***
	(0.0252)	(0.0213)	(0.0158)	(0.0272)	(0.0232)	(0.0174)
N	63,500	63,500	63,500	53,000	53,000	53,000
R^2	0.548	0.599	0.656	0.557	0.605	0.655

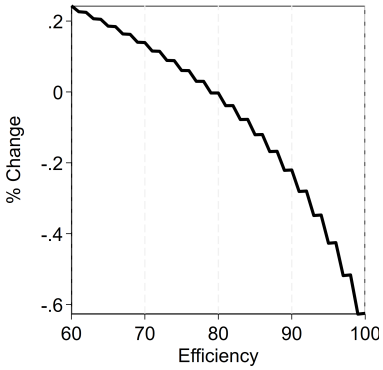
Notes: Standard errors clustered by firm in parentheses. *, **, and *** indicate significance at 10, 5, and 1%, respectively. N is rounded to prevent disclosure of confidential information. Instrumental variables are in the left-most column.

Figure A1: The evolution of firm outcomes before and after the China trade shock

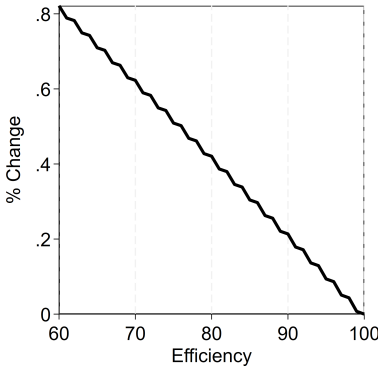


Source: LBD, USPTO Trademark Database, LFTTD
Notes: Panels (a), (b), and (c) show average employment, revenue, and labor productivity (revenue/employment) by year for firms with registered trademarks in 2001 and their controls. Panel (d) plots the relative value (ratio) of each average outcome for trademark firms with respect to their controls. The vertical bar indicates the year of China trade shock (2001).

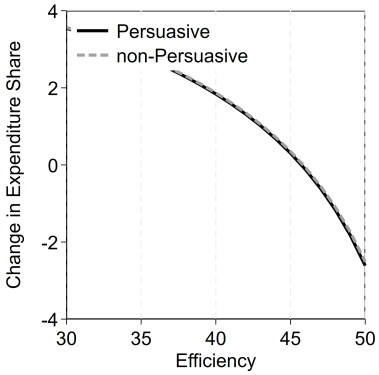
Figure A2: High-diversion economy with no trademarks



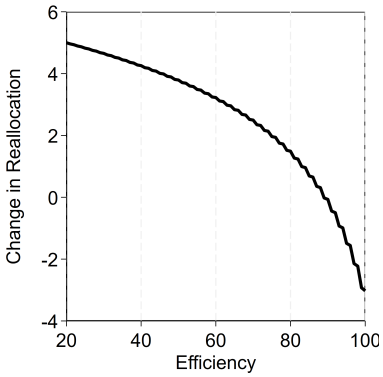
(a) Quality Policy



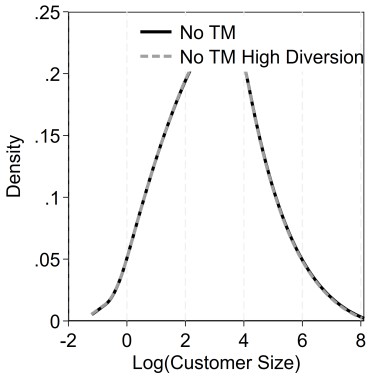
(b) Probability of Sale



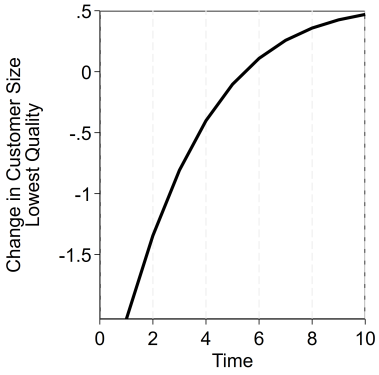
(c) Share of Expenditure



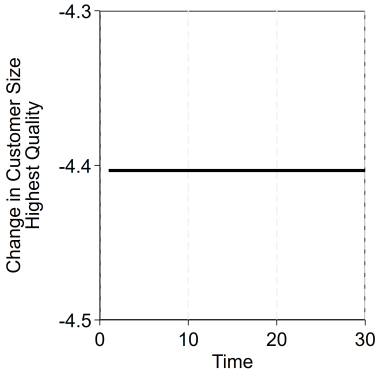
(d) Reallocation of Employment



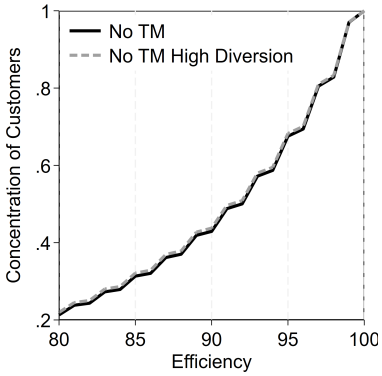
(e) Firm Size Distribution



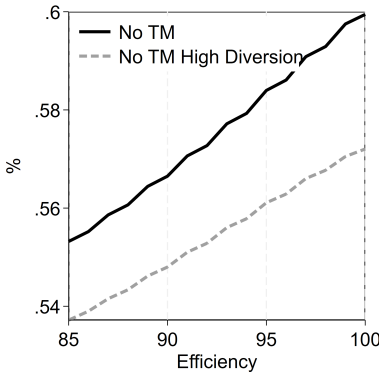
(f) Evolution of Firm Size (Lowest Quality)



(g) Evolution of Firm Size (Highest Quality)



(h) Concentration (Cumulative Share of Customer Size)



(i) Advertising Intensity

B Proofs

Proof of Proposition 1. Consider the pricing decision of a firm with actual quality q^o and without persuasive advertising

$$\max_p \; x(p,q^o)\left(p-\frac{(q^o)^\sigma}{\mathcal{M}}\right).$$

Substituting for $x(p,q^o)$ using (23) yields

$$\max_p \; \left[\frac{(q^o)^\xi}{p}\right]^{\frac{1}{1-\theta}}\left[\frac{p(\underline{q})}{\underline{q}^\xi}\right]^{\frac{1}{1-\theta}} x(p(\underline{q}),\underline{q})\left[p-\frac{(q^o)^\sigma}{\mathcal{M}}\right] \equiv \max_p \; \left[\frac{1}{p}\right]^{\frac{1}{1-\theta}}\left[p-\frac{(q^o)^\sigma}{\mathcal{M}}\right].$$

The first order condition for p is

$$\frac{\theta}{\theta-1}p^{\frac{\theta}{\theta-1}-1}-\frac{1}{1-\theta}\frac{(q^o)^\sigma}{\mathcal{M}}p^{\frac{1}{1-\theta}-1}=0,$$

which has the solution

$$p(q^o)=\frac{1}{\theta}\frac{(q^o)^\sigma}{\mathcal{M}}.$$

■

Proof of Proposition 2. *i*) Consider two products with perceived qualities $q_i > q_j$. Note that $\frac{x(p_i,q_i)}{x(p_j,q_j)} = \left(\frac{p_j}{p_i}\left[\frac{q_i}{q_j}\right]^\xi\right)^{\frac{1}{1-\theta}}$, $p_i=\frac{1}{\theta}\frac{q_i^\sigma}{\mathcal{M}}$, and $p_j=\frac{1}{\theta}\frac{q_j^\sigma}{\mathcal{M}}$. Therefore, $\widetilde{x}(q_i)=\left[\frac{q_i}{q_j}\right]^{\frac{\xi-\sigma}{1-\theta}}\widetilde{x}(q_j)$, where $\widetilde{x}(q)=x(p(q),q)$. The consumer's period surplus from these two products can then be written as

$$\widetilde{u}(q_i)=q_i^\xi\left[\frac{q_i}{q_j}\right]^{\theta\frac{\xi-\sigma}{1-\theta}}[\widetilde{x}(q_j)]^\theta-\upsilon\frac{1}{\theta}\frac{q_i^\sigma}{\mathcal{M}}\left[\frac{q_i}{q_j}\right]^{\frac{\xi-\sigma}{1-\theta}}\widetilde{x}(q_j),$$

and

$$\widetilde{u}(q_j)=q_j^\xi[\widetilde{x}(q_j)]^\theta-\upsilon\frac{1}{\theta}\frac{q_j^\sigma}{\mathcal{M}}\widetilde{x}(q_j).$$

where υ represents the Lagrange multiplier for the consumer's budget constraint.

Let $r_q=\frac{q_i}{q_j}>1$. The ratio of the two surplus terms is

$$\frac{\widetilde{u}(q_i)}{\widetilde{u}(q_j)}=\frac{q_i^\xi r_q^{\theta\frac{\xi-\sigma}{1-\theta}}[\widetilde{x}(q_j)]^\theta-\upsilon\frac{q_i^\sigma}{\mathcal{M}}r_q^{\frac{\xi-\sigma}{1-\theta}}\widetilde{x}(q_j)}{q_j^\xi[\widetilde{x}(q_j)]^\theta-\upsilon\frac{1}{\theta}\frac{q_j^\sigma}{\mathcal{M}}\widetilde{x}(q_j)}.$$

Substituting $q_i=r_qq_j$ yields

$$\frac{\widetilde{u}(q_i)}{\widetilde{u}(q_j)}=\frac{r_q^\xi q_j^\xi r_q^{\theta\frac{\xi-\sigma}{1-\theta}}[\widetilde{x}(q_j)]^\theta-\upsilon\frac{1}{\theta}\frac{r_q^\sigma q_j^\sigma}{\mathcal{M}}r_q^{\frac{\xi-\sigma}{1-\theta}}\widetilde{x}(q_j)}{q_j^\xi[\widetilde{x}(q_j)]^\theta-\upsilon\frac{1}{\theta}\frac{q_j^\sigma}{\mathcal{M}}\widetilde{x}(q_j)}.$$

Now divide through first by $\widetilde{x}(q_j)$ and then by q_j^ξ to obtain

$$\frac{\widetilde{u}(q_i)}{\widetilde{u}(q_j)}=\frac{r_q^{\xi+\theta\frac{\xi-\sigma}{1-\theta}}-\upsilon\frac{1}{\theta}r_q^{\sigma+\frac{\xi-\sigma}{1-\theta}}\frac{q_j^{\sigma-\varepsilon}}{\mathcal{M}}[\widetilde{x}(q_j)]^{1-\theta}}{1-\upsilon\frac{1}{\theta}\frac{q_j^{\sigma-\xi}}{\mathcal{M}}[\widetilde{x}(q_j)]^{1-\theta}}.$$

Since $\xi + \theta\frac{\xi-\sigma}{1-\theta} = \sigma + \frac{\xi-\sigma}{1-\theta}$, one can write

$$\frac{\widetilde{u}(q_i)}{\widetilde{u}(q_j)} = \frac{r_q^{\xi+\theta\frac{\xi-\sigma}{1-\theta}}\left(1 - \upsilon\frac{1}{\theta}\frac{q_j^{\sigma-\varepsilon}}{\mathcal{M}}\left[\widetilde{x}(q_j)\right]^{1-\theta}\right)}{1 - \upsilon\frac{1}{\theta}\frac{q_j^{\sigma-\xi}}{\mathcal{M}}\left[\widetilde{x}(q_j)\right]^{1-\theta}} = r_q^{\xi+\theta\frac{\xi-\sigma}{1-\theta}}.$$

Note that $r_q^{\xi+\theta\frac{\xi-\sigma}{1-\theta}} > 1$ under the assumption $\xi > \sigma$. Hence, $\widetilde{u}(q_i) > \widetilde{u}(q_j)$, implying that $\widetilde{u}(q)$ is increasing. Deterministic dynamic programming techniques (see Stokey and Lucas (1989)) then imply that $W(q)$ is increasing. Therefore, so are $\omega(q)$ and $z(q)$.

ii) The variable profit per customer is

$$\pi(q, q^o) = \left(p(q) - \frac{1}{\mathcal{M}}(q^o)^\sigma\right)x(p(q), q) = \left(\frac{1}{\theta\mathcal{M}}(q)^\sigma - \frac{1}{\mathcal{M}}(q^o)^\sigma\right)x(p(q), q).$$

Using the envelope theorem and the fact that $\frac{dq}{dq^o}$, differentiation with respect to q^o yields

$$\frac{d\pi(q, q^o)}{dq^o} = (-\sigma\frac{1}{\mathcal{M}}(q^o)^{\sigma-1})x(p(q), q) + \left(p(q) - \frac{1}{\mathcal{M}}(q^o)^\sigma\right)\frac{\partial x(p(q), q)}{\partial q^o}.$$

Next, note that $x(p(q), q) = \left[\frac{q^\xi}{p(q)}\right]^{\frac{1}{1-\theta}}\left[\frac{p(q)}{\underline{q}^\xi}\right]^{\frac{1}{1-\theta}}x(p(\underline{q}), \underline{q})$, and therefore

$$\frac{\partial x(p(q), q)}{\partial q^o} = \frac{\xi}{1-\theta}\frac{1}{q}x(p(q), q).$$

Thus,

$$\begin{aligned}
\frac{d\pi(q, q^o)}{dq^o} &= x(p(q), q)(-\sigma\frac{1}{\mathcal{M}}(q^o)^{\sigma-1} + \frac{\xi}{(1-\theta)\theta\mathcal{M}}q^{\sigma-1} - \frac{\xi}{(1-\theta)\mathcal{M}}(q^o)^{\sigma-1}\frac{q^o}{q}) \\
&\geq x(p(q), q)(-\sigma\frac{1}{\mathcal{M}}(q^o)^{\sigma-1} + \frac{\xi}{(1-\theta)\theta\mathcal{M}}q^{\sigma-1} - \frac{\xi}{(1-\theta)\mathcal{M}}(q^o)^{\sigma-1}) \\
&\geq x(p(q), q)(\frac{\xi}{(1-\theta)\theta\mathcal{M}}(q^o)^{\sigma-1} - \frac{\xi+\sigma(1-\theta)}{(1-\theta)\mathcal{M}}(q^o)^{\sigma-1}) \\
&= x(p(q), q)(\frac{(\xi-\theta\sigma)}{\theta\mathcal{M}}(q^o)^{\sigma-1}) \\
&> 0,
\end{aligned}$$

where the first and second inequalities follow from the fact that $q \geq q^o$, and the last from $\xi > \sigma$ and $\theta \in (0, 1)$. Thus, $\pi(q, q^o)$ is increasing in q^o. Therefore,

$$\widetilde{\Pi}^\Gamma(q, q^o, m) = \max_{p,n,n_a}\Pi^\Gamma(p, n, n_a; q, q^o, m) = \max_{n_a} m^\Gamma(\Phi^\Gamma(n_a); q, q^o, m)\pi(q, q^o) - n_a$$

is also increasing in q^o. To see this, note that the Envelope Theorem implies

$$\frac{d\widetilde{\Pi}^\Gamma(q, q^o, m)}{dq^o} = \frac{dm^\Gamma}{dq^o}\pi(q, q^o) + m^\Gamma\frac{d\pi(q, q^o)}{dq^o},$$

where

$$\frac{dm^\Gamma}{dq^o} = m[1 - (1 - I(\Gamma = T))\delta_N - I(\Gamma = T)\delta_T]z'(q) + \Phi^\Gamma(n_a)\omega'(q) + d'(q) > 0,$$

by part (i) and because $d(q)$ is non-decreasing. Furthermore, $\frac{d\pi(q,q^o)}{dq^o} > 0$, as shown above. Next, observe that

$$\frac{dm^\Gamma}{dm} = [1 - (1 - I(\Gamma = T))\delta_N - I(\Gamma = T)\delta_T]z(q) > 0.$$

As a result, by the deterministic dynamic programming techniques in Stokey and Lucas (1989) $V^\Gamma(q, q^o, m)$ is increasing in q^o and m.

The above proof uses the assumption that $d(q)$ is non-decreasing in stationary equilibrium. When does this property hold? The definition of $d(q)$ in (22) implies

$$d(q) \quad = \quad \delta_N(1 - s_T(q))\bar{m}^N(q) + \delta_T s_T(q)\bar{m}^T(q)$$

where $s_T(q)$ is the fraction of trademark firms at q, and $\bar{m}^\Gamma(q), \Gamma = N, T$ is the average size for trademark and non-trademark firms at the beginning of the period *before* any customer diversion and redistribution takes place.

Over the range of q for which there are no trademark firms (i.e. $s_T(q) = 0$), $d(q) = \delta_N \bar{m}^N(q)$, which is increasing in q - see Lemma 1 at the end of the proofs. Similarly, over the range of q where all firms have trademarks (i.e. $s_T(q) = 1$), $d(q) = \delta_T \bar{m}^T(q)$, which is, again, increasing in q – see Lemma 1. Over the range where only some firms have trademarks, $s_T(q) \in (0, 1)$, there are two effects as q increases: both $\bar{m}^N(q)$ and $\bar{m}^T(q)$ increase, but $s_T(q)$ also increases. The first effect leads to an increase $d(q)$. The second effect implies that some non-trademark firms become trademarks firms, and their customer depreciation rate decreases, since $\delta_T < \delta_N$. This effect can lead to a decrease in $d(q)$, depending on the relative magnitudes of δ_T and δ_N. To see this, consider any region of equilibrium as described in section 4.5 where $s_T(q) \in (0, 1)$. Differentiation of (35) yields

$$\begin{aligned} d'(q) \quad &= \quad s'_T(q)[\delta_T \bar{m}^T(q) - \delta_N \bar{m}^N(q)] + \delta_T s_T(q)\bar{m}^{T\prime}(q) + \delta_N(1 - s_T(q))\bar{m}^{N\prime}(q) \\ &> \quad s'_T(q)[\delta_T \bar{m}^T(q) - \delta_N \bar{m}^N(q)] + \delta_T s_T(q)[\bar{m}^{T\prime}(q) - \bar{m}^{N\prime}(q)] + \delta_N \bar{m}^{N\prime}(q) \\ &> \quad s'_T(q)[\delta_T \bar{m}^T(q) - \delta_N \bar{m}^N(q)] + \delta_T s_T(q)[\bar{m}^{T\prime}(q) + ((1/s_T(q)) - 1)\bar{m}^{N\prime}(q)] \end{aligned}$$

The second term in the last expression is positive, because $\bar{m}^T(q)$ and $\bar{m}^N(q)$ are increasing in q. In the first term, $s'_T(q) \geq 0$ - i.e. fraction of trademark firms is non-decreasing in q, as an implication of Proposition 5. The only term remaining to be signed is then $[\delta_T \bar{m}^T(q) - \delta_N \bar{m}^N(q)]$, which is non-negative if $\delta_T/\delta_N \geq \bar{m}^N(q)/\bar{m}^T(q)$. Thus, a sufficient condition for $d(q)$ to be non-decreasing is $\delta_T/\delta_N > sup_q[\bar{m}^N(q)/\bar{m}^T(q)]$ for all q such that $s_T(q) \in (0, 1)$ – i.e., δ_T is not too small with respect to δ_N. A similar analysis is applicable to going from one region of equilibrium where no firm of a given type (persuasive or non-persuasive) does not trademark to a region where some trademark – see Lemma 1. In our numerical analysis, we find that for the parameter ranges we experiment with, $d(q)$ is always non-decreasing.

iii) Note that

$$\pi(q^o + \Delta, q^o) > \pi(q^o + \Delta, q^o + \Delta) > \pi(q^o, q^o)$$

where the first inequality follows because

$$\pi(q^o + \Delta, q^o) - \pi(q^o + \Delta, q^o + \Delta) \quad = \frac{1}{A}[(q^o + \Delta)^\sigma - (q^o)^\sigma] > 0,$$

and the second inequality follows because $\pi(q^o, q^o)$ is increasing in q^o by part *(ii)*. In addition, m_t^Γ is increasing in q. Therefore, $\widetilde{\Pi}^\Gamma(q^o + \Delta, q^o, m) > \widetilde{\Pi}^\Gamma(q^o, q^o, m)$. It follows that $V^\Gamma(q^o + \Delta, q^o, m) > V^\Gamma(q^o, q^o, m)$.

■

Proof of Proposition 3. Take two efficiency levels $e_1 > e_2$. Consider the optimal quality choices $q^o(e_1)$

and $q^o(e_2)$. By the optimality of these choices, using (19) one can write

$$\begin{array}{rcl}
\varphi V_t^N(q^o(e_1)+\Delta, q^o(e_1), 0) + (1-\varphi)V_t^N(q^o(e_1), q^o(e_1), 0) - C(q^o(e_1); e_1) & & \\
-[V_t^N(q^o(e_2)+\Delta, q^o(e_2), 0) + (1-\varphi)V_t^N(q^o(e_2), q^o(e_2), 0) - C(q^o(e_2); e_1)] & \geq & 0, \\
\varphi V_t^N(q^o(e_2)+\Delta, q^o(e_2), 0) + (1-\varphi)V_t^N(q^o(e_2), q^o(e_2), 0) - C(q^o(e_2); e_2) & & \\
-[V_t^N(q^o(e_1)+\Delta, q^o(e_1), 0) + (1-\varphi)V_t^N(q^o(e_1), q^o(e_1), 0) - C(q^o(e_1); e_2)] & \geq & 0.
\end{array}$$

Adding the two inequalities, one obtains

$$-C(q^o(e_1); e_1) - (-C(q^o(e_2); e_1)) - [-C(q^o(e_1); e_2) - (-C(q^o(e_2); e_2))] \geq 0$$

Note that $\left(-\frac{\partial C(q^o,e)}{\partial q^o \partial e}\right) > 0$ by the assumption that C is strictly convex and decreasing in e. Therefore, the function $(-C)$ satisfies increasing differences and the above holds with strict inequality. It follows that $q^o(e_1) > q^o(e_2)$. ■

Proof of Proposition 4. *i)* Consider the first order condition (26). The left hand side is strictly decreasing in n_a because $\rho < 1$. Since $\frac{\partial q}{\partial q^o} = 1$, the left hand side is also increasing in q^o, because $\omega(q)$, $\pi(q, q^o)$, and $z(q)$ (hence, $l^{\Gamma(\tau)}$ and $L(q, \tau)$) are all strictly increasing in q^o. Therefore, a higher q^o implies a higher value for the left hand side. To maintain the first order condition, n_a must then increase. Therefore, $n_a^{\Gamma(\tau)}(q, q^o, \tau)$ is strictly increasing in q^o.

ii) The result for $\tau \geq 0$ follows from (30). For $-\infty < \tau < 0$, note that

$$\begin{array}{rcl}
L(q,\tau) & = & \displaystyle\sum_{t=0}^{-\tau}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t + \left[\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau I(\tau<0)}\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^T(q)}{1+r}\right)^t\right], \\
& > & \displaystyle\sum_{t=0}^{-\tau}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t + \left[\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau I(\tau<0)}\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t\right], \\
& = & \displaystyle\sum_{t=0}^{-\infty}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t \\
& = & L(q,-\infty),
\end{array}$$

where the inequality follows from $l^T(q) > l^N(q)$. Hence, by (26) $n_a^{\Gamma(\tau)}(q, q^o, \tau) > n_a^{\Gamma(\tau)}(q, q^o, -\infty)$.

iii) For $\tau \geq 0$ and $\tau = -\infty$, $n_a^{\Gamma(t)}(q, q^o, \tau)$ is a constant, which follows directly from (29). For $-\infty < \tau < 0$, the difference between the left hand side of (26) evaluated at $\tau + 1$ and at τ is

$$D(\tau) = \rho A^{\Gamma(\tau)}(n_a)^{\rho-1}\omega(q)\pi(q, q^o)\left[L(q, \tau+1) - L(q, \tau)\right].$$

The sign of $D(\tau)$ is determined by the sign of $[L(q,\tau+1)-L(q,\tau)]$, which can be simplified to

$$\begin{aligned}
&\sum_{t=0}^{-(\tau+1)}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t+\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-(\tau+1)}\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^T(q)}{1+r}\right)^t\\
&-\sum_{t=0}^{-\tau}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t-\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau}\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^T(q)}{1+r}\right)^t\\
=&\sum_{t=0}^{-(\tau+1)}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t-\sum_{t=0}^{-\tau}\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^t+\\
&\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-(\tau+1)}\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^T(q)}{1+r}\right)^t-\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau}\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^T(q)}{1+r}\right)^t,\\
=&-\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau}+\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau}\left[\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-1}-1\right]\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^T(q)}{1+r}\right)^t,\\
=&\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau}\left\{-1+\left[\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-1}-1\right]\sum_{t=1}^{\infty}\left(\frac{(1-\chi)l^T(q)}{1+r}\right)^t\right\},\\
=&\left(\frac{(1-\chi)l^N(q)}{1+r}\right)^{-\tau}\left\{-1+\left(\frac{1+r-(1-\chi)l^N(q)}{l^N(q)}\right)\left(\frac{l^T(q)}{1+r-(1-\chi)l^T(q)}\right)\right\}>0,
\end{aligned}$$

where the last inequality follows because $l^T(q)>l^N(q)$ and the term in curly brackets is thus positive. Therefore, $D(\tau)>0$, and n_a must increase to maintain the first order condition going from τ to $\tau+1$. Hence, $n_a^N(q,q^o;\tau)$ is strictly increasing in τ, for $-\infty<\tau<0$.

iv) This result follows directly from the first order condition (26) and the facts that per-consumer profit is higher for a firm with persuasive advertising $\pi(q^o+\Delta,q^o)>\pi(q^o,q^o)$, and so are the probabilities of sale, $\omega(q^o+\Delta)>\omega(q^o)$, and $z(q^o+\Delta)>z(q^o)$. ■

Proof of Proposition 5. *i*) Consider a firm with perceived and actual quality (q,q^o) and age A that has no trademark. Let m be the size of that firm at the end of the previous period. The difference between the flow profits from registering a trademark versus not registering at age A is given by

$$\begin{aligned}
R(q,q^o,m) &= \widetilde{\Pi}^T(q,q^o,m)-\widetilde{\Pi}^N(q,q^o,m)\\
&= m^T(n_a^T;q,q^o,m)\pi(q,q^o)-n_a^T(q,q^o,m)-m^N(n_a^N;q,q^o,m)\pi(q,q^o)+n_a^N(q,q^o,m)\\
&= [m^T(\cdot)-m^N(\cdot)]\pi(q,q^o)-[n_a^T(q,q^o,m)-n_a^N(q,q^o,m)],
\end{aligned}$$

where the difference between the end-of-period customer size is

$$\begin{aligned}
m^T(\cdot)-m^N(\cdot) &= m(1-\delta_T)z(q)+\Phi^T(n_a^T)\omega(q)+d(q)-m(1-\delta_N)z(q)-\Phi^N(n_a^N)\omega(q)-d(q),\\
&= (\delta_N-\delta_T)mz(q)+\left(\Phi^T(n_a^T)-\Phi^N(n_a^N)\right)\omega(q)>0.
\end{aligned}$$

The last inequality follows because $\delta_N>\delta_T$ and $\Phi^T(n_a^T)>\Phi^N(n_a^N)$.

First, it will be shown that $R(q,q^o,m)$ is increasing in q^o and m. By the Envelope Theorem, $\frac{\partial R}{\partial n_a^T}\frac{dn_a^T}{dq^o}=\frac{\partial R}{\partial n_a^N}\frac{dn_a^N}{dq^o}=0$. Hence,

$$\begin{aligned}
\frac{dR}{dq^o} &= \frac{d\widetilde{\Pi}^T}{dq^o}-\frac{d\widetilde{\Pi}^N}{dq^o}\\
&= \frac{d\pi(q,q^o)}{dq^o}\left[m^T(\cdot)-m^N(\cdot)\right]+\left[\frac{dm^T}{dq^o}-\frac{dm^N}{dq^o}\right]\pi(q,q^o).
\end{aligned}$$

Note that $m^T(\cdot) - m^N(\cdot) > 0$ and $\frac{d\pi(q,q^o)}{dq^o} > 0$. Next, consider the term

$$\frac{dm^T}{dq^o} - \frac{dm^N}{dq^o} = (\delta_N - \delta_T)mz'(q) + \omega'(q)(\Phi^T(n_a^T) - \Phi^N(n_a^N)).$$

Here, $z'(q), \omega'(q) > 0$, since the probabilities of sale increase with quality. In addition, $\Phi^T(n_a^T) - \Phi^N(n_a^N) > 0$. Therefore, $R(q, q^o, m)$ is strictly increasing in q^o.

Next, consider the change in $R(q, q^o, m)$ as age A, or equivalently, size m increases. Again, by the Envelope Theorem,

$$\frac{dR}{dm} = (\delta_N - \delta_T)z(q)\pi(q, q^o) > 0.$$

Therefore, $R(q, q^o, m)$ is increasing in m (or A).

Let $\beta^f = (1-\chi)/(1-r)$. Now, consider the difference

$$\begin{aligned}
&\mathrm{V}_{\mathrm{A}}^D(q, q^o) = V_{\mathrm{A}}^T(q, q^o) - V_{\mathrm{A}}^{NN}(q, q^o), \\
&= \widetilde{\Pi}^T(q, q^o, m) - \widetilde{\Pi}^N(q, q^o, m) + \beta^f[V^T(q, q^o, m^T) - V^N(q, q^o, m^N)], \\
&= \widetilde{\Pi}^T(q, q^o, m) - \widetilde{\Pi}^N(q, q^o, m) + \beta^f[V^T(q, q^o, m^T) - \max\{V^T(q, q^o, m^N) - c_T, V^{NN}(q, q^o, m^N)\}], \\
&= \widetilde{\Pi}^T(q, q^o, m) - \widetilde{\Pi}^N(q, q^o, m) \\
&+ \beta^f[V^T(q, q^o, m^T) - V^T(q, q^o, m^N) + V^T(q, q^o, m^N) - \max\{V^T(q, q^o, m^N) - c_T, V^{NN}(q, q^o, m^N)\}], \\
&= \widetilde{\Pi}^T(q, q^o, m) - \widetilde{\Pi}^N(q, q^o, m) + \beta^f[V^T(q, q^o, m^T) - V^T(q, q^o, m^N)] + \beta^f \min\{V^T(q, q^o, m^N) - V^{NN}(q, q^o, m^N), c_T\}, \\
&= \widetilde{\Pi}^T(q, q^o, m) - \widetilde{\Pi}^N(q, q^o, m) + \beta^f V_{\mathrm{A}+1}^{T,D}(q, q^o) + \beta^f \min\{V_{\mathrm{A}+1}^D(q, q^o), c_T\}, \\
&= R(q, q^o, m) + \beta^f V_{\mathrm{A}+1}^{T,D}(q, q^o) + \beta^f \min\{V_{\mathrm{A}+1}^D(q, q^o), c_T\}
\end{aligned}$$

Now note that $V_{\mathrm{A}+1}^{T,D}(q, q^o)$ is the difference between the values of an age A+1 trademark firm when it has $m^T(m)$ customers versus $m^N(m)$ customers to start the period. Define

$$V^{T,D,n+1} \equiv \mathbf{T_1} V^{T,D,n} = R_1(q, q^o, m^T, m^N) + \beta^f V^{T,D,n},$$

where $R_1(q, q^o, m^T, m^N) = \widetilde{\Pi}^T(q, q^o, m^T) - \widetilde{\Pi}^T(q, q^o, m^N) = (m^T - m^N)(1 - \delta_T)z(q)\pi(q, q^o)$.

The operator $\mathbf{T_1}$ has a fixed point $V^{T,D}$ by Banach fixed point theorem, as it satisfies Blackwell's sufficient conditions for a contraction mapping. $\mathbf{T_1}$ maps any bounded function $V^{T,D,n}$ into a bounded function $\mathbf{T_1}V^{T,D,n}$, and $\mathbf{T_1}V^{T,D,n}$ is continuous in its arguments because $V^{T,D,n}$ and R_1 are continuous. For any two functions $V_1^{T,D,n} \geq V_2^{T,D,n}$, it holds that $\mathbf{T_1}V_1^{T,D,n} \geq \mathbf{T_1}V_2^{T,D,n}$. Finally, it is straightforward to show that $\mathbf{T_1}$ satisfies the discounting hypothesis. Furthermore, $V^{T,D}$ is increasing q^o and m because R_1 is.

Next, let $\mathbf{T}$ be the operator such that

$$V^{D,n+1} \equiv \mathbf{T}V^{D,n} = R(q, q^o, m) + \beta^f V^{T,D} + \beta^f \min\{V^{D,n}, c_T\}.$$

Again, $\mathbf{T}$ has a fixed point $V^{T,D}$ by Banach fixed point theorem. $\mathbf{T}$ maps any bounded function $V^{D,n}$ into a bounded function $\mathbf{T}V^{D,n}$, and $\mathbf{T}V^{D,n}$ is continuous in its arguments, since $V^{D,n}$, $V^{T,D}$, and R are continuous. Given $V_1^{D,n} \geq V_2^{D,n}$, it holds that $\mathbf{T}V_1^{D,n} \geq \mathbf{T}V_2^{D,n}$ because $\min\{V_1^{D,n}, c_T\} \geq \min\{V_2^{D,n}, c_T\}$. Moreover, $\mathbf{T}$ satisfies the discounting hypothesis:

$$\begin{aligned}
\mathbf{T}(V^{D,n} + b) &= R(q, q^o, m) + \beta^f V^{T,D} + \beta^f \min\{V^{D,n} + b, c_T\}, \\
&\leq R(q, q^o, m) + \beta^f V^{T,D} + \beta^f \min\{V^{D,n} + b, c_T + b\}, \\
&= R(q, q^o, m) + \beta^f V^{T,D} + \beta^f[\min\{V^{D,n}, c_T\} + b], \\
&= \mathbf{T}V^{D,n} + \beta^f b.
\end{aligned}$$

Assume now that $V^{D,n}$ is non-decreasing in q^o. It will now be shown that this implies $\mathbf{T}V^{D,n}$ is increasing in q^o. If $V^{D,n}$ is non-decreasing in q^o, so is $\min\{V^{D,n}, c_T\}$. Furthermore, R and $V^{T,D}$ are increasing in q^o. Therefore, $V^{D,n+1}$ is increasing in q. Hence, $\mathbf{T}$ maps non-decreasing functions of q^o into increasing functions of q^o. By Theorem 3.2 in Stokey and Lucas (1989), the fixed point of $\mathbf{T}$, V^D, must also be increasing in q^o. Similarly, assume that $V^{D,n}$ is non-decreasing in m. Then, $\min\{V^{D,n}, c_T\}$ is also non-decreasing in m. Because R and $V^{T,D}$ are increasing in m, $V^{D,n+1}$ is increasing in m, or equivalently in A. The fixed point V^D is then also increasing in A.

ii) The proof follows from similar steps in part (*i*) above and the facts that $R(q^o + \Delta, q^o, m) > R(q^o, q^o, m)$ and $R_1(q^o + \Delta, q^o, m^T, m^N) > R_1(q^o, q^o, m^T, m^N)$. ■

Lemma 1 *$d(q)$ is increasing in the range of q where $s_T(q) = 0$ and $s_T(q) = 1$. It is also increasing over the ramge where $s_T(q) \in (0,1)$ as long as δ_T is not too small compared to δ_N.*

Proof of Lemma 1. Consider regions I and II in equilibrium, as described in section 4.5. In these regions, no firm trademarks ($s_N(q) = 0$). For region I, using (33) for age-A size of a firm without persuasive advertising that never trademarks, and writing out the average size of firms at quality q yields

$$\begin{aligned} d^I(q) = \delta_N \bar{m}^N(q) &= \frac{\delta_N \chi \Phi^N(n_a^N(q, q^o, -\infty))\omega(q) \sum_{\text{A}=0}^{\infty} \frac{1-[l^N(q)]^{\text{A}+1}}{1-l^N(q)}(1-\chi)^{\text{A}}}{1 - \delta_N \chi \sum_{\text{A}=0}^{\infty} \frac{1-[l^N(q)]^{\text{A}+1}}{1-l^N(q)}(1-\chi)^{\text{A}}} \\ &= \frac{\delta_N \Phi^N(n_a^N(q, q^o, -\infty))\omega(q)}{(1-\delta_N)(1-z(q)(1-\chi))}, \end{aligned}$$

Note that the numerator is increasing in q because $\Phi^N((n_a^N(q, q^o, -\infty))$ and $\omega(q)$ are increasing in q, and the denominator is decreasing in q because $z(q)$ is increasing. Hence, $d(q)$ is increasing.

In region II, there are firms with and without persuasive advertising and they never trademark. In this case,

$$d^{II}(q) = \frac{\delta_N[(1-\varphi)\Phi^N(n_a^N(q, q^o, -\infty)) + \varphi\Phi^N(n_a^N(q, q^o - \Delta, -\infty))]\omega(q)}{(1-\delta_N)(1-z(q)(1-\chi))},$$

which is again increasing in q. Note also that $d^{II}(q) > d^I(q)$, because $\Phi^N(n_a^N(q, q^o - \Delta, -\infty)) > \Phi^N(n_a^N(q, q^o, -\infty))$, and $\varphi \in (0,1)$. Therefore, $d(q)$ is increases from region I to II.

In regions VII and VI, all firms trademark ($s_T(q) = 1$). In region VII, all firms have persuasive advertising. Using (32) for age-A size of a firm that trademarks at age zero and writing out the average size of firms at quality q yields

$$d^{VII}(q) = \frac{\delta_T \Phi^T(n_a^T(q, q^o - \Delta, 0))\omega(q)}{(1-\delta_T)(1-z(q)(1-\chi))},$$

In region VI, both firms with and without persuasive advertising trademark, and

$$d^{VI}(q) = \frac{\delta_T[(1-\varphi)\Phi^T(n_a^T(q, q^o, 0)) + \varphi\Phi^T(n_a^T(q, q^o - \Delta, 0))]\omega(q)}{(1-\delta_T)(1-z(q)(1-\chi))}.$$

Again, for both cases, the numerator is increasing in q because $\Phi^T(\cdot)$ and $\omega(q)$ are increasing in q, and the denominator is decreasing in q because $z(q)$ is increasing. Therefore, $d(q)$ is increasing. Note also that $d^{VII}(q) > d^{VI}(q)$, and $d(q)$ increases from region VI to VII.

Finally, consider the range of q where $s_T(q) \in (0,1)$ – corresponding to regions III, IV, and V. In this range, for any given q either some firms with persuasive advertising, or some without persuasive advertising, or some of both types will have trademarks. For a given type, all firms with age $\text{A} \geq \text{A}(q)$

for some threshold age $\text{A}(q)$ will have trademarks. Thus, in this range the change in $d(q)$ depends on the change in the trademark age, $\text{A}(q)$, as quality increases. As q increases, the size of each firm increases. At the same time, non-trademark firms at age $\text{A}(q)$ become trademark firms, and their customer diversion rate changes from δ_N to δ_T. Because $\delta_T < \delta_N$, this effect can reduce $d(q)$ if it dominates the increase in firm size for all firms. However, this effect will be small if δ_T is not too small compared to δ_N.

To fix these arguments, consider region III, for example. In this region, firms with persuasive advertising will trademark above some age $\text{A}(q)$ whereas firms without persuasive advertising will not trademark at all. Thus,

$$d^{III}(q) = \frac{(1-\varphi)\delta_N \chi \Phi^N(n_a^N(q, q^o, -\infty)\omega(q) + \varphi\chi\omega(q)[\delta_N S^N(q) + \delta_T S^T(q)]}{1-(1-\varphi)(1-\delta_N)(1-z(q)(1-\chi)) - \varphi\chi[\delta_N J^N(q) + \delta_T J^T(q)]},$$

where

$$S^N(q) = \sum_{\text{A}=0}^{\text{A}(q)-1} \Phi^N(n_a^N(q, q^o - \Delta, \text{A} - \text{A}(q))) \frac{1-[l^N(q)]^{\text{A}+1}}{1-l^N(q)}(1-\chi)^{\text{A}},$$

$$S^T(q) = \sum_{\text{A}=\text{A}(q)}^{\infty} \Phi^T(n_a^T(q, q^o - \Delta, 0)) \frac{1-[l^T(q)]^{\text{A}+1}}{1-l^T(q)}(1-\chi)^{\text{A}},$$

$$J^N(q) = \sum_{\text{A}=0}^{\text{A}(q)-1} \frac{1-[l^N(q)]^{\text{A}+1}}{1-l^N(q)}(1-\chi)^{\text{A}},$$

$$J^T(q) = \sum_{\text{A}=\text{A}(q)}^{\infty} \frac{1-[l^T(q)]^{\text{A}+1}}{1-l^T(q)}(1-\chi)^{\text{A}}.$$

Note that as q increases the threshold trademark age $\text{A}(q)$ decreases. Hence, firms at age $\text{A}(q)-1$ will register trademarks. This effect increases $S_T(q)$ and decreases $S_N(q)$, holding Φ_N, Φ_T, ω and z constant. However, if δ_T is sufficiently small relative to δ_N the net effect can be a reduction in the term $[\delta_N S^N(q) + \delta_T S^T(q)]$. Similarly, this effect can also increase the denominator if $[\delta_N J^N(q) + \delta_T J^T(q)]$ decreases. Overall, then, $d(q)$ can decrease. If, however, δ_T is not too small relative to δ_N, this effect will be small compared to the effects of the increases in Φ_N, Φ_T, ω, and z. As a result, $d(q)$ will increase. ■